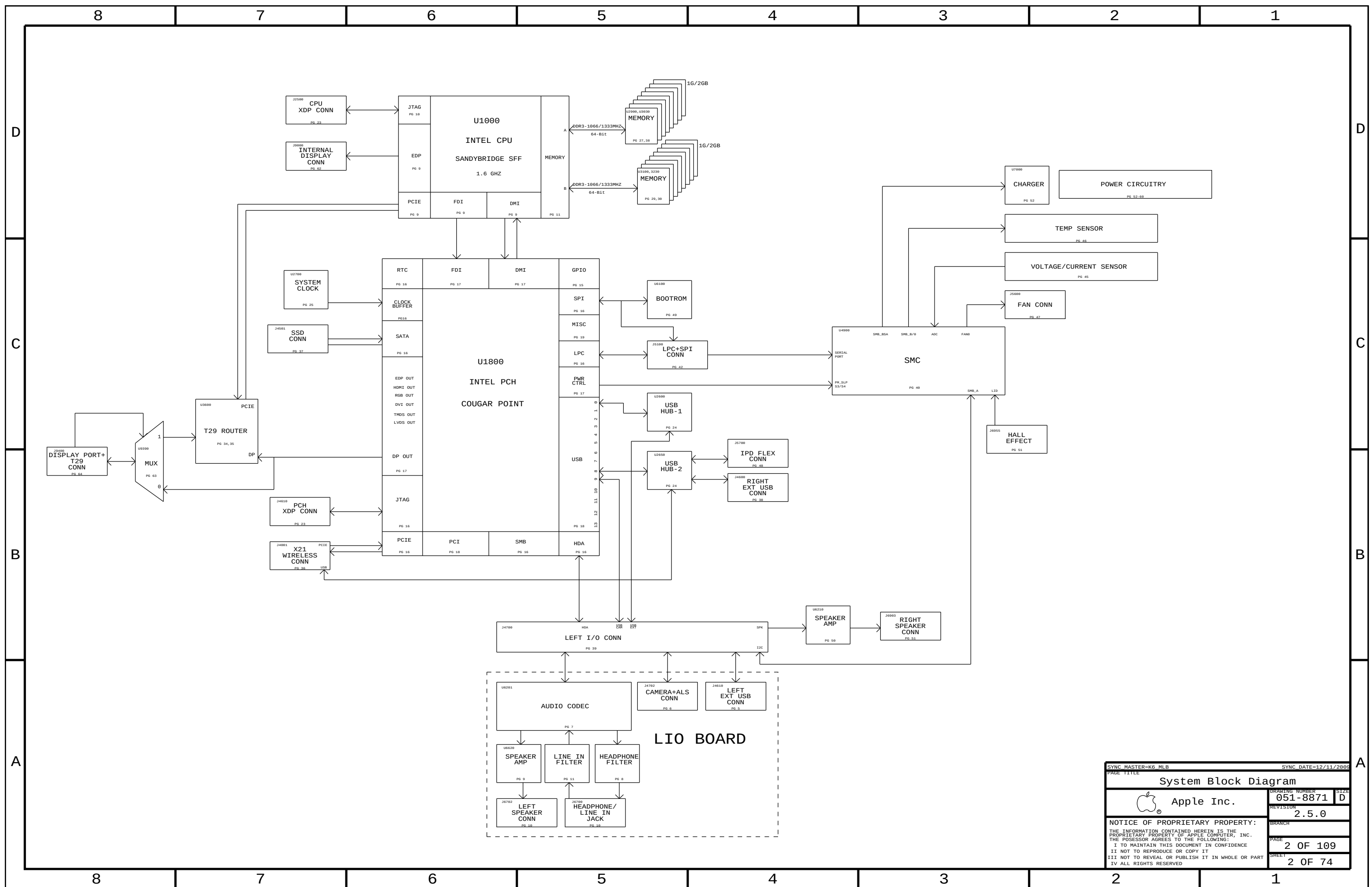
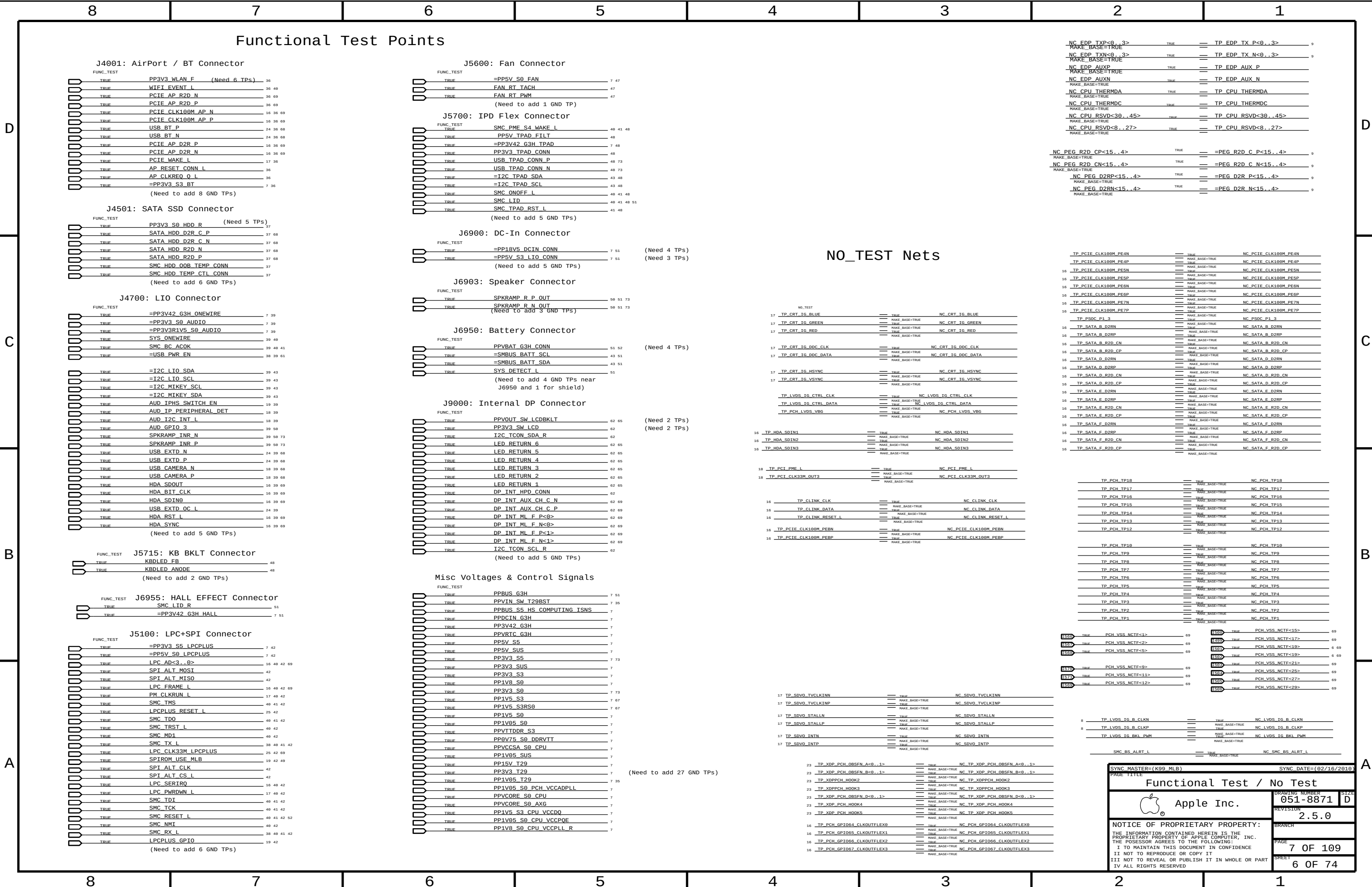




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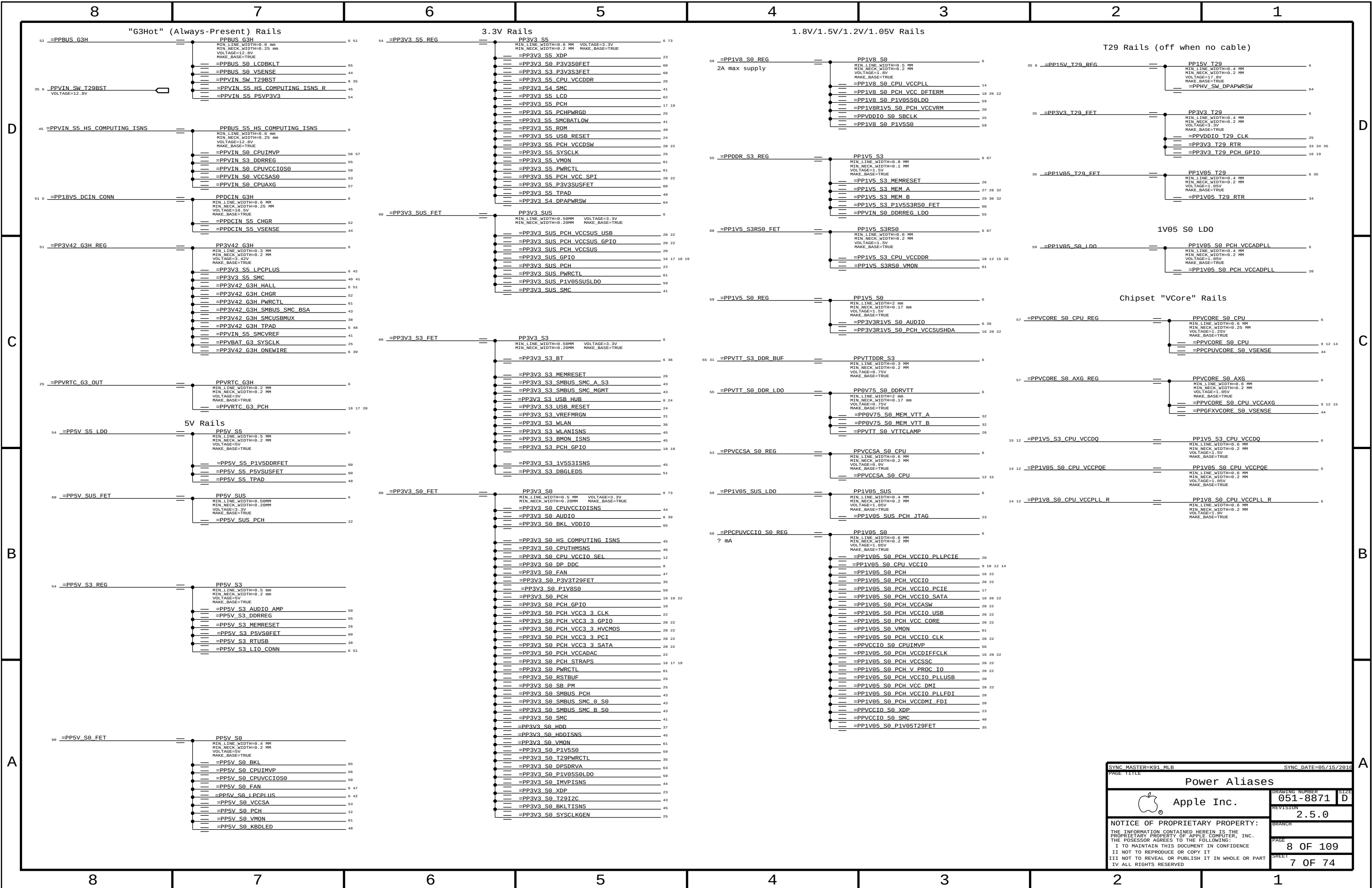
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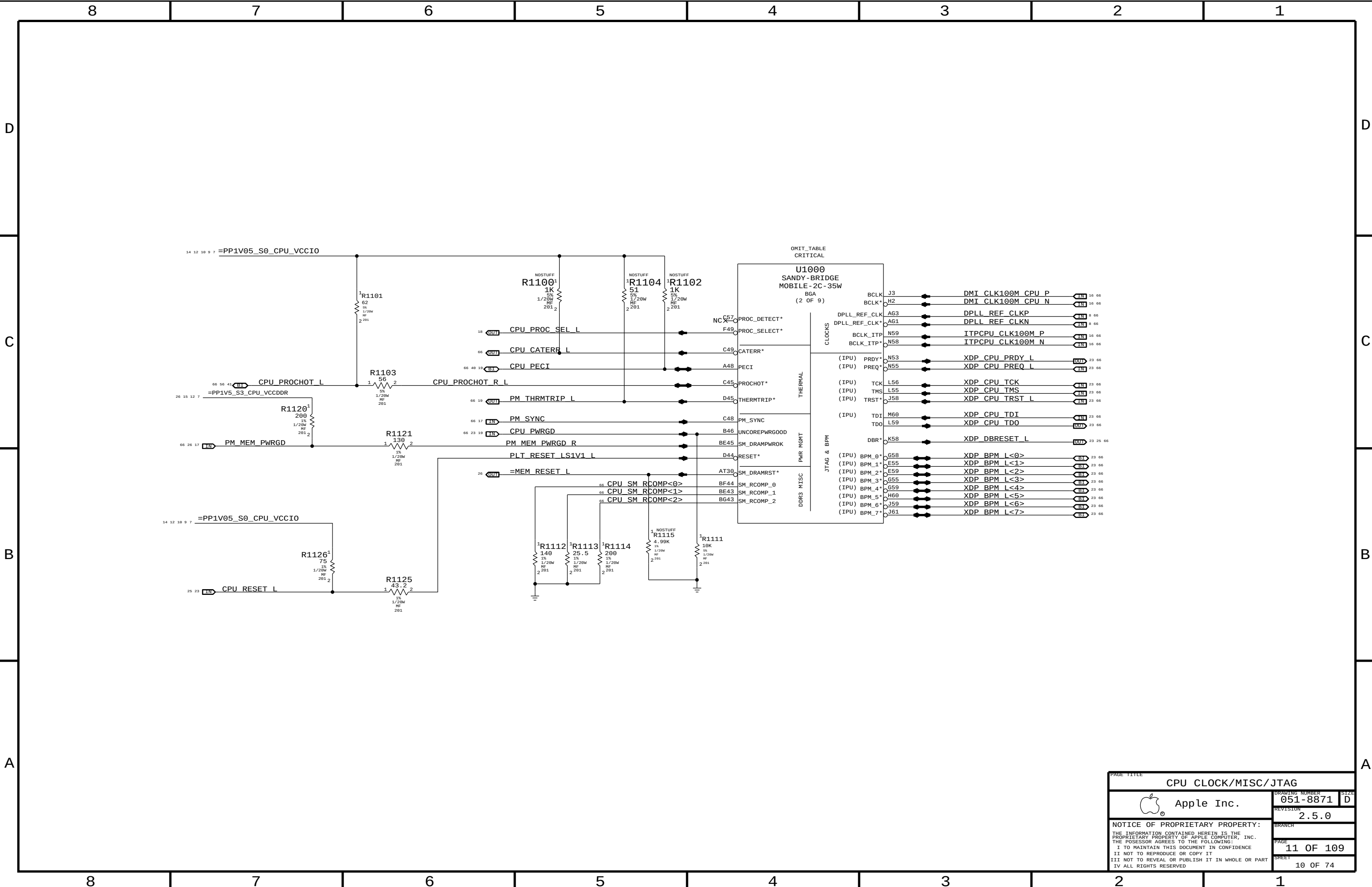
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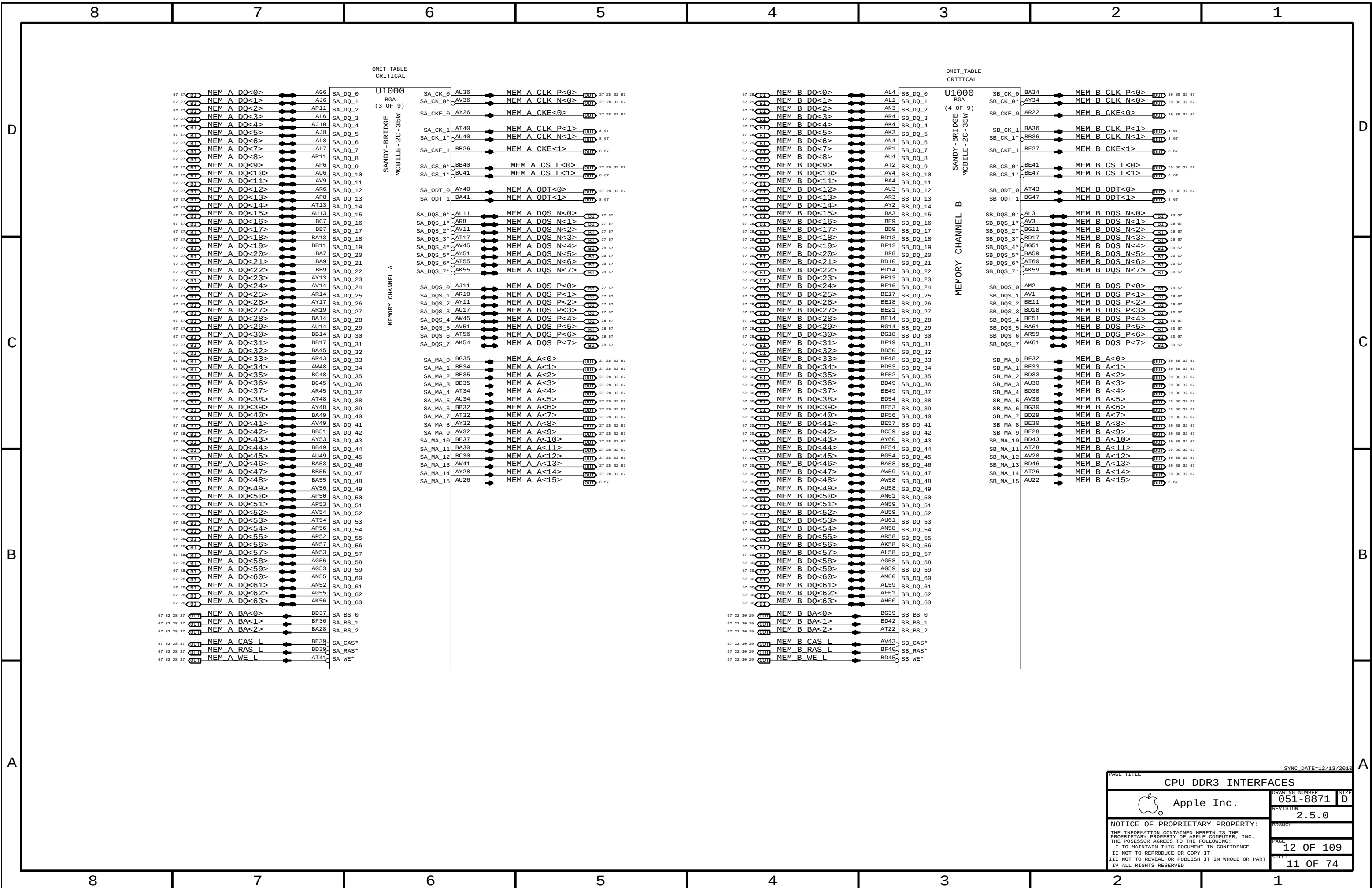
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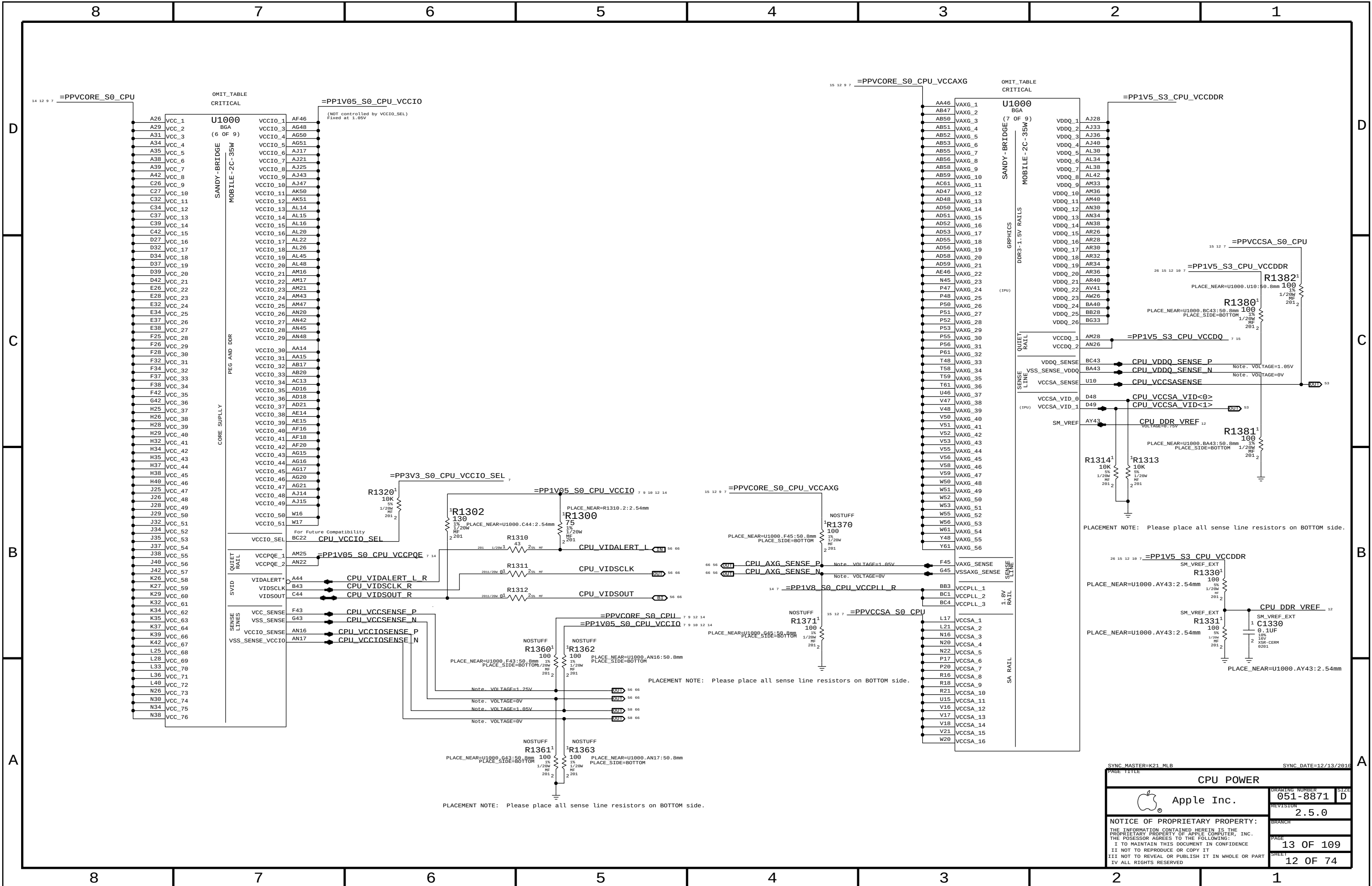
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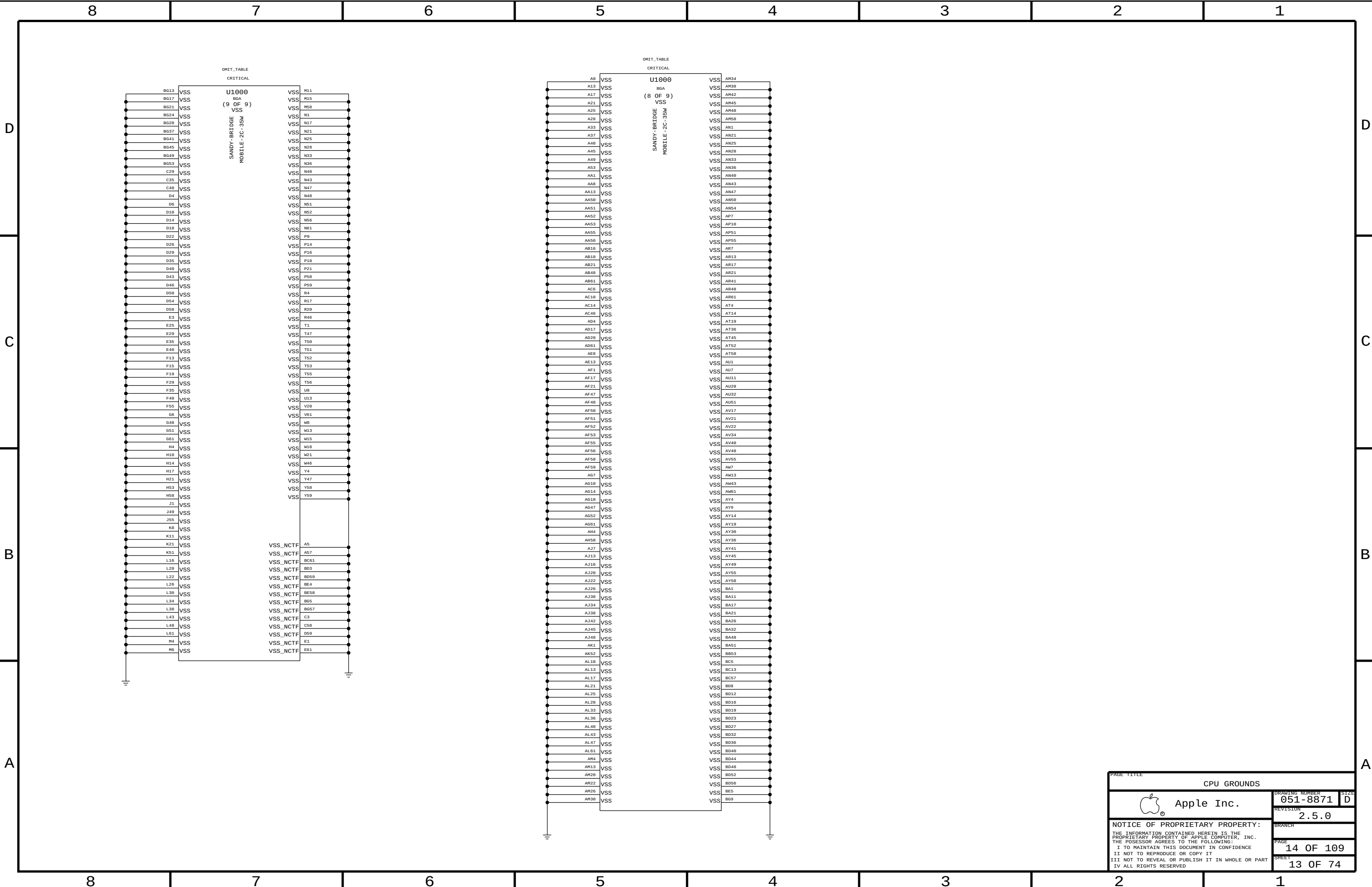


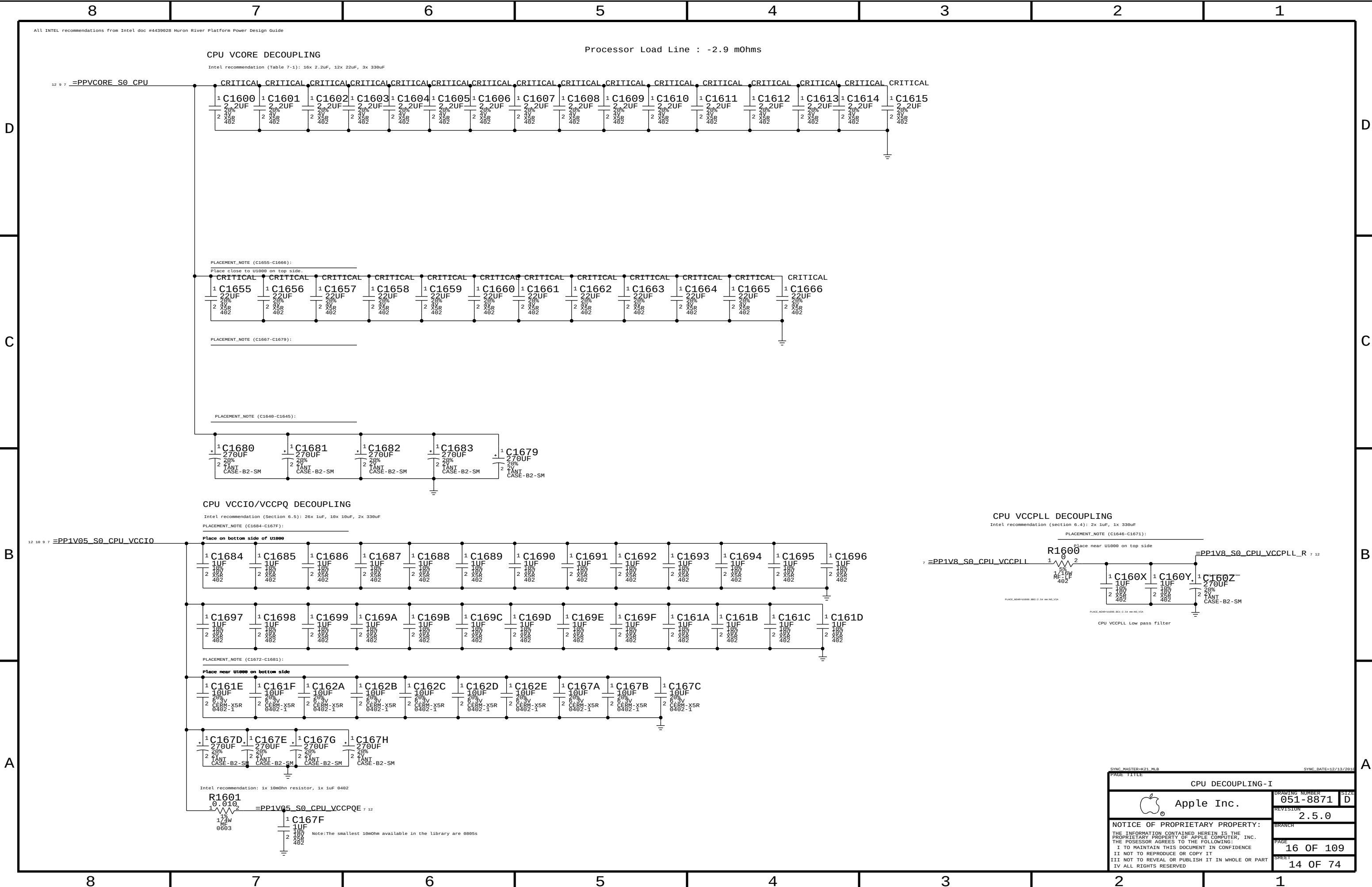






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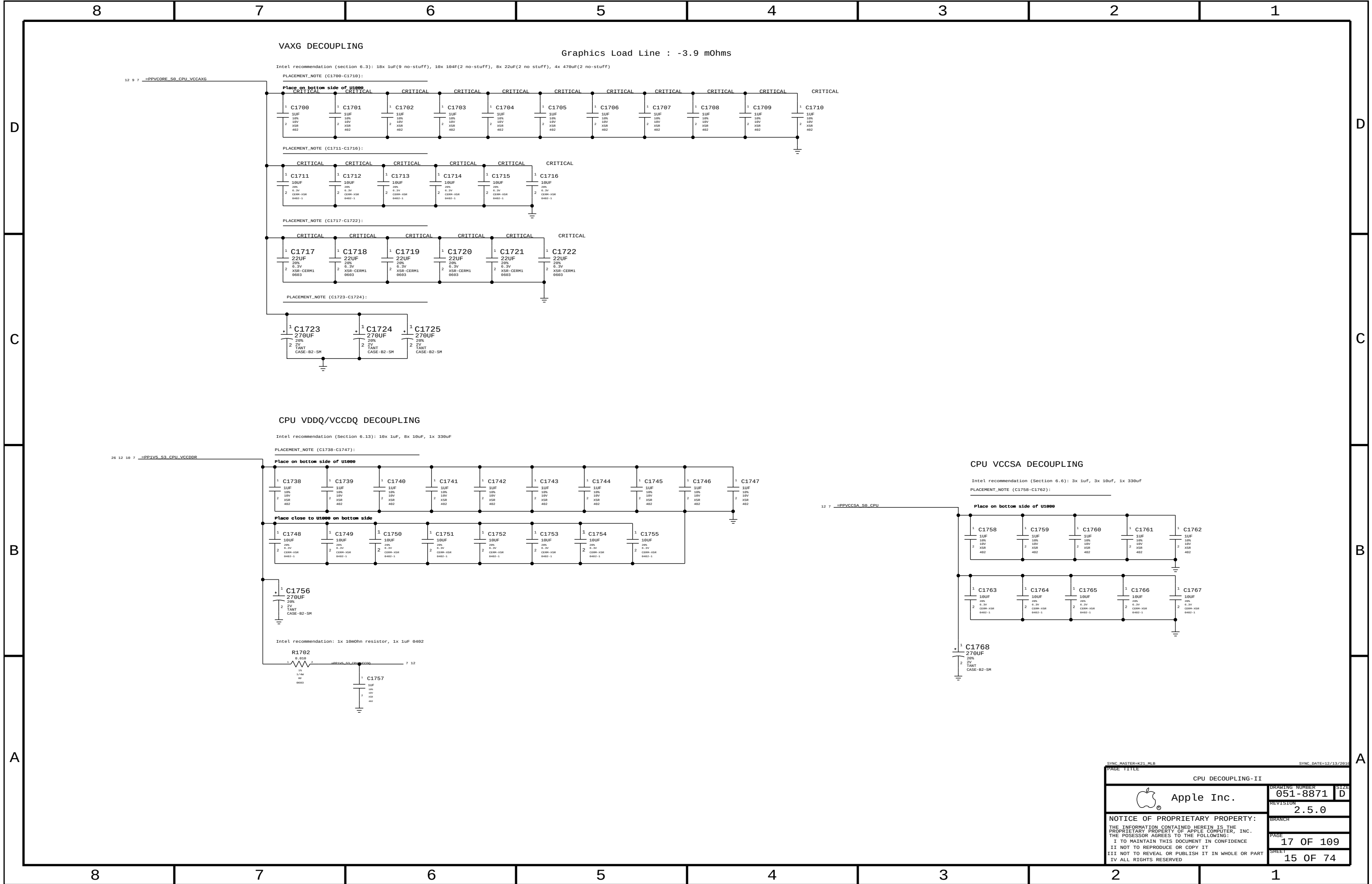
PAGE TITLE		CPU DECOUPLING-I	
DRAWING NUMBER		051-8871	SIZE D
REVISION		2.5.0	BRANCH
PAGE		16 OF 109	SHEET
SHEET		14 OF 74	

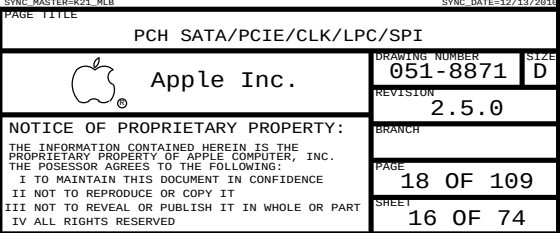
SYNC MASTER=k21_MLB

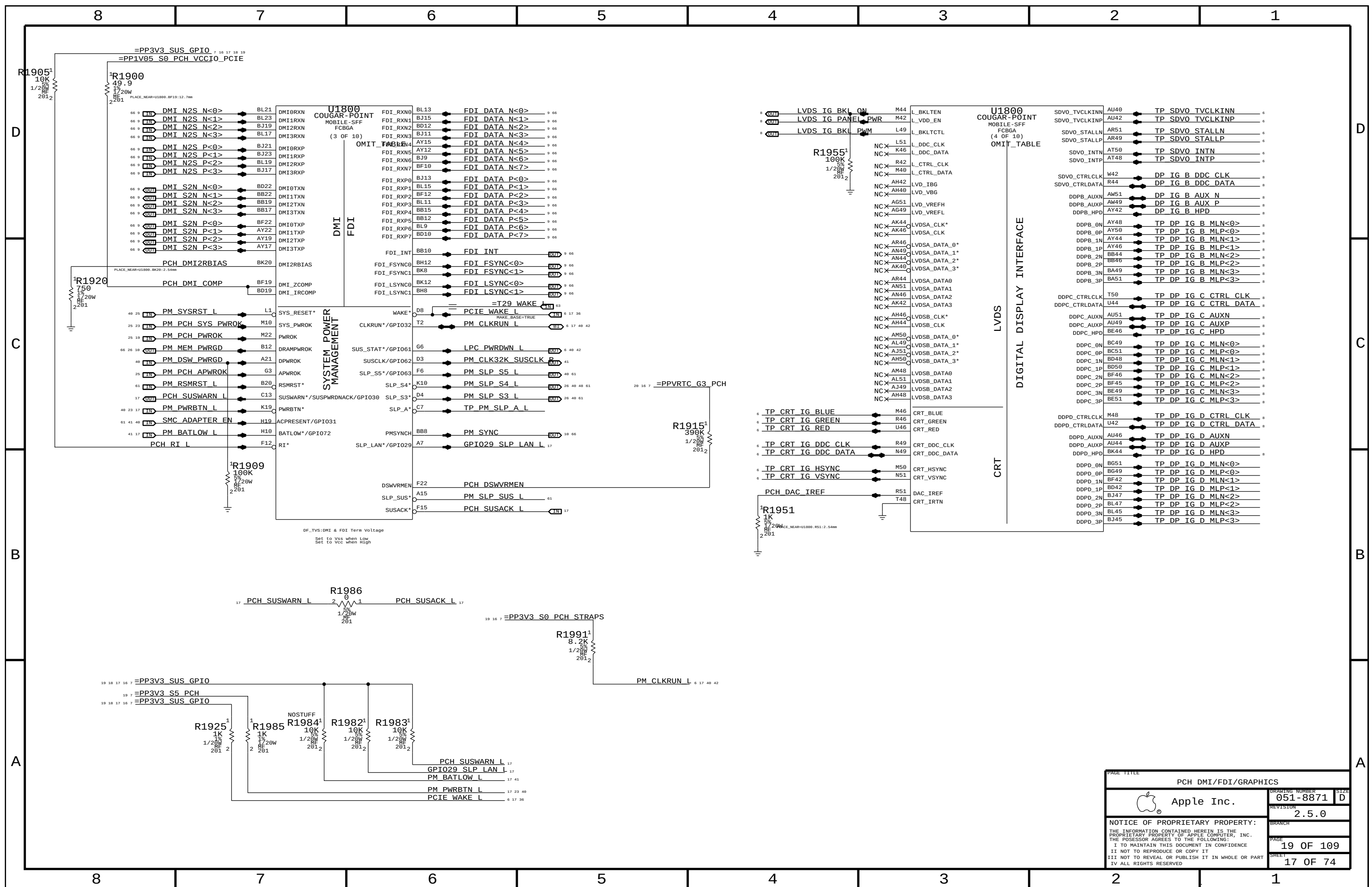
SYNC DATE=12/13/2016

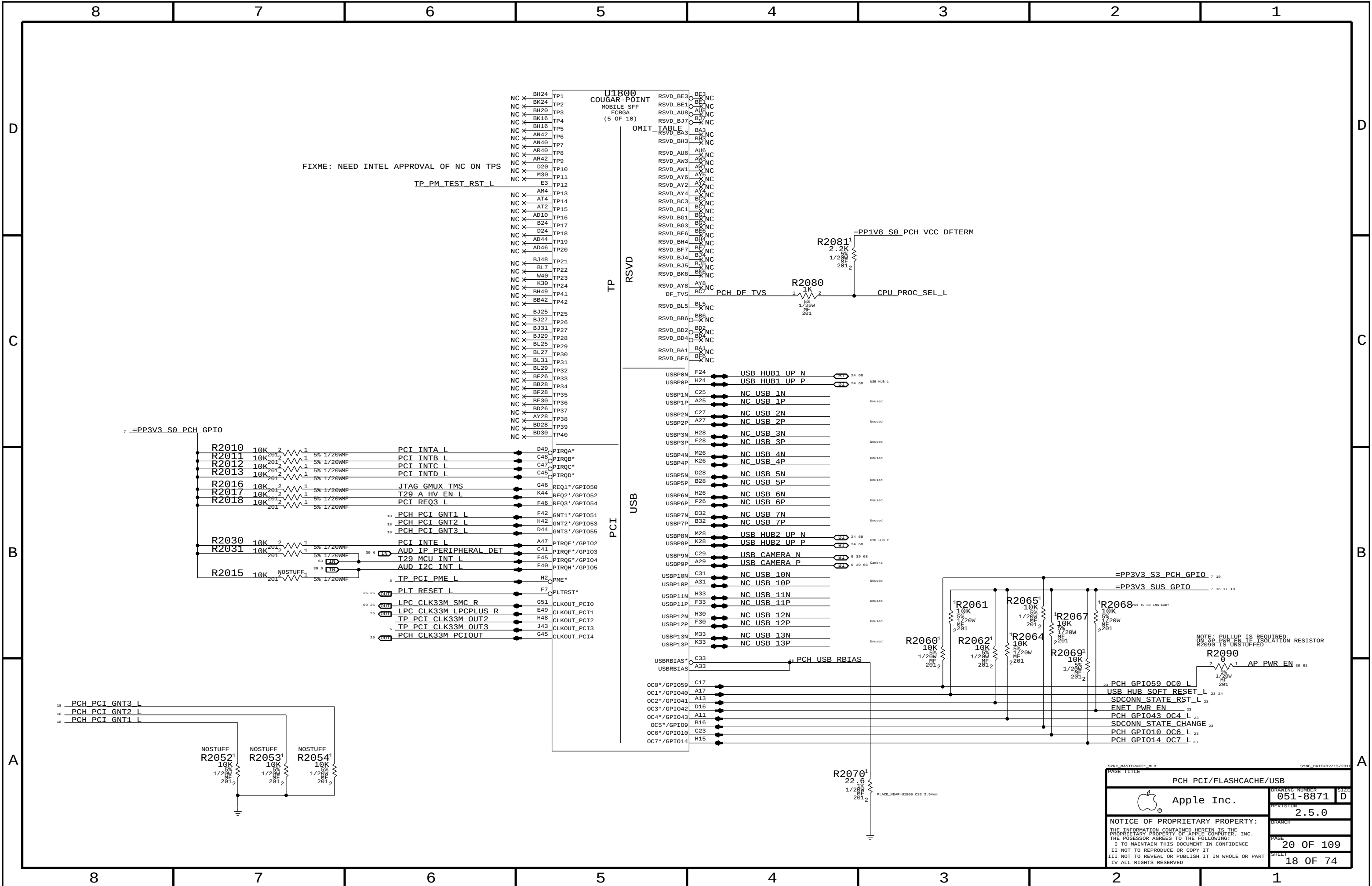
Apple Inc.

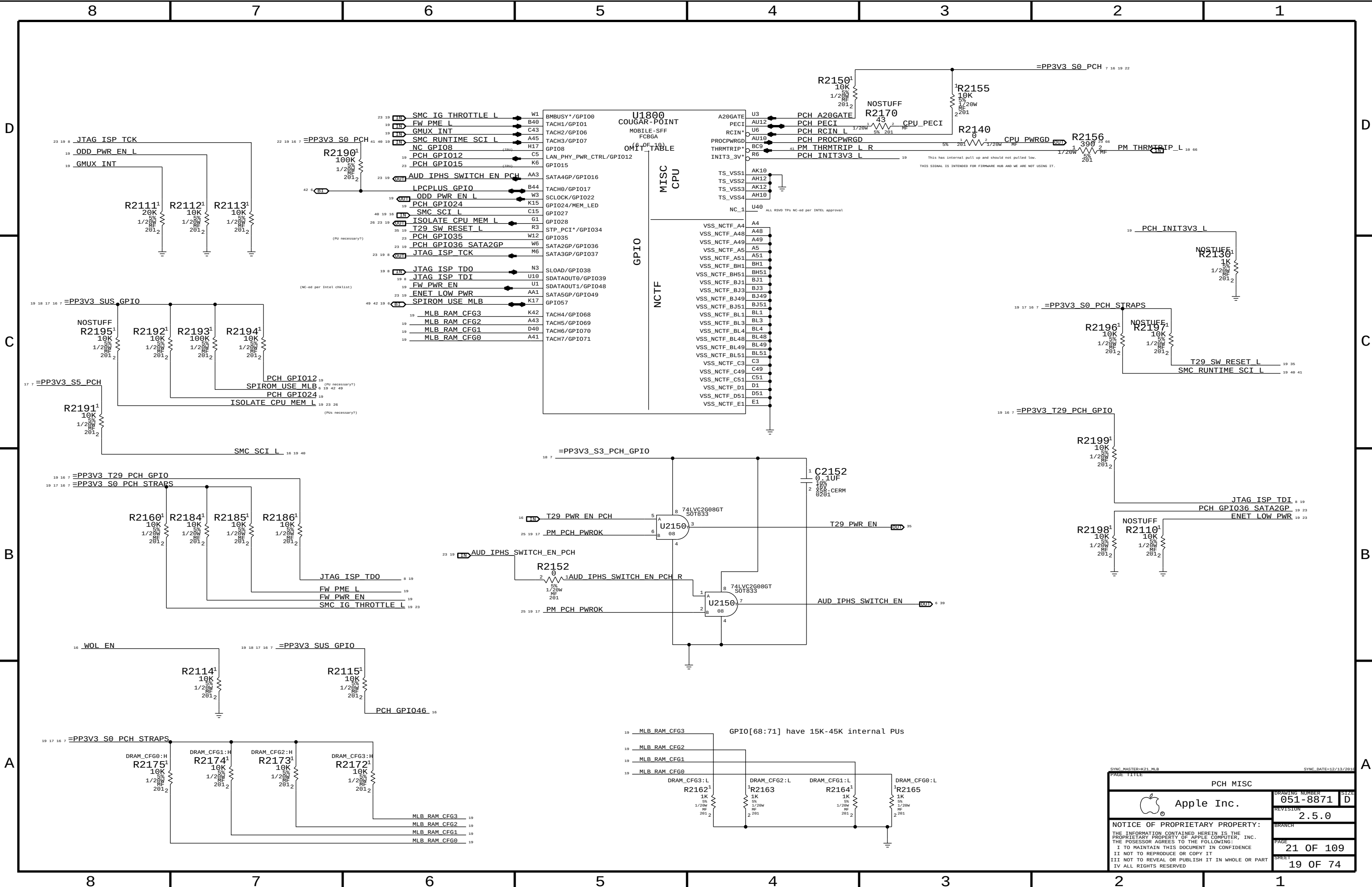
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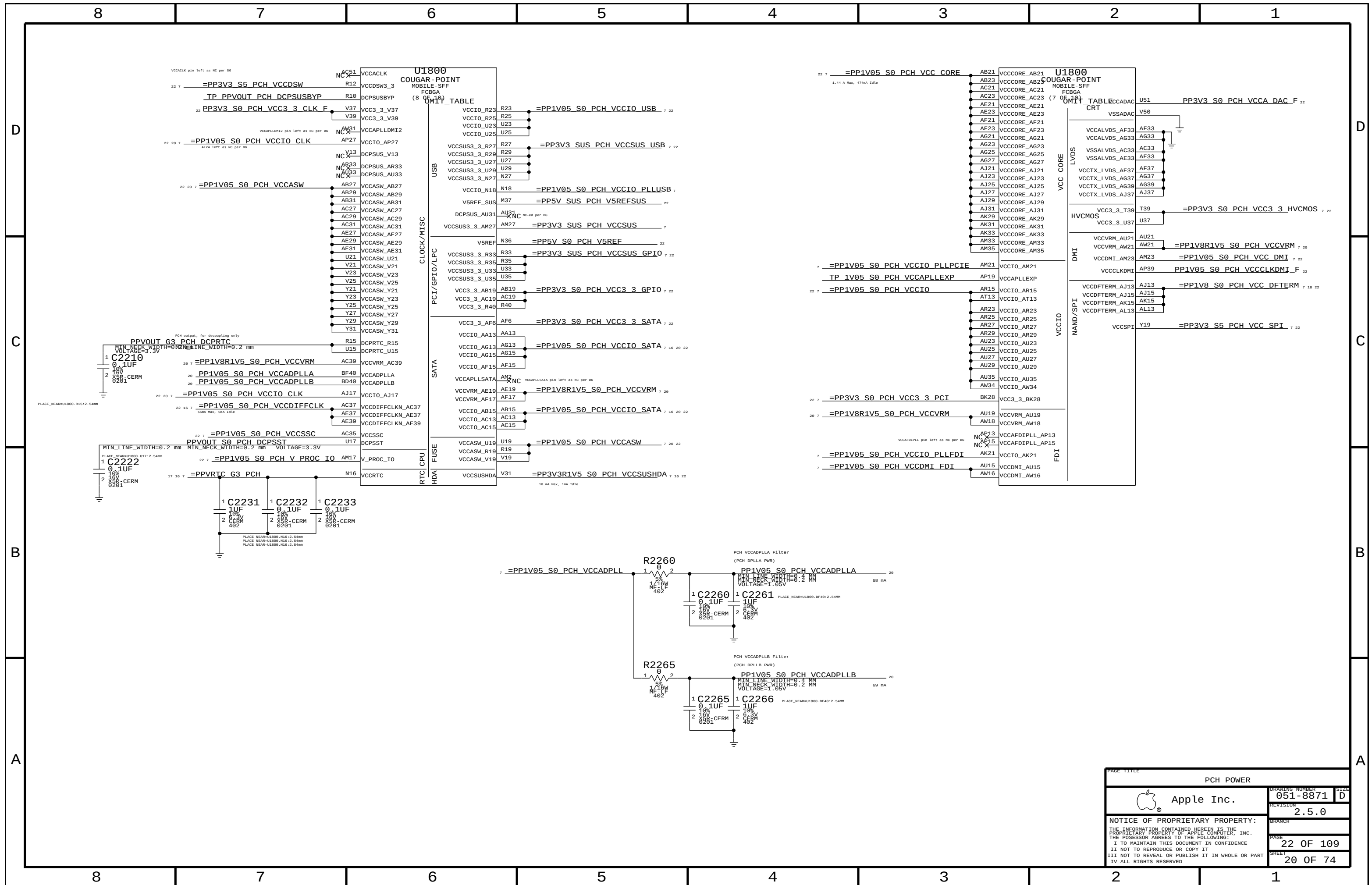


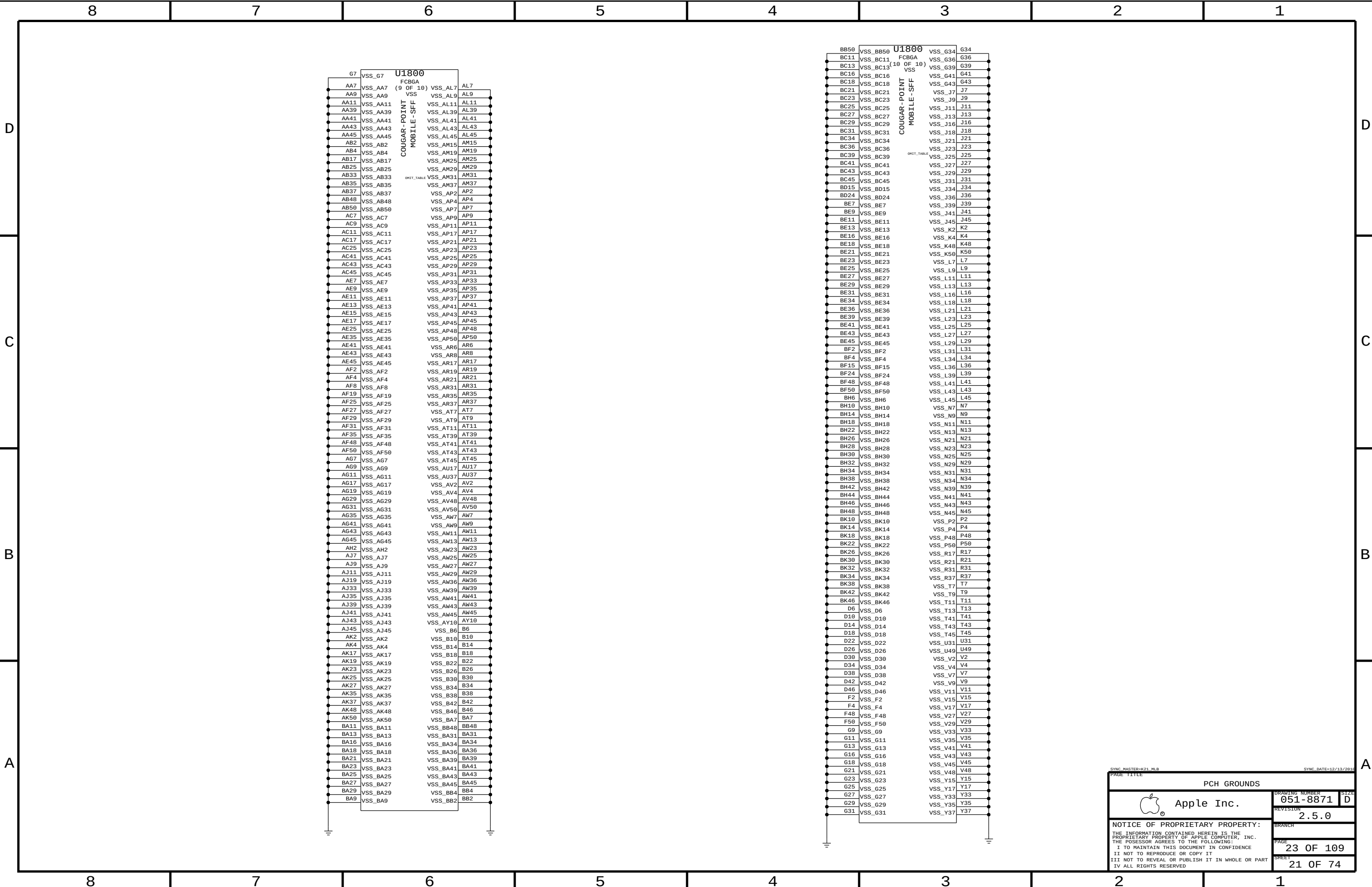


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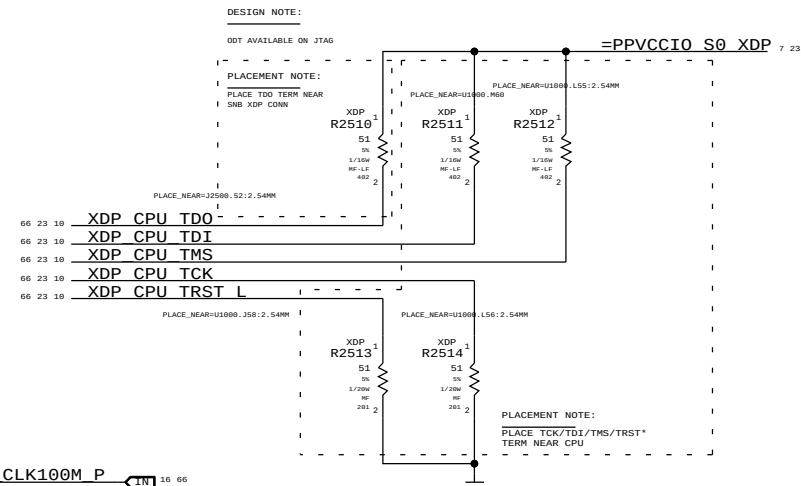
SYNC DATE=12/13/2011

PAGE TITLE		PCH MISC	
		DRAWING NUMBER	051-8871
		REVISION	2.5.0
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		PAGE	21 OF 109
		SHEET	19 OF 74





NOTE: This is not the standard XDP pinout
Use with 920-0782 Adapter Flex to support chipset debug



PLACEMENT NOTE:
 PLACE TDRM TERM NEAR
 PCH XDP CONN

XDP R2550 51 1/16 1/16 402 2

XDP R2551 51 1/250 1/250 201 2

XDP R2551 51 1/250 1/250 201 2

PLACE_NEAR=12550.52;2.54MM

XDP PCH TDRM

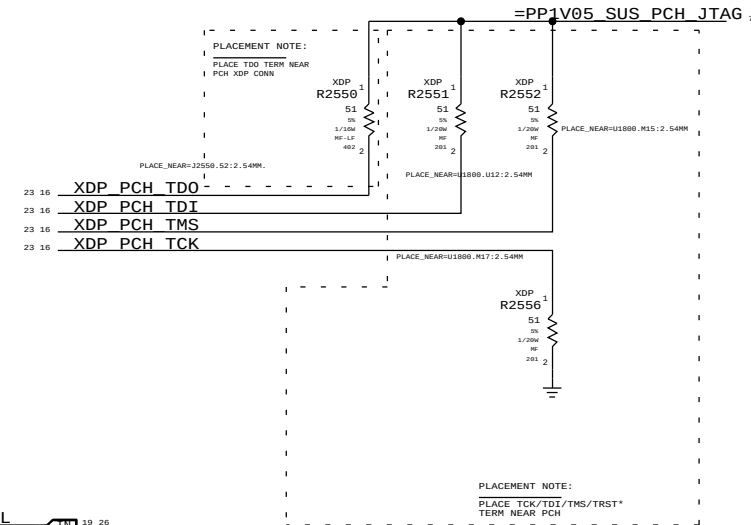
XDP PCH TDI

XDP PCH TMS

XDP PCH TCK

=PP1V05_SUS_PCH_JTAG

NOTE: This is not the standard XDP pinout
Use with 920-0782 Adapter Flex to support chipset debug



SYNC MASTER=K21 MLB		SYNC DATE=12/13/2016	
PAGE TITLE			
CPU & PCH XDP			
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		25 OF 109	
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CPU & PCH XDP



Apple Inc.

DRAWING NUMBER
051-8871

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SHEET

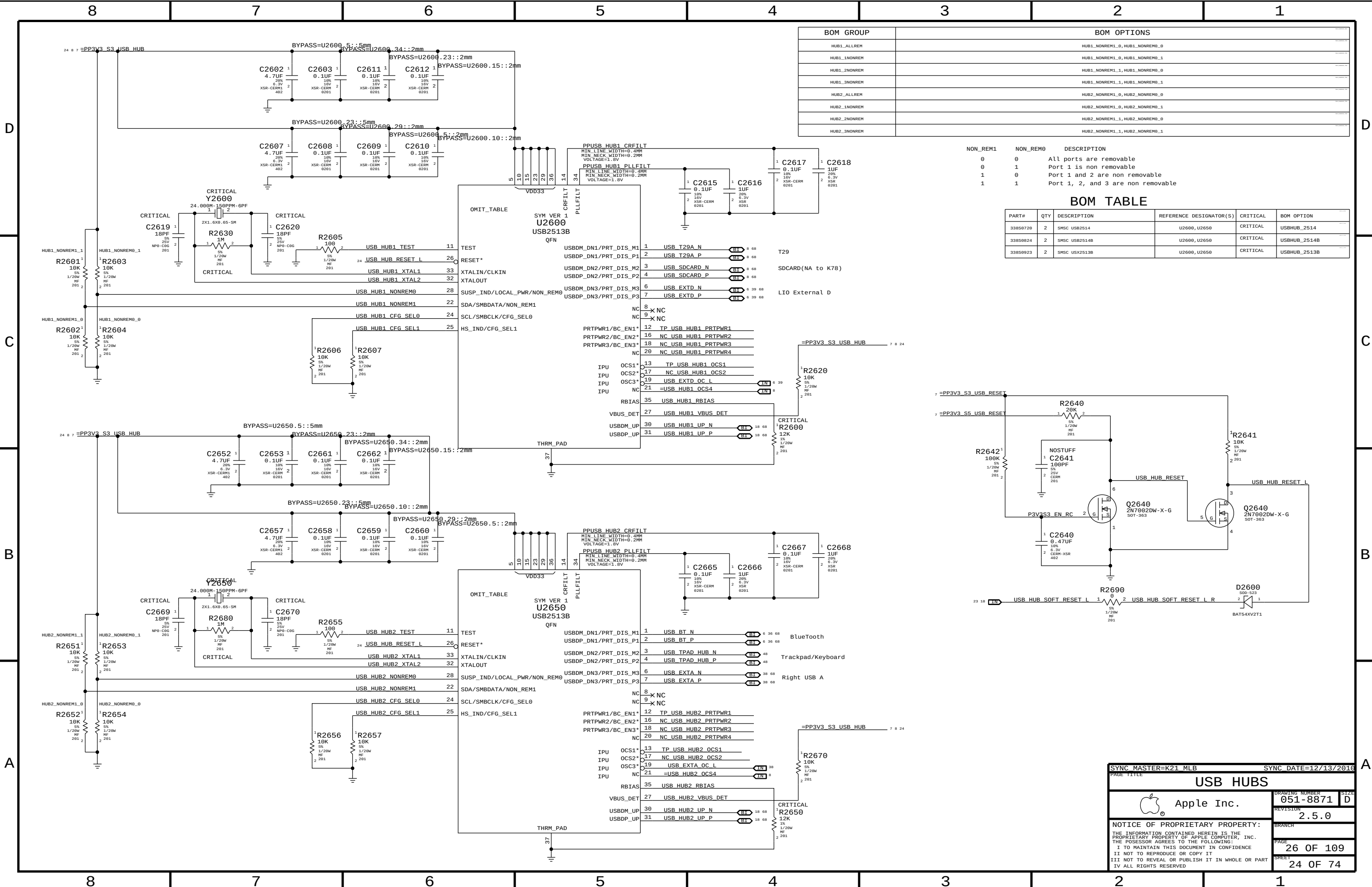
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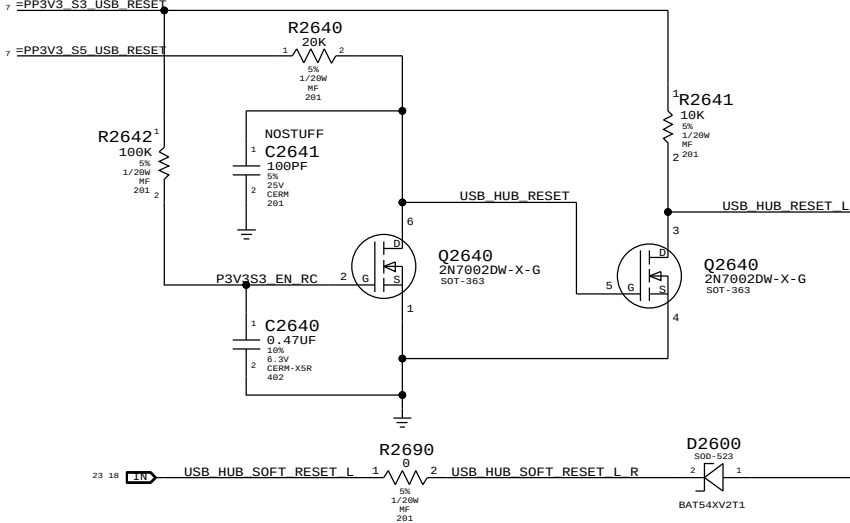


BOM GROUP		BOM OPTIONS	
HUB1_ALLREM		HUB1_NONREM1_0, HUB1_NONREM0_0	
HUB1_1NONREM		HUB1_NONREM1_0, HUB1_NONREM0_1	
HUB1_2NONREM		HUB1_NONREM1_1, HUB1_NONREM0_0	
HUB1_3NONREM		HUB1_NONREM1_1, HUB1_NONREM0_1	
HUB2_ALLREM		HUB2_NONREM1_0, HUB2_NONREM0_0	
HUB2_1NONREM		HUB2_NONREM1_0, HUB2_NONREM0_1	
HUB2_2NONREM		HUB2_NONREM1_1, HUB2_NONREM0_0	
HUB2_3NONREM		HUB2_NONREM1_1, HUB2_NONREM0_1	

NON_REM1	NON_REM0	DESCRIPTION
0	0	All ports are removable
0	1	Port 1 is non removable
1	0	Port 1 and 2 are non removable
1	1	Port 1, 2, and 3 are non removable

BOM TABLE

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
33850728	2	SMSC USB2514	U2600, U2650	CRITICAL	USBHUB_2514
33850824	2	SMSC USB2514B	U2600, U2650	CRITICAL	USBHUB_2514B
33850923	2	SMSC USB2513B	U2600, U2650	CRITICAL	USBHUB_2513B



SYNC MASTER=K21 MLB

SYNC DATE=12/13/2016

USB HUBS

Apple Inc.

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051-8871

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REVISION

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PAGE

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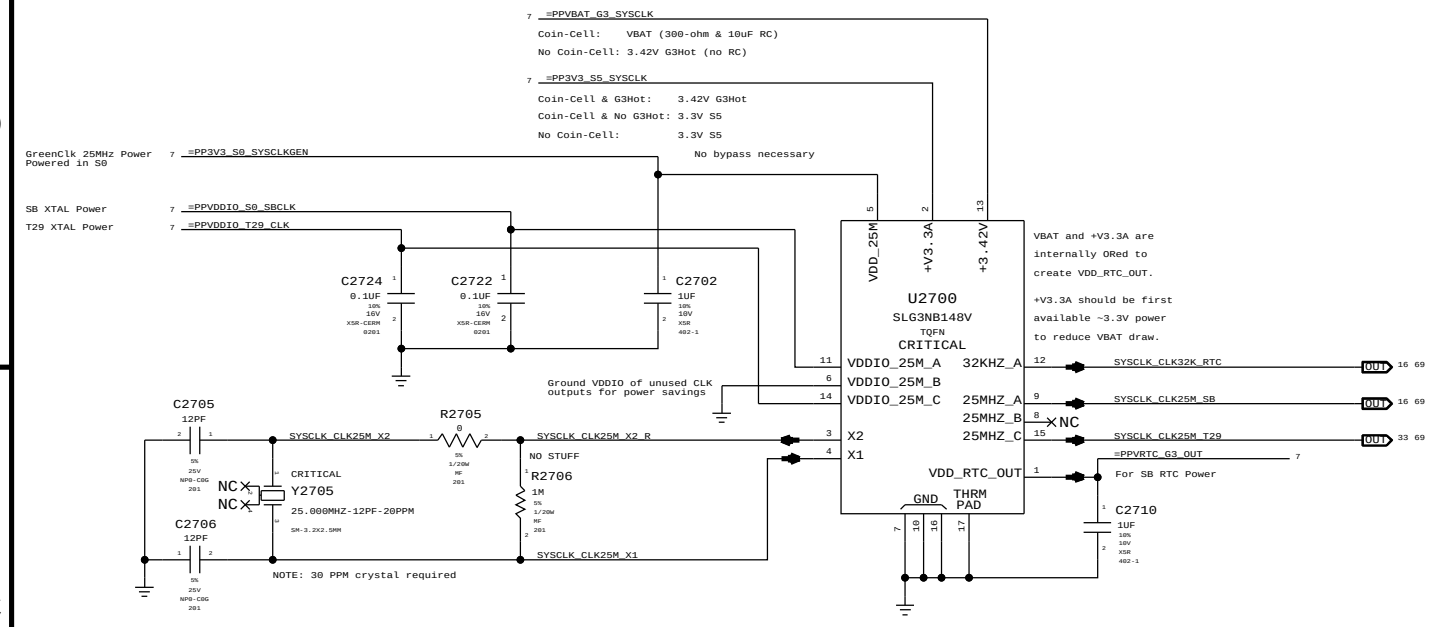
System RTC Power Source & 32kHz / 25MHz Clock Generator

D

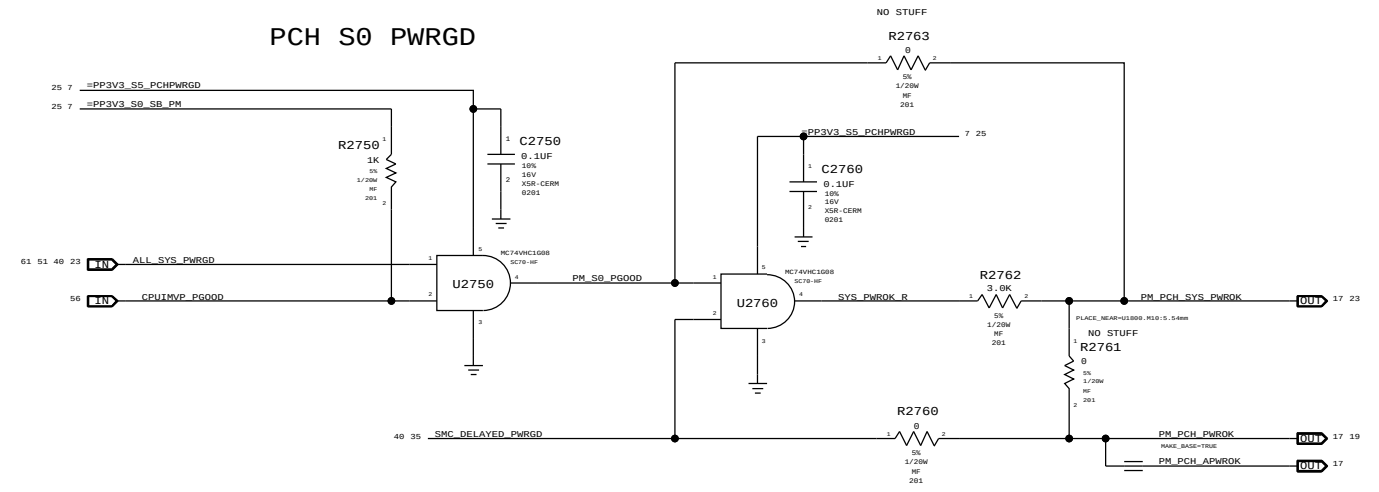
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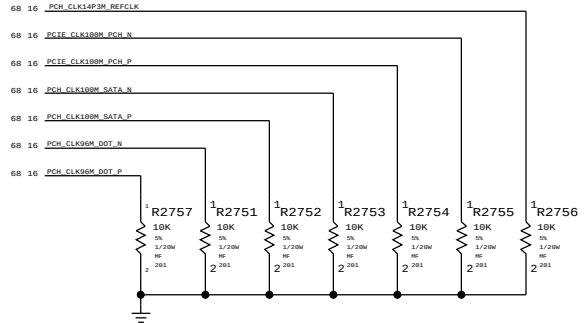


PCH S0 PWRGD

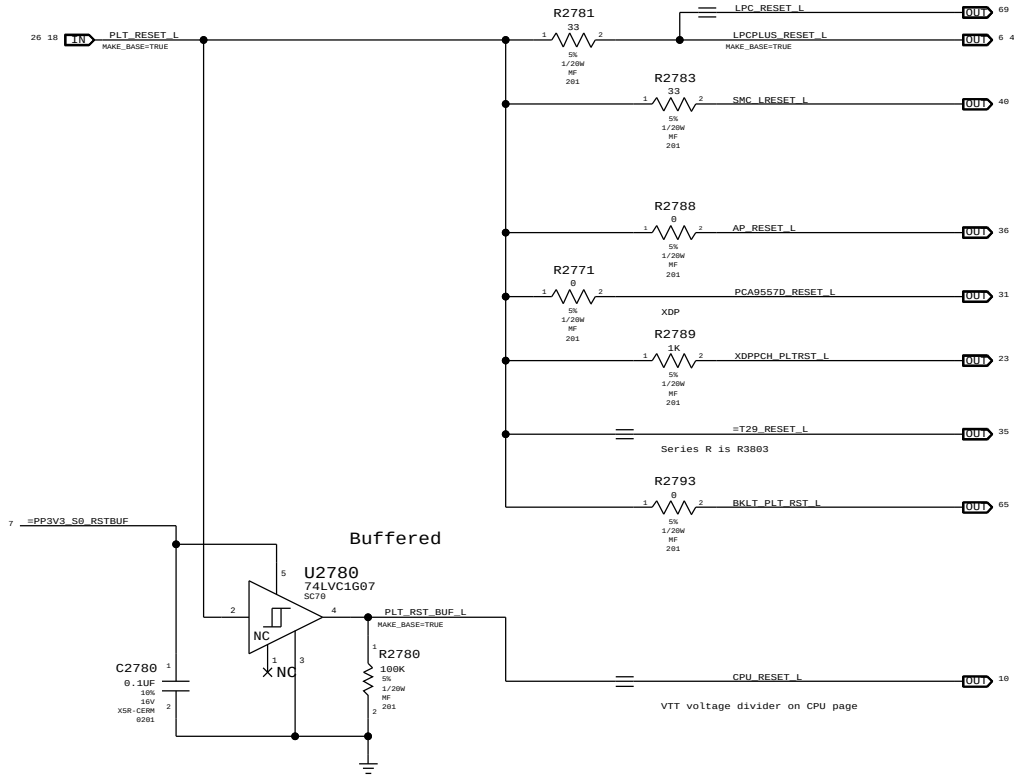


CLOCK (CK505)

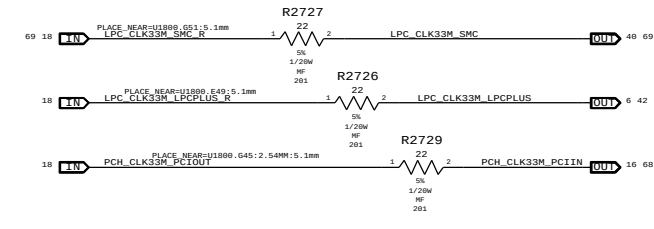
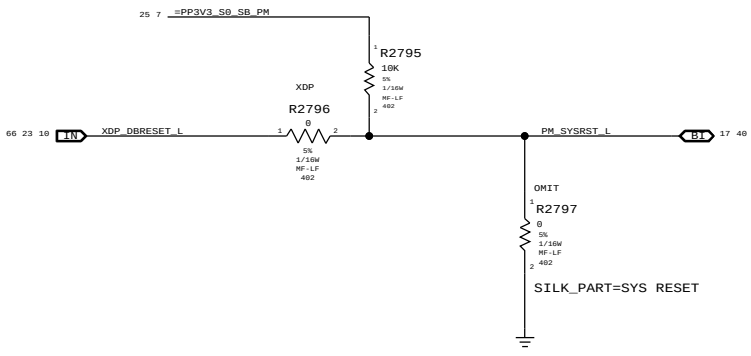
UNUSED clock terminations for FCIM MODE



Platform Reset Connections
Unbuffered



PCH Reset Button



Clock (CK505) and Chipset Support

Apple Inc.

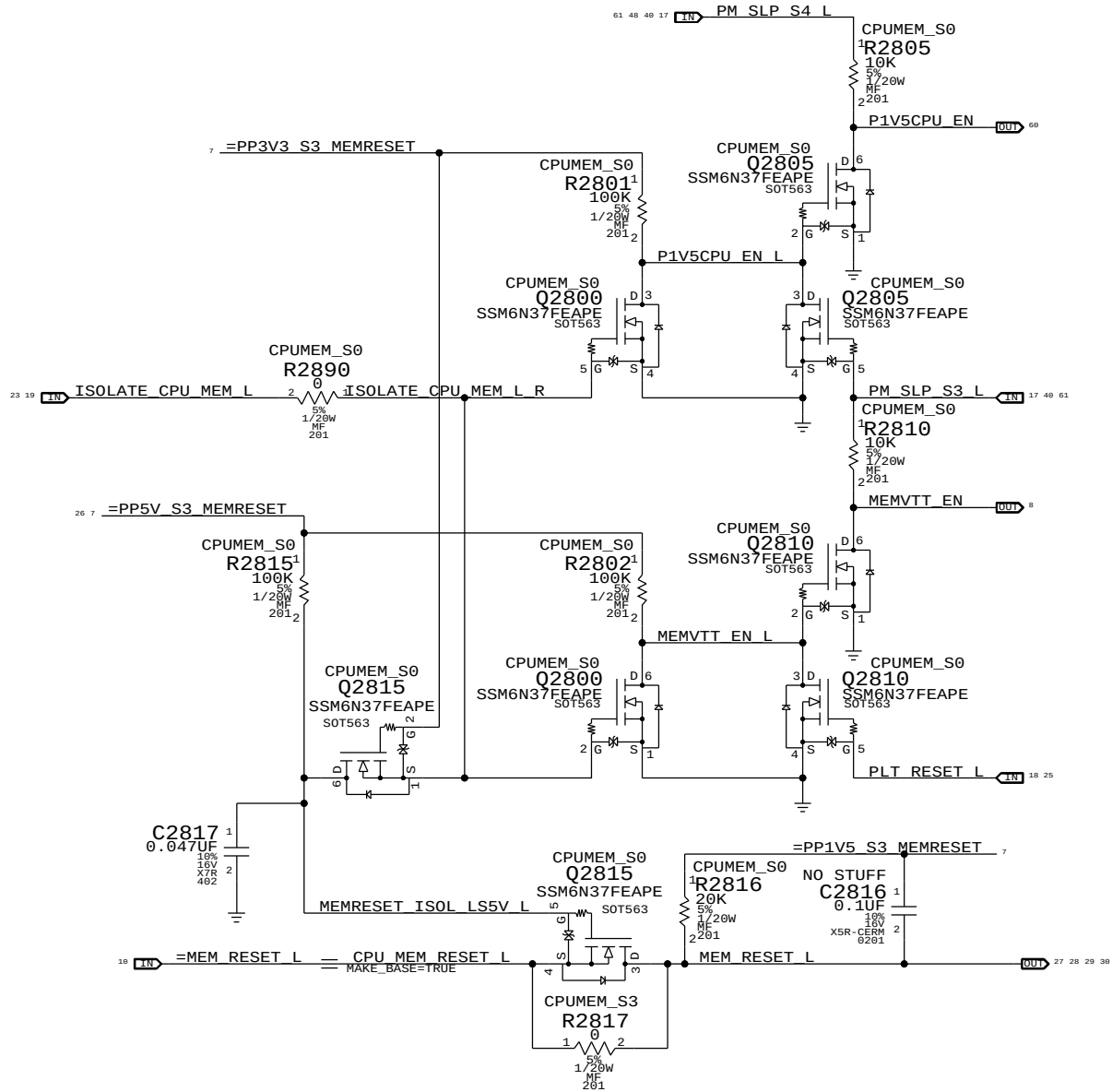
DRAWING NUMBER 051-8871
REVISION 2.5.0
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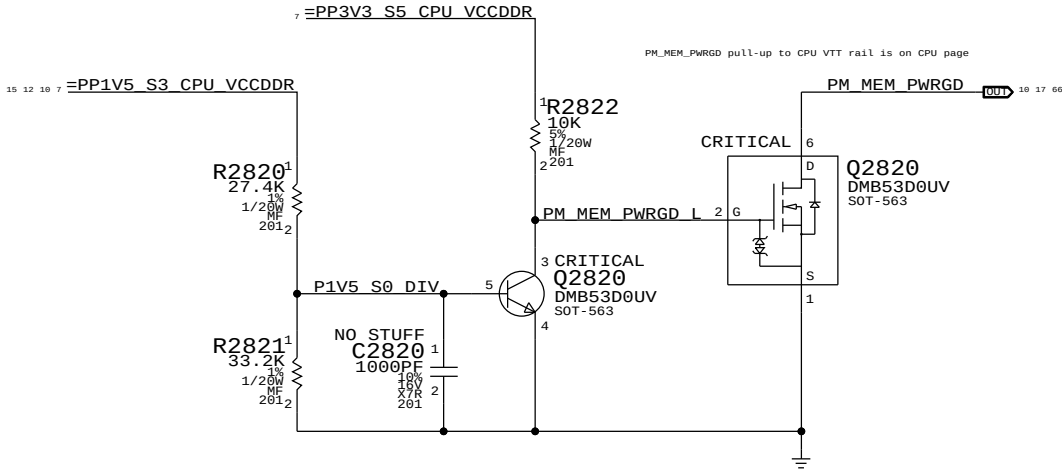
The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the S0-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3->S0 transitions determines behavior of signals.
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.
WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

P1V5CPU_EN = (ISOLATE_CPU_MEM_L + PM_SLP_S3_L) * PM_SLP_S4_L
MEMVTT_EN = (ISOLATE_CPU_MEM_L + PLT_RST_L) * PM_SLP_S3_L
MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L

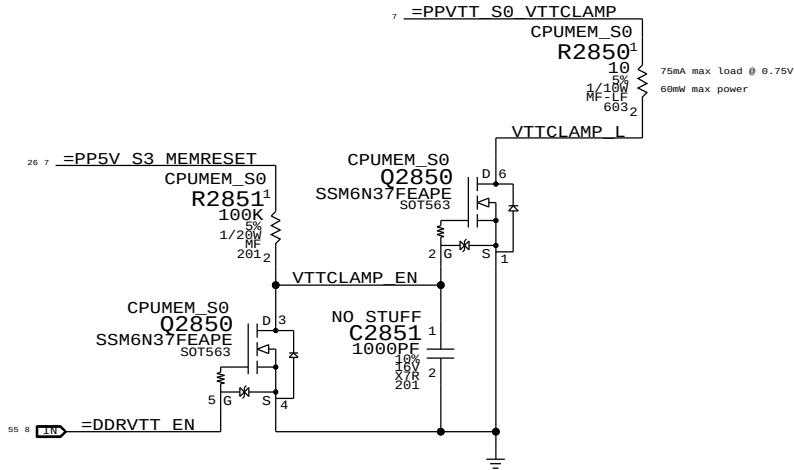


1V5 S0 "PGOOD" for CPU



MEMVTT Clamp

Ensures CKE signals are held low in S3

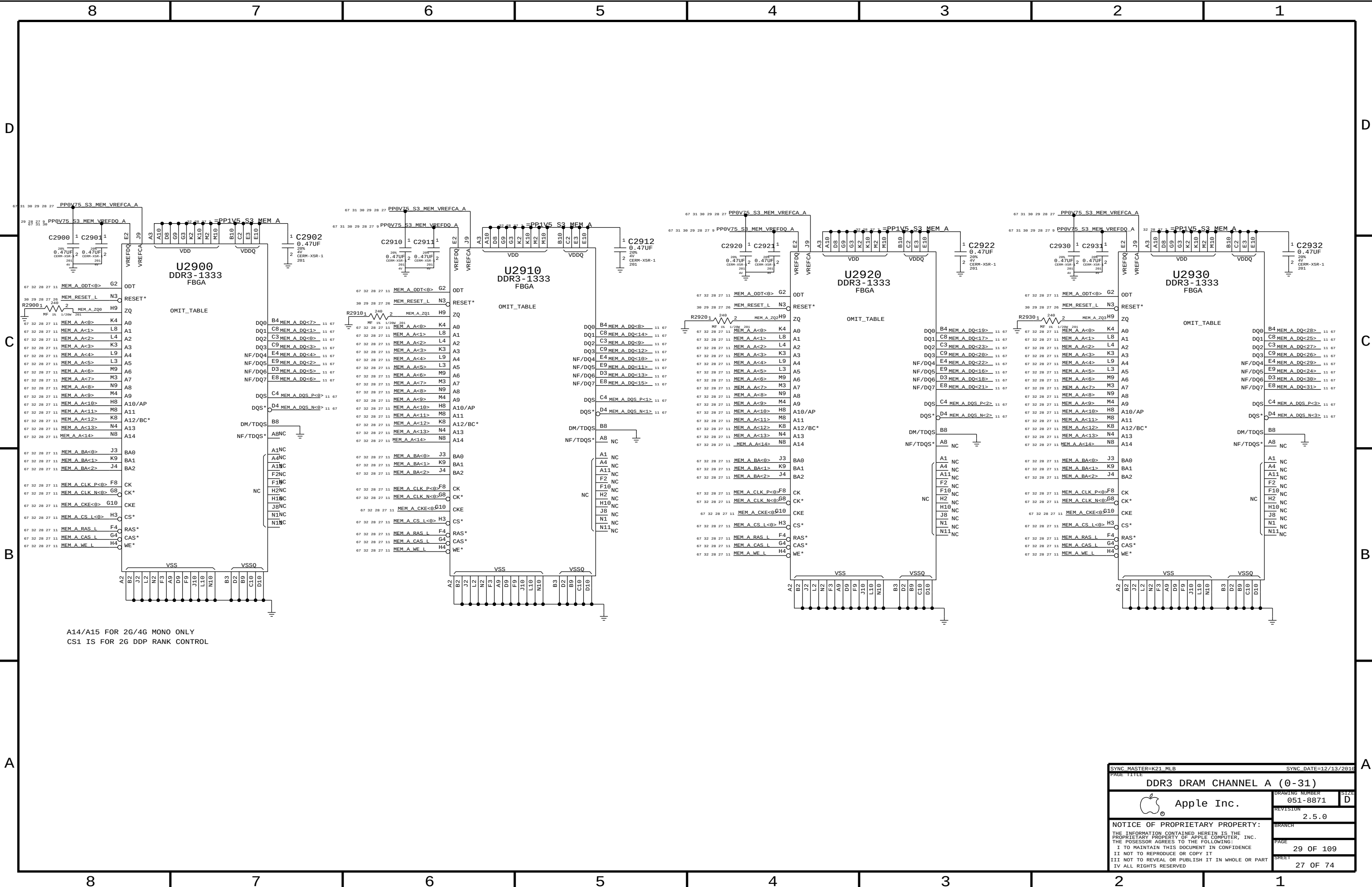


Step	ISOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	PM_SLP_S4_L	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN	P1V5CPU_EN
S0	0	1	1	1	1	1	1	1
1	0	1	1	1	1	1	1	1
2	0	0	1	1	1	1	0	1
3	0	0	0	1	X	1	0	0
4	0	0	1	1	X	1	0	1
5	0	1	1	1	0 (*)	1	1	1
6	0	1	1	1	1	1	1	1
S0	1	1	1	1	1	1	1	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

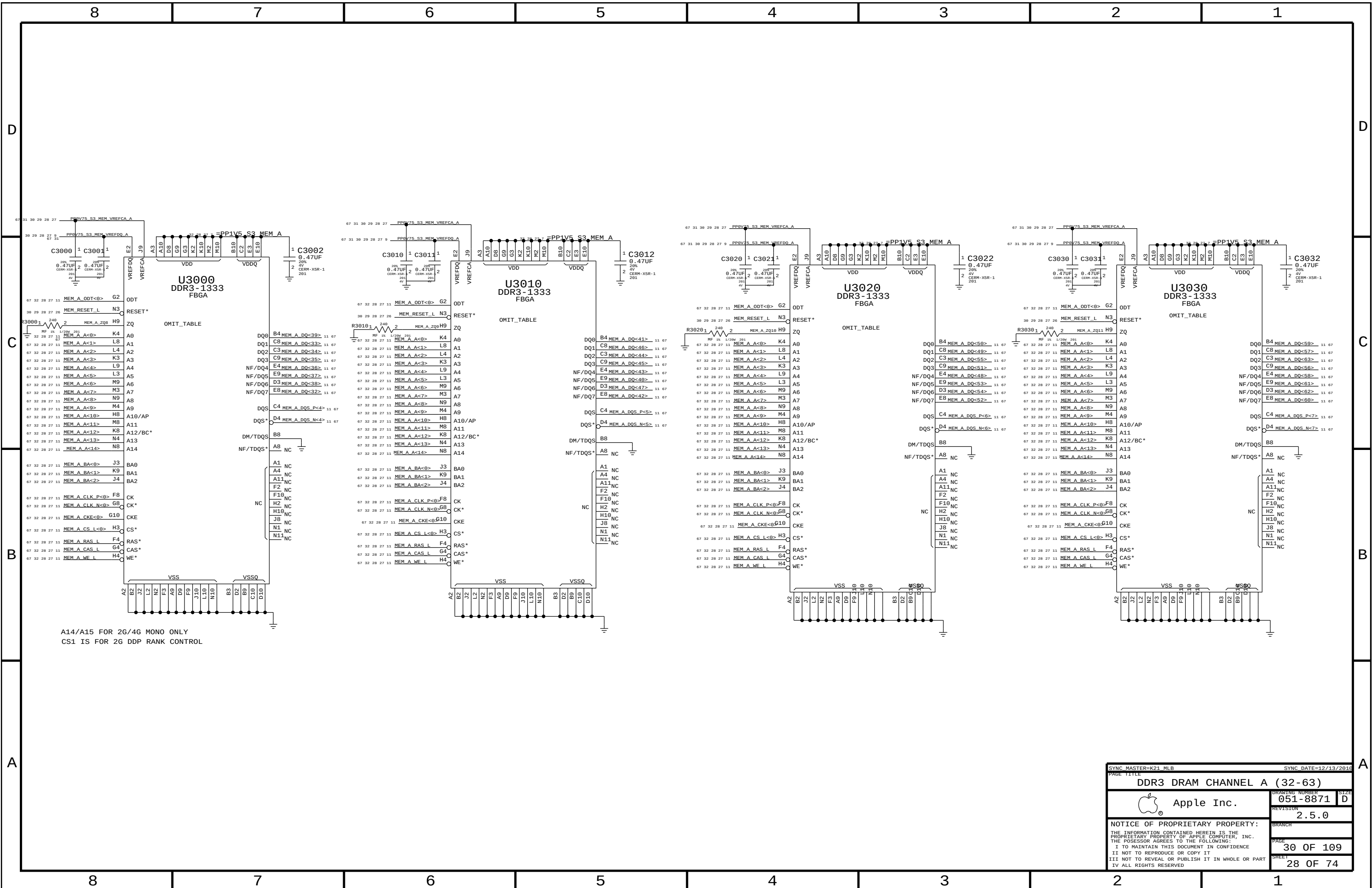
NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must deassert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

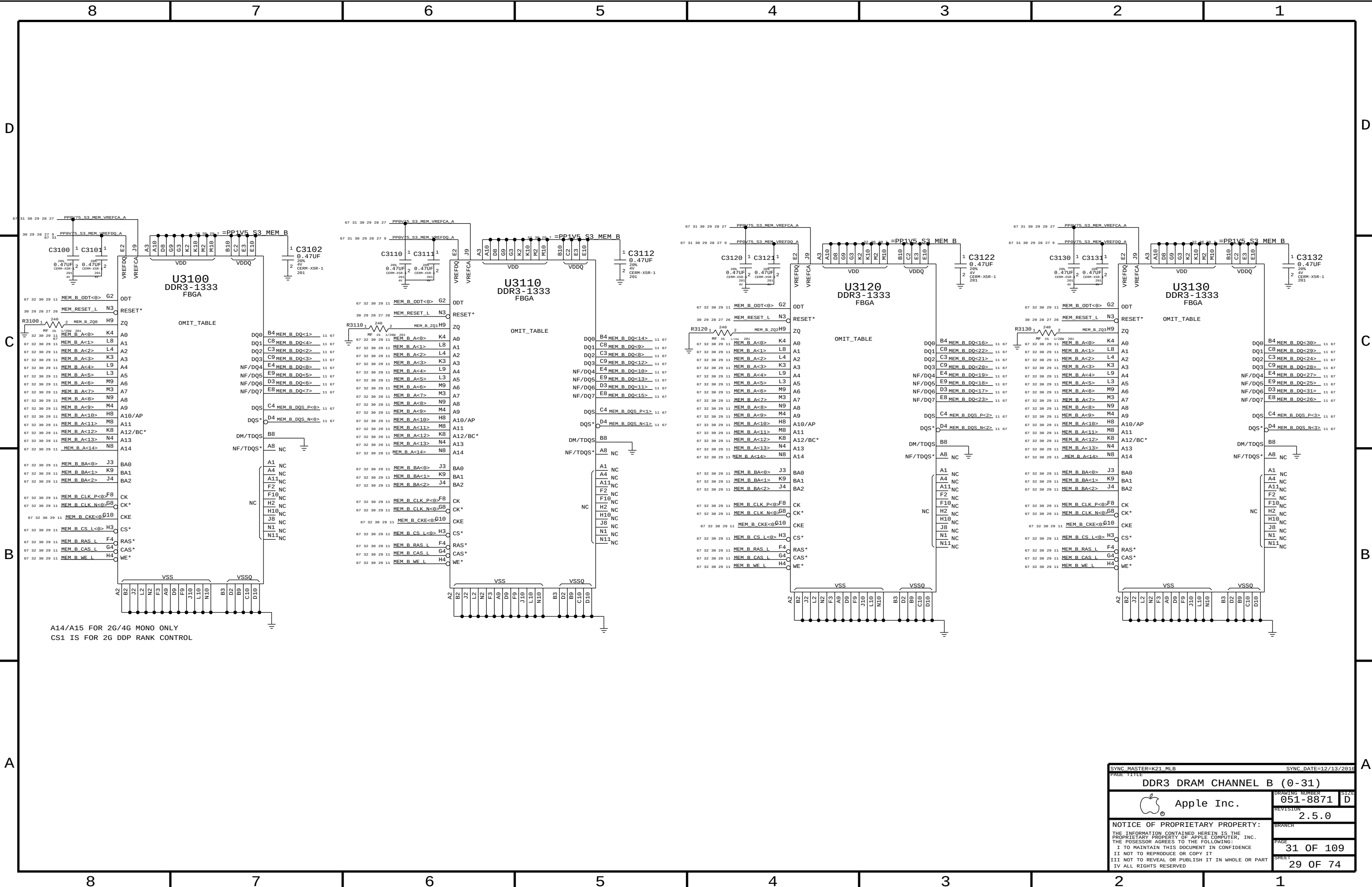
CPU Memory S3 Support	
Apple Inc.	051-8871
REVISION	2.5.0
BRANCH	
PAGE	28 OF 109
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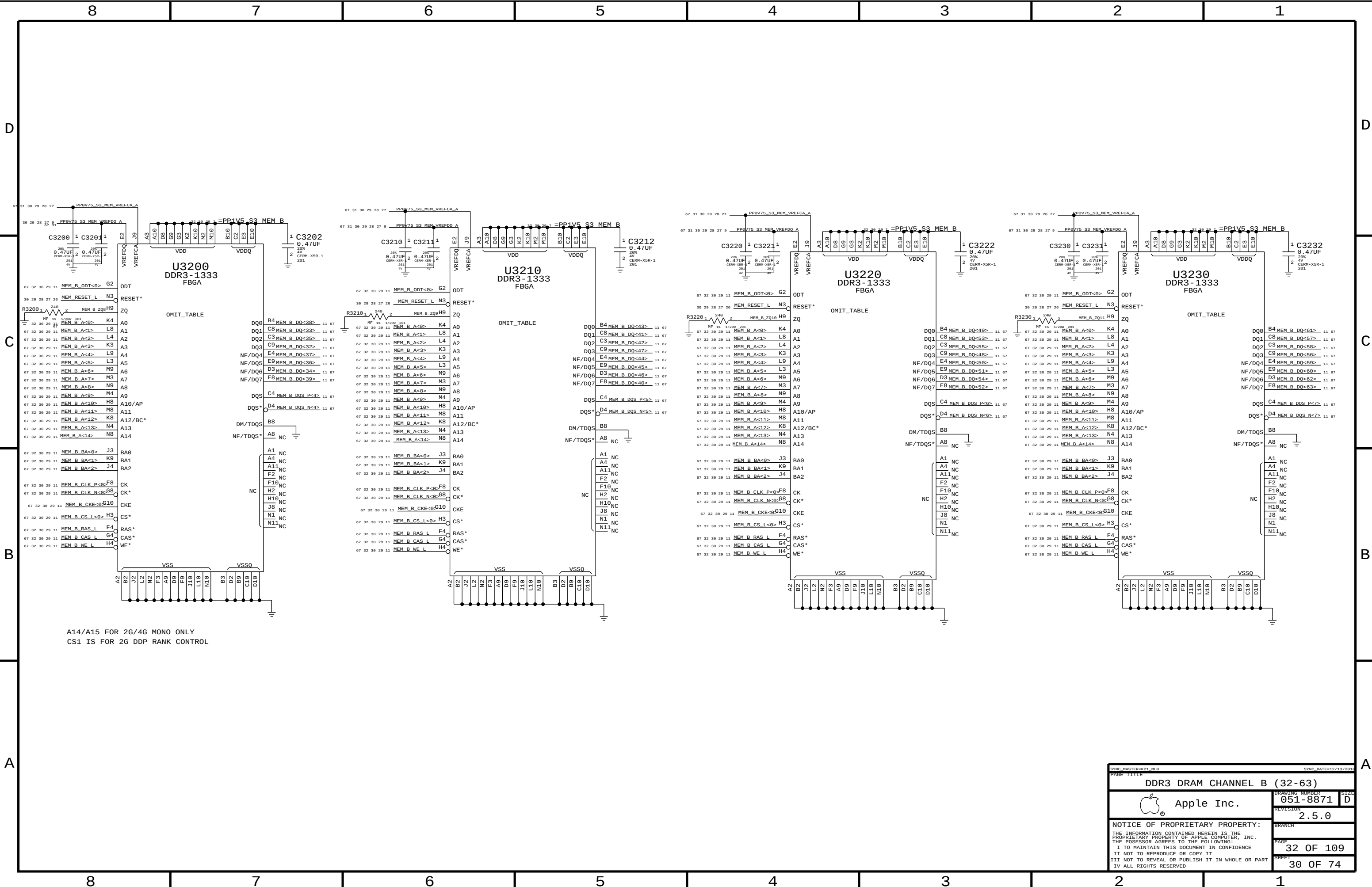


A14/A15 FOR 2G/4G MONO ONLY
CS1 IS FOR 2G DDP RANK CONTROL

SYNC MASTER=K21 MLB		SYNC DATE=12/13/2010	
PAGE TITLE			
DDR3 DRAM CHANNEL A (0-31)			
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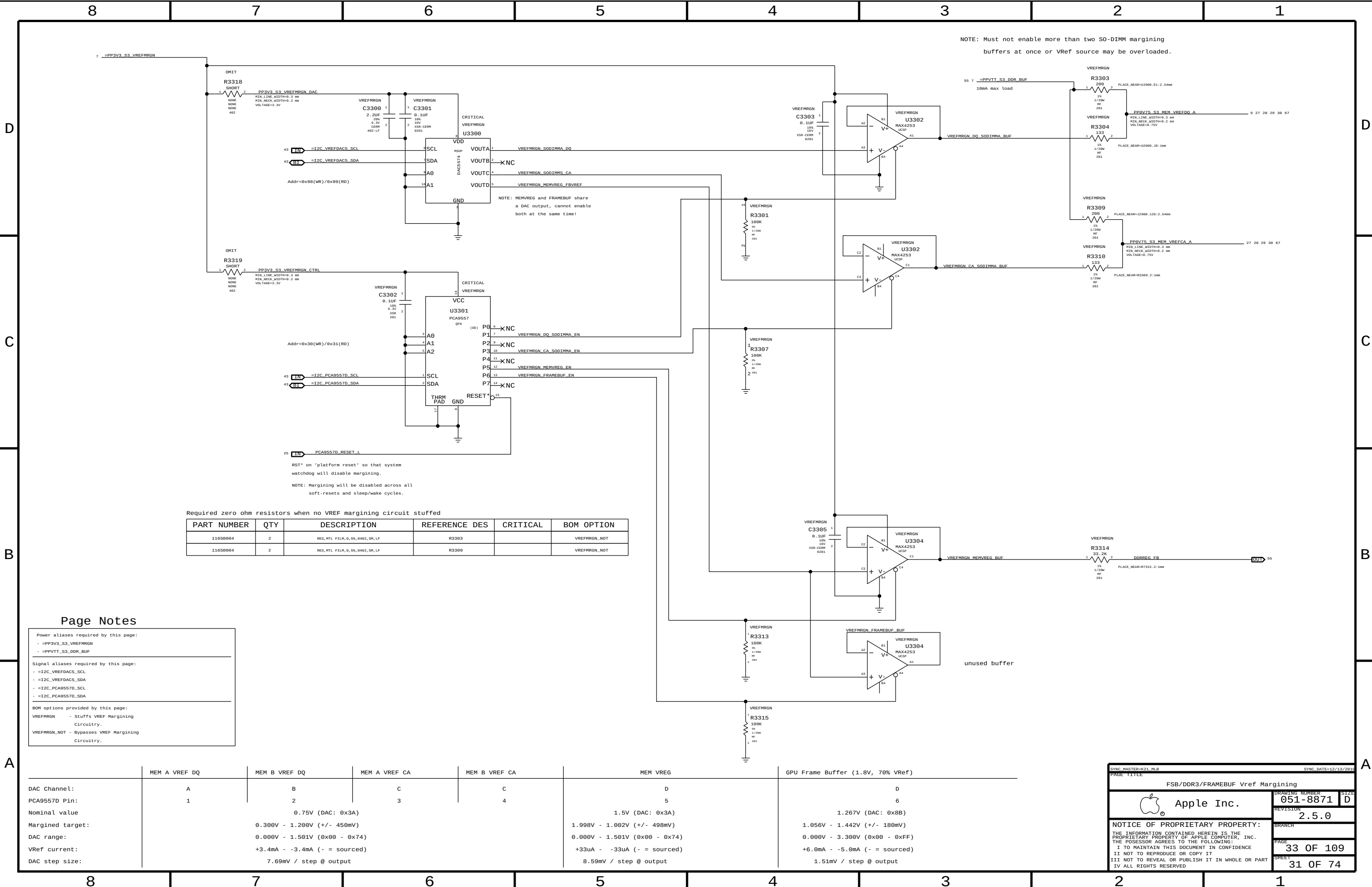






A14/A15 FOR 2G/4G MONO ONLY
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SYNC MASTER=K21_MLB		SYNC DATE=12/13/2011	
PAGE TITLE			
DDR3 DRAM CHANNEL B (32-63)			
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		SHEET	30 OF 74



Required zero ohm resistors when no VREF margining circuit stuffed

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
11650004	2	RES,MTL F1LM,0.5N,0402,SM,LF	R3303		VREFMGRN_NOT
11650004	2	RES,MTL F1LM,0.5N,0402,SM,LF	R3309		VREFMGRN_NOT

Page Notes

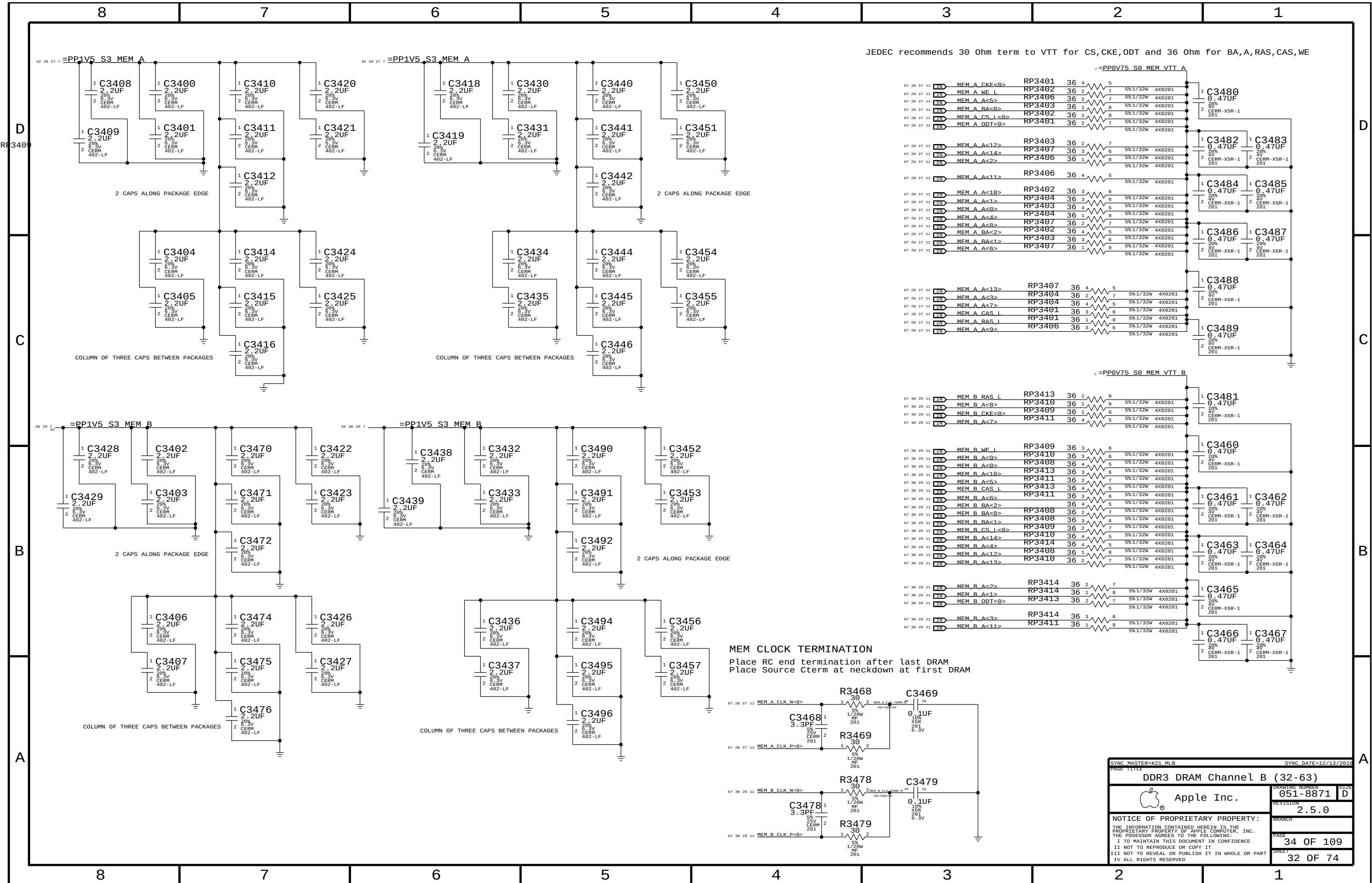
Power aliases required by this page:
- =PP3V3_S3_VREFMGRN
- =PPVTT_S3_DDR_BUF

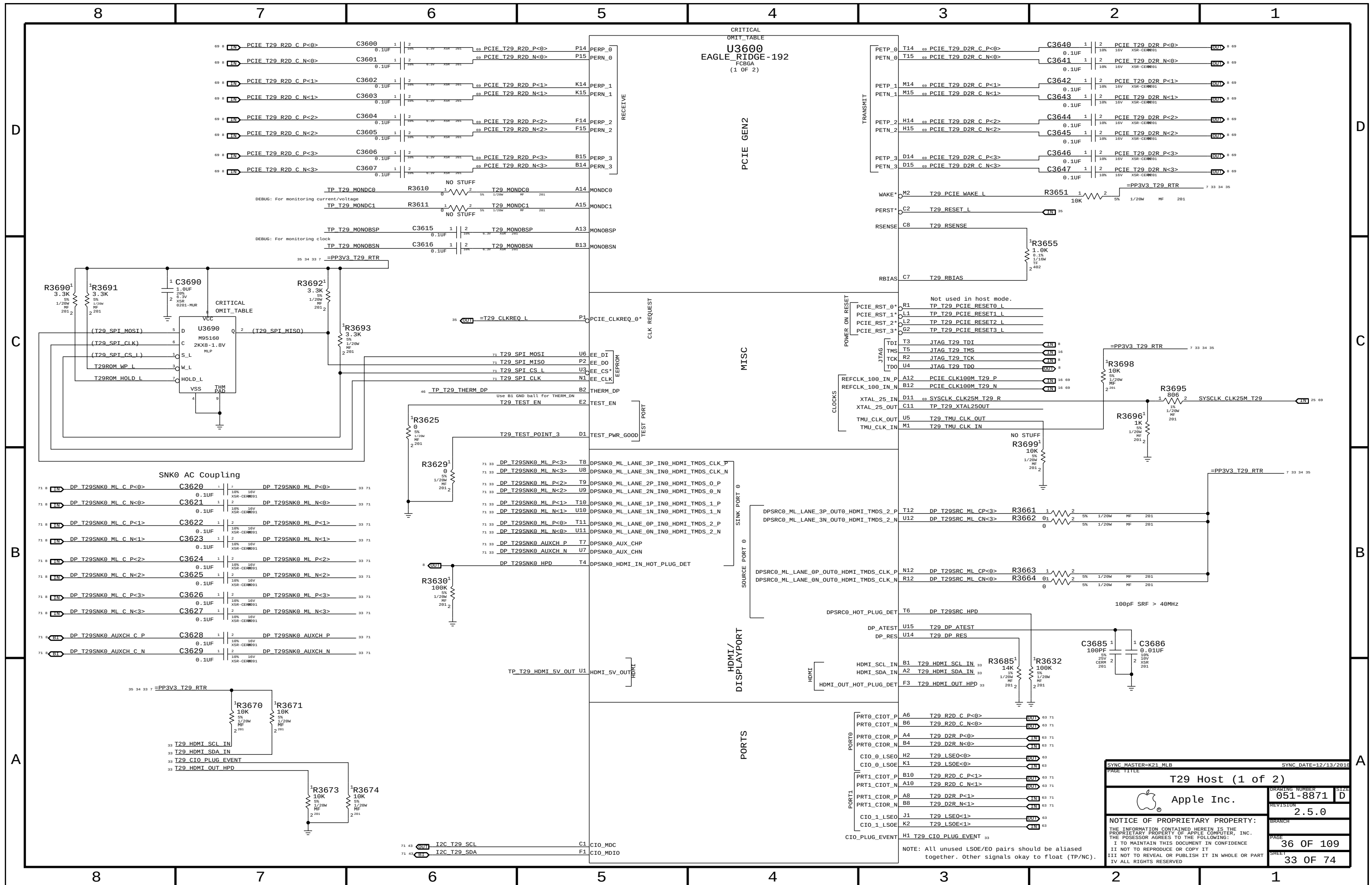
Signal aliases required by this page:
- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:
VREFMGRN - Stuffs VREF Margining Circuitry.
VREFMGRN_NOT - Bypasses VREF Margining Circuitry.

	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG	GPU Frame Buffer (1.8V, 70% VRef)
DAC Channel:	A	B	C	C	D	D
PCA9557D Pin:	1	2	3	4	5	6
Nominal value		0.75V (DAC: 0x3A)			1.5V (DAC: 0x3A)	1.267V (DAC: 0x8B)
Margined target:		0.300V - 1.200V (+/- 450mV)			1.998V - 1.002V (+/- 498mV)	1.056V - 1.442V (+/- 180mV)
DAC range:		0.000V - 1.501V (0x00 - 0x74)			0.000V - 1.501V (0x00 - 0x74)	0.000V - 3.300V (0x00 - 0xFF)
VRef current:		+3.4mA - -3.4mA (- = sourced)			+33uA - -33uA (- = sourced)	+6.0mA - -5.0mA (- = sourced)
DAC step size:		7.69mV / step @ output			8.59mV / step @ output	1.51mV / step @ output

PAGE TITLE		DRAWING NUMBER		SIZE	
FSB/DDR3/FRAMBUF Vref Margining		051-8871		D	
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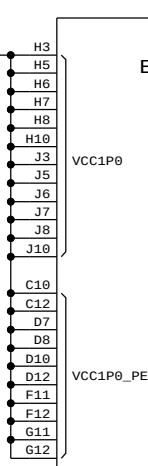
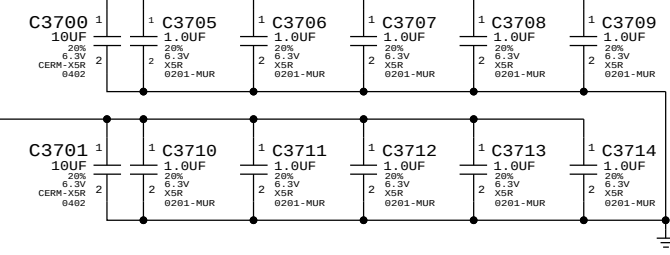
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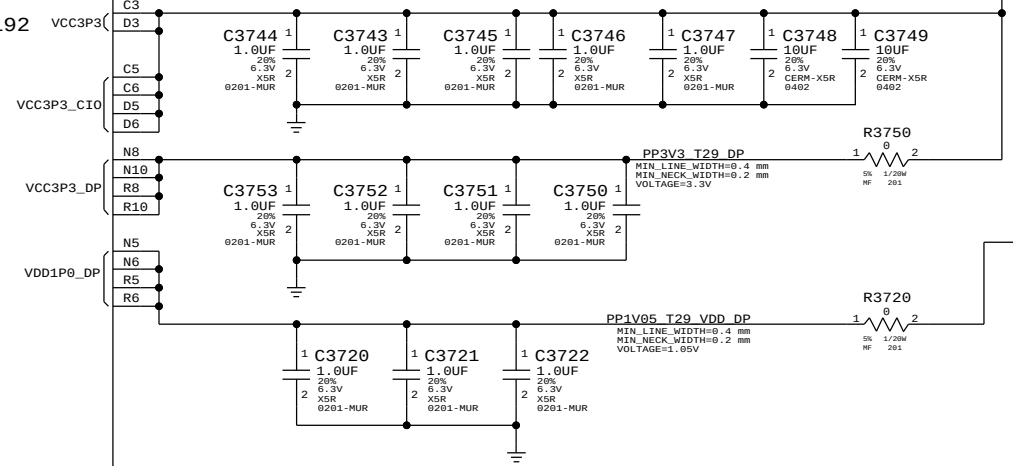
³⁴ 7 =PP1V05_T29_RTR
2100 mA (Single Port)
2250 mA (Dual Port)
EDP: 3000 mA



CRITICAL
OMIT_TABLE
U3600
EAGLE_RIDGE-192
FCBGA
(2 OF 2)

VCC

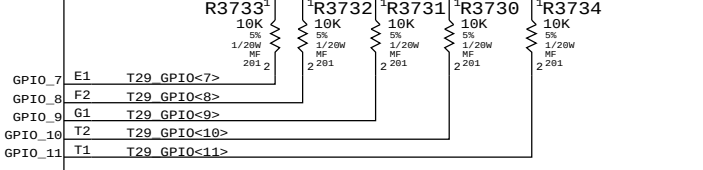
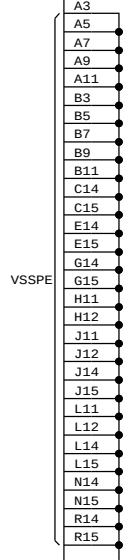
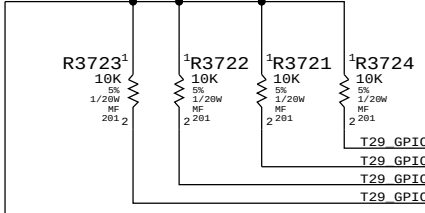
GND



=PP3V3_T29_RTR
135 mA (Single-Port)
152 mA (Dual-Port)
EDP: 200 mA

0-ohms are placeholders for now, replace with proper values after characterization.

=PP1V05_T29_RTR ⁷ 34
2100 mA (Single Port)
2250 mA (Dual Port)
EDP: 3000 mA



GPIO_7
GPIO_8
GPIO_9
GPIO_10
GPIO_11

E1
F2
G1
T2
T1

T29_GPIO<7>
T29_GPIO<8>
T29_GPIO<9>
T29_GPIO<10>
T29_GPIO<11>

SYNC MASTER=K21 MLB		SYNC DATE=12/13/2010	
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		PAGE	37 OF 109
		SHEET	34 OF 74

Page Notes

Power aliases required by this page:

- =PPVIN_SW_T29BST (8-13V Boost Input)
- =PP18V_T29_REG (18V Boost Output)
- =PP3V3_T29_P3V3T29FET (3.3V FET Input)
- =PP3V3_T29_FET (3.3V FET Output)
- =PP3V3_S0_T29PWRCTL
- =PP1V05_T29_P1V05T29FET (1.05V FET Input)
- =PP1V05_T29_FET (1.05V FET Output)

Signal aliases required by this page:

- =T29_CLKREQ_L
- =T29_RESET_L

BOM options provided by this page:

T29BST:Y - Stuffs 18V boost circuitry.

T29 18V Boost Regulator

SI8409DB:
Vds(max): -30V
Vgs(max): +/-12V
Vgs(th): -1.4V
Rds(on): 46m0hm @ 4.5V Vgs
Id(max): 3.7A @ 70C

CRITICAL
T29BST:Y
Q3880
SI8409DB
BGA

CRITICAL
T29BST:Y
L3895
6.8uH-4.0A
PIMB062D-SM

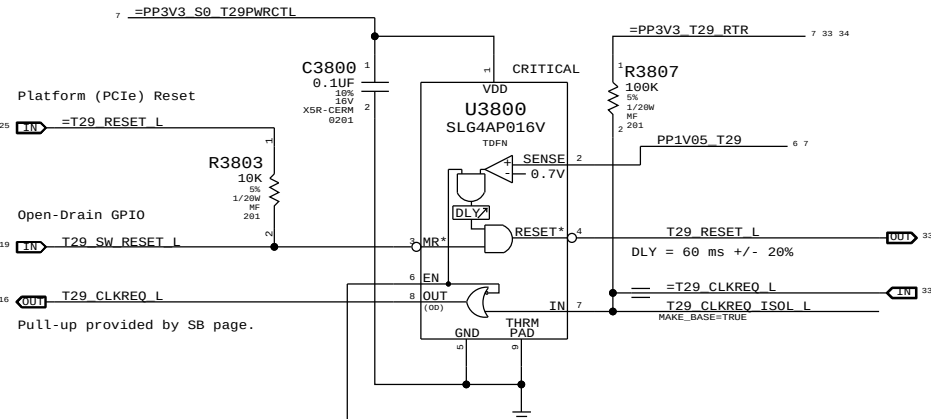
CRITICAL
T29BST:Y
D3895
POWER01-123
DFLS230L

Vout = 15.1V
Max Current = 1.0A
Freq = 300KHz

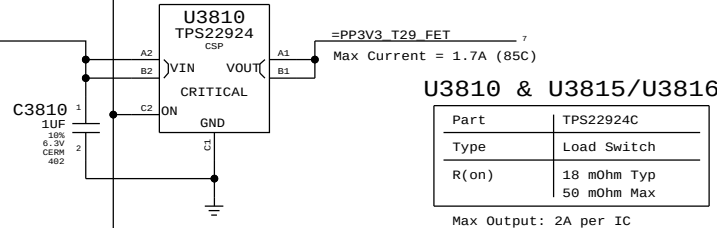
$$V_{out} = 1.6V * (1 + R_a / R_b)$$

$$UVLO(falling) = 1.22 * (R_1 + R_2) / R_2$$
$$UVLO(rising) = UVLO(falling) + (2uA * R_1)$$
$$UVLO = 4.55V (falling), 4.95 (rising)$$

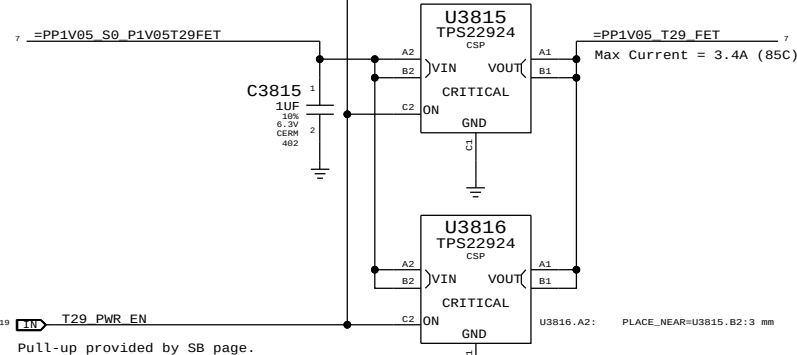
Supervisor & CLKREQ# Isolation



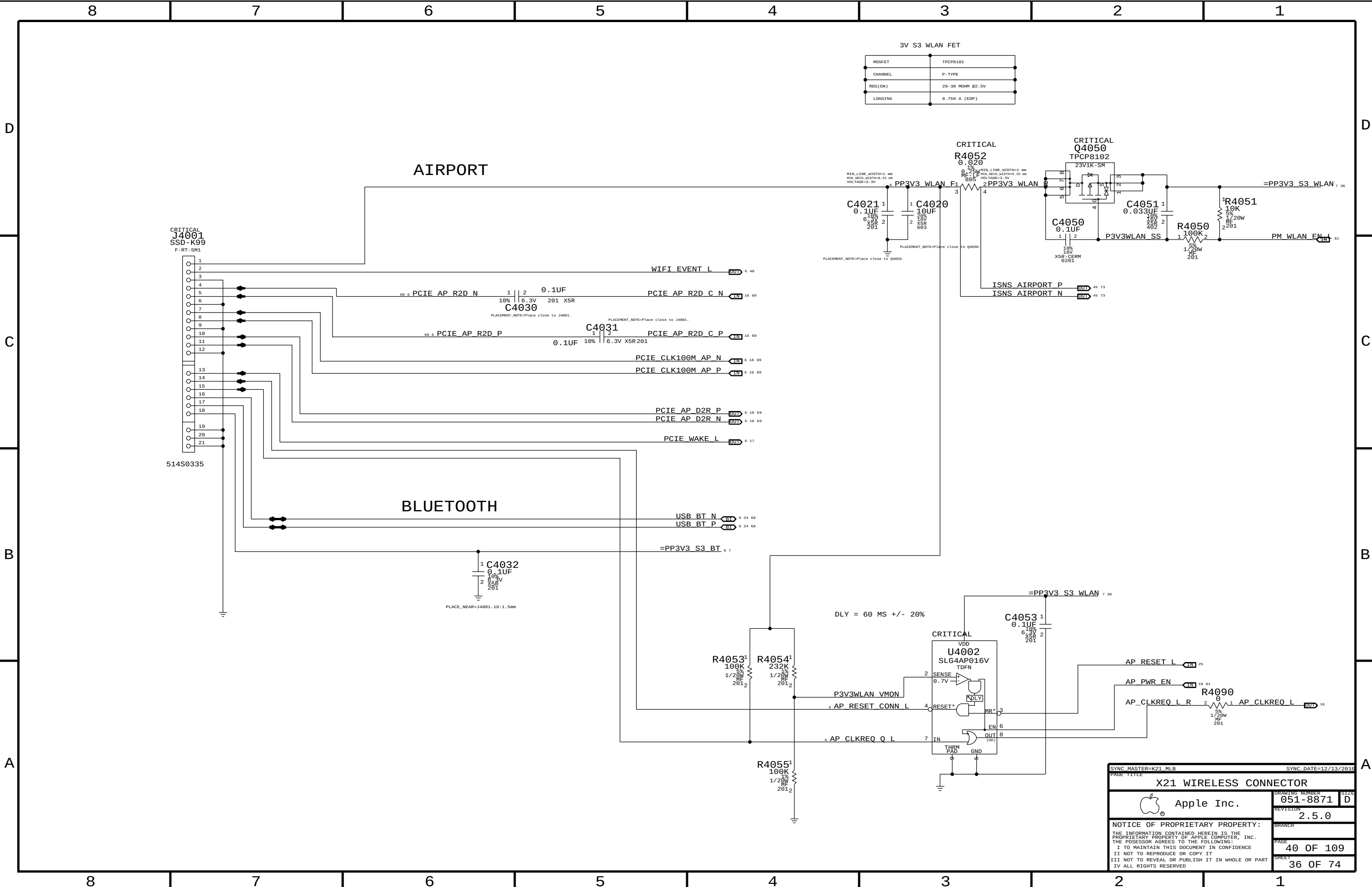
3.3V T29 Switch



1.05V T29 Switch



SYNC MASTER=K21 MLB		SYNC DATE=12/13/2010	
PAGE TITLE		T29 Power Support	
Apple Inc.		DRAWING NUMBER	051-8871
		REVISION	2.5.0
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3V S3 WLAN FET	
MOSFET	TPCP8102
CHANNEL	P-TYPE
RDS(ON)	28-30 MOHM @2.5V
LOADING	0.750 A (EDP)

SYNC MASTER=K21 MLB

SYNC DATE=12/13/2010

X21 WIRELESS CONNECTOR

Apple Inc.

051-8871

2.5.0

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36 OF 74

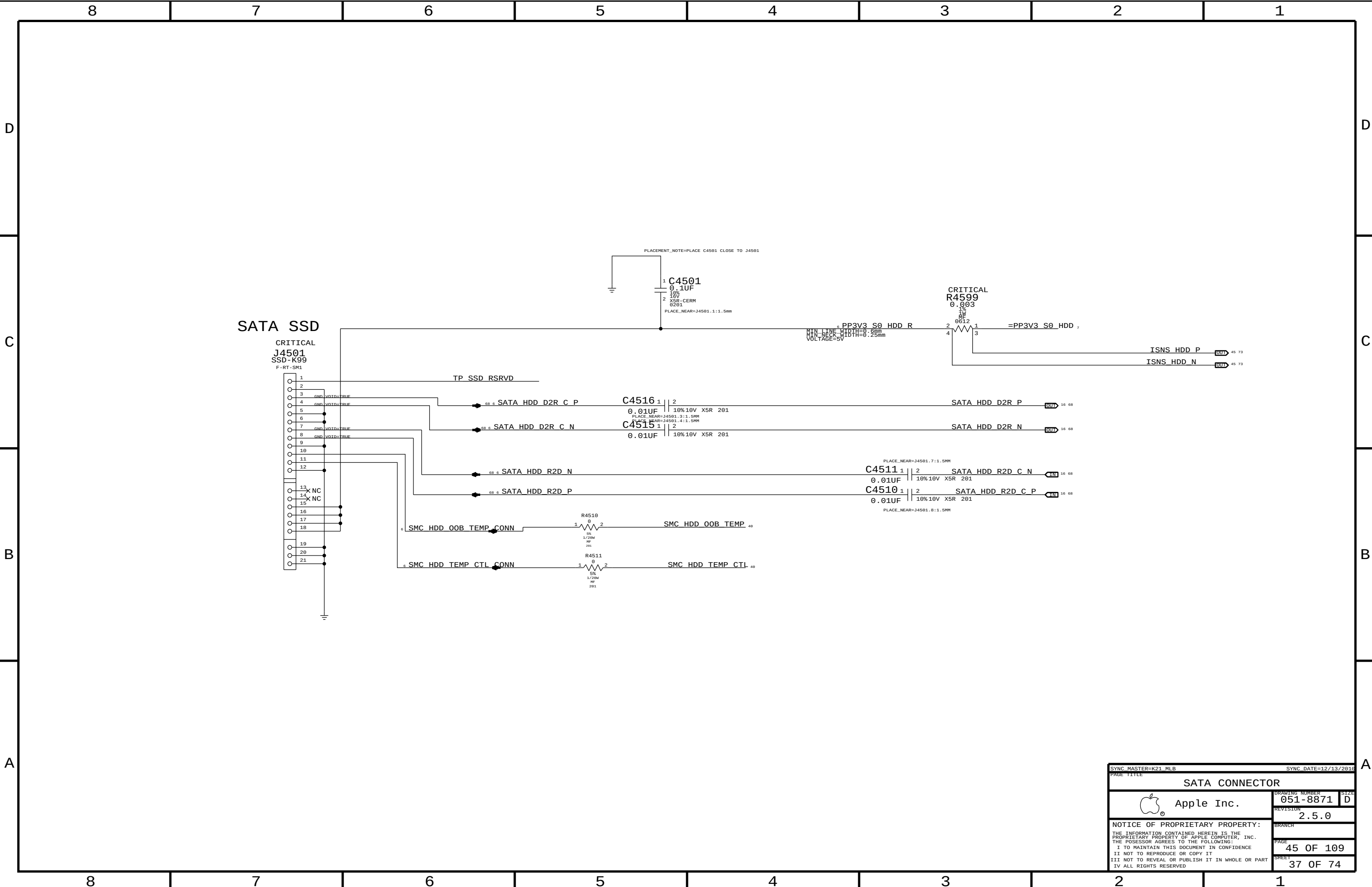
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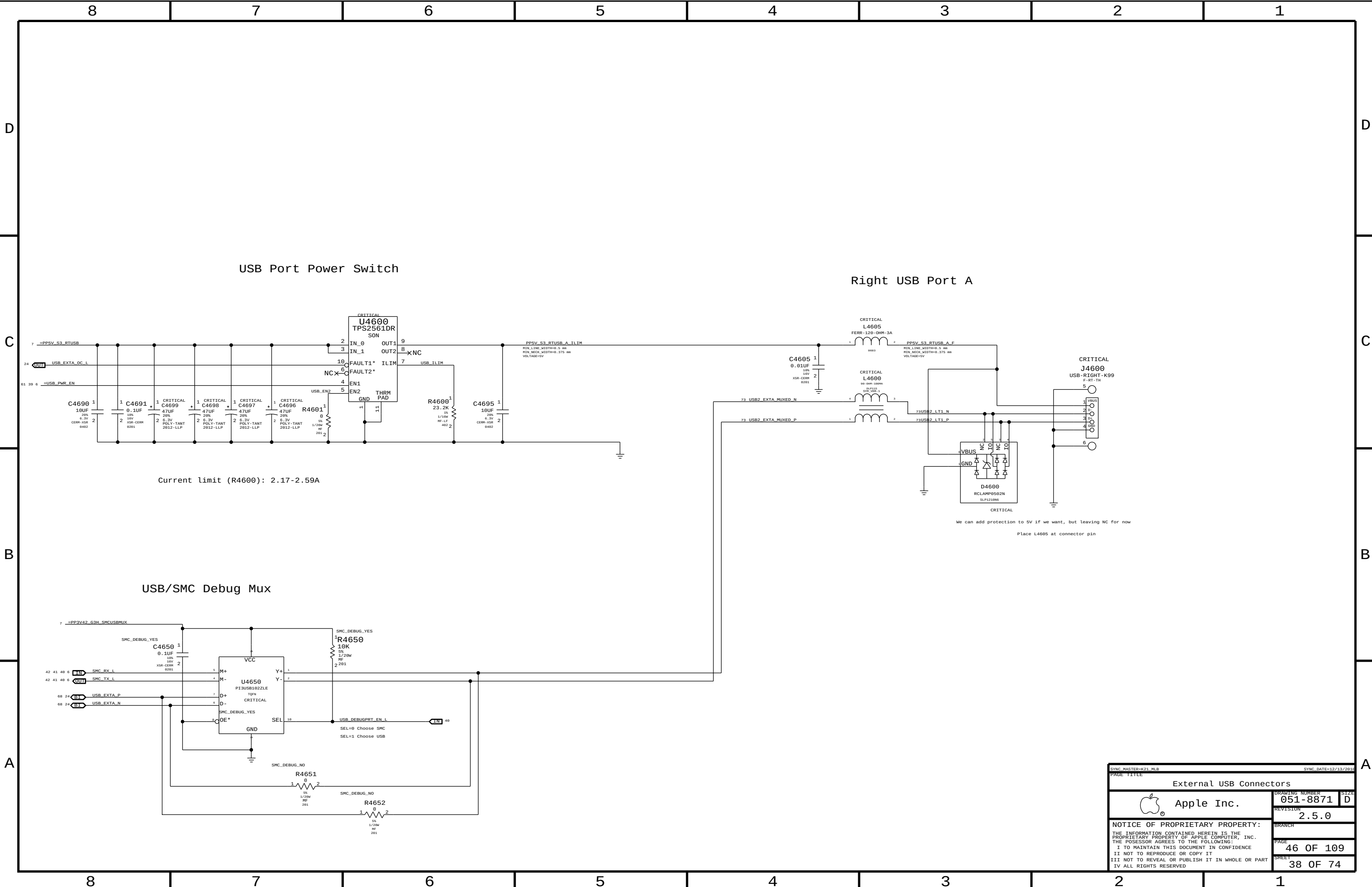
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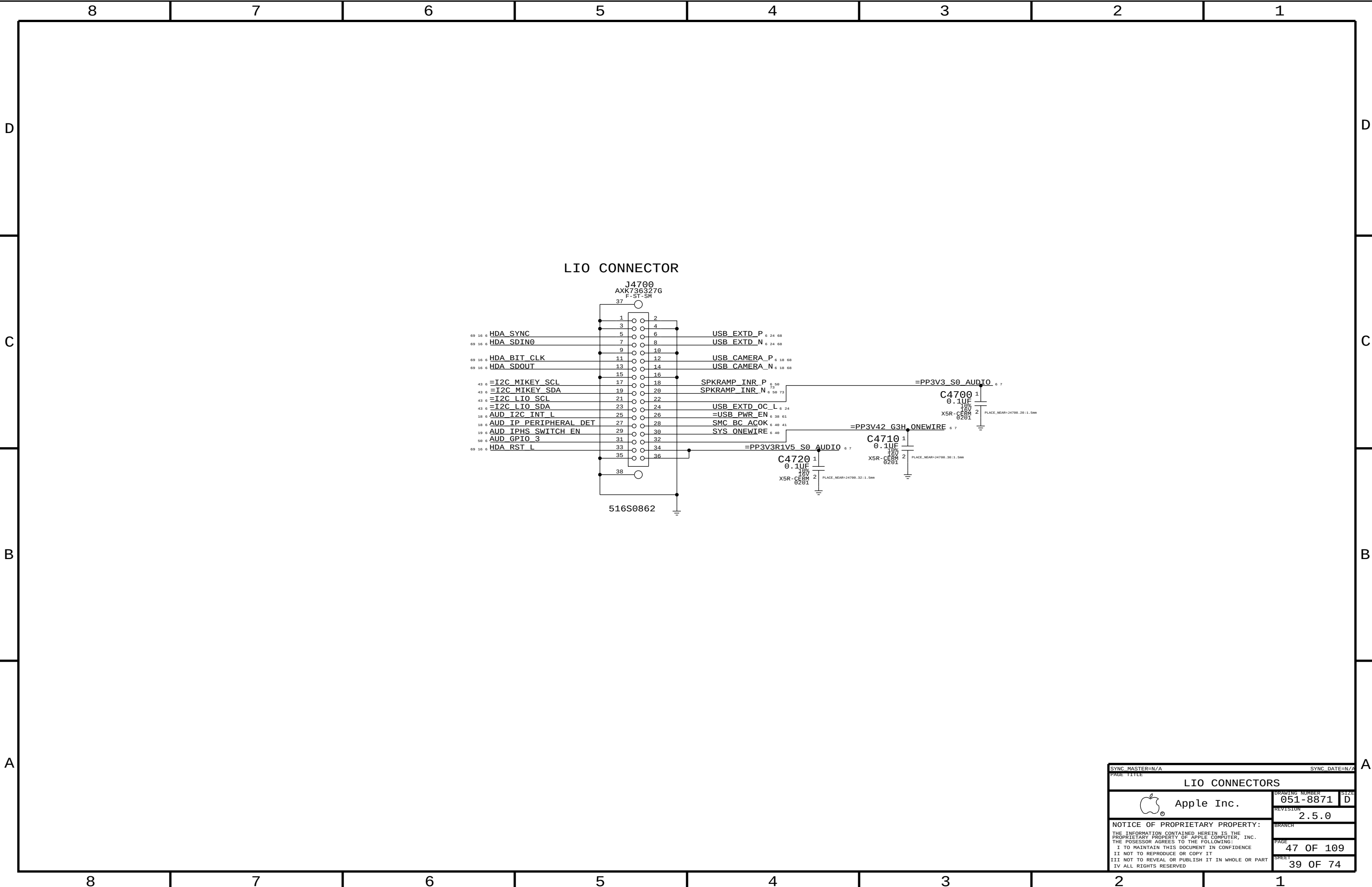
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

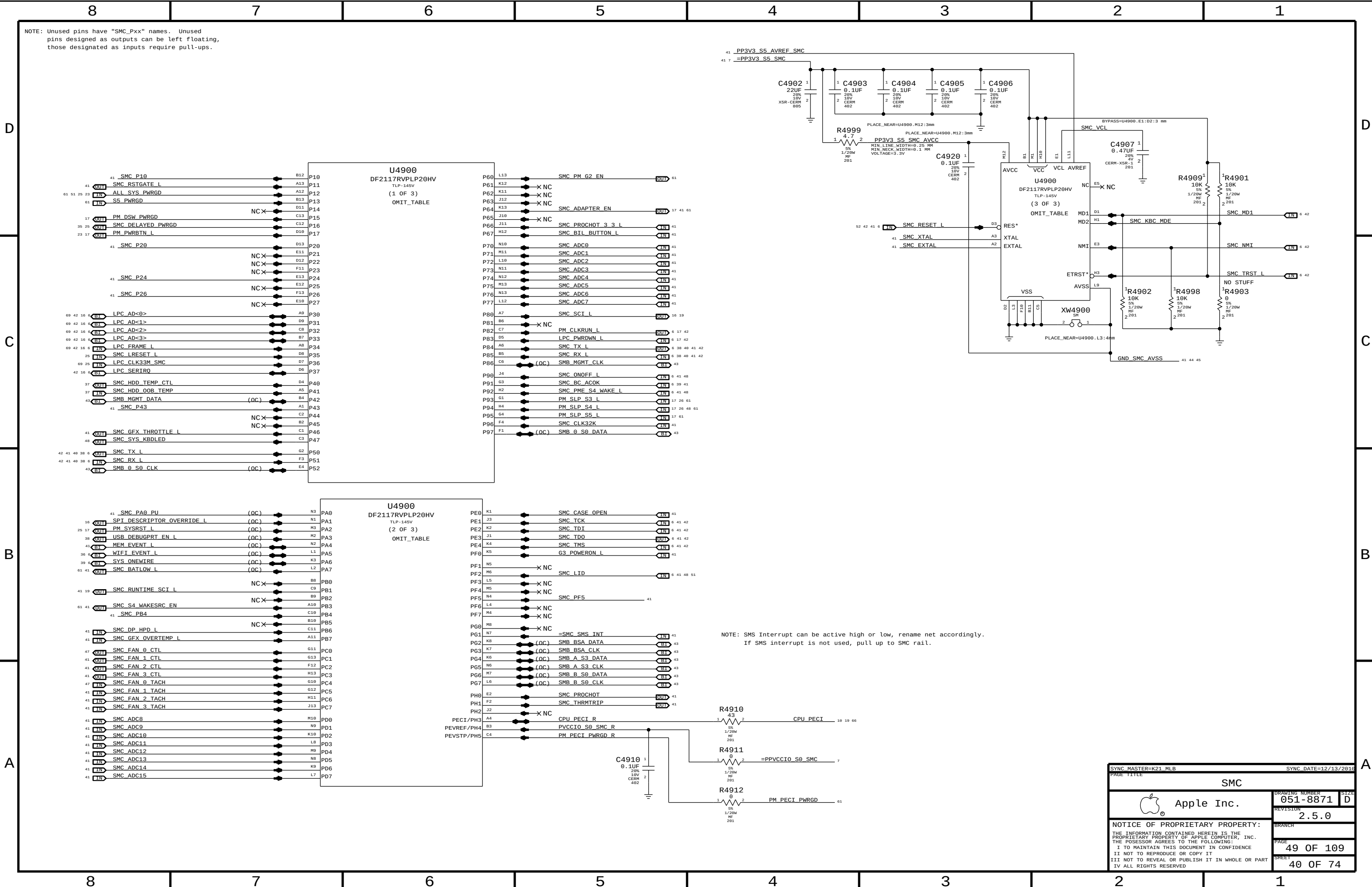
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SYNC MASTER=K21 MLB		SYNC DATE=12/13/2010	
PAGE TITLE			
SATA CONNECTOR			
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		SHEET	37 OF 74



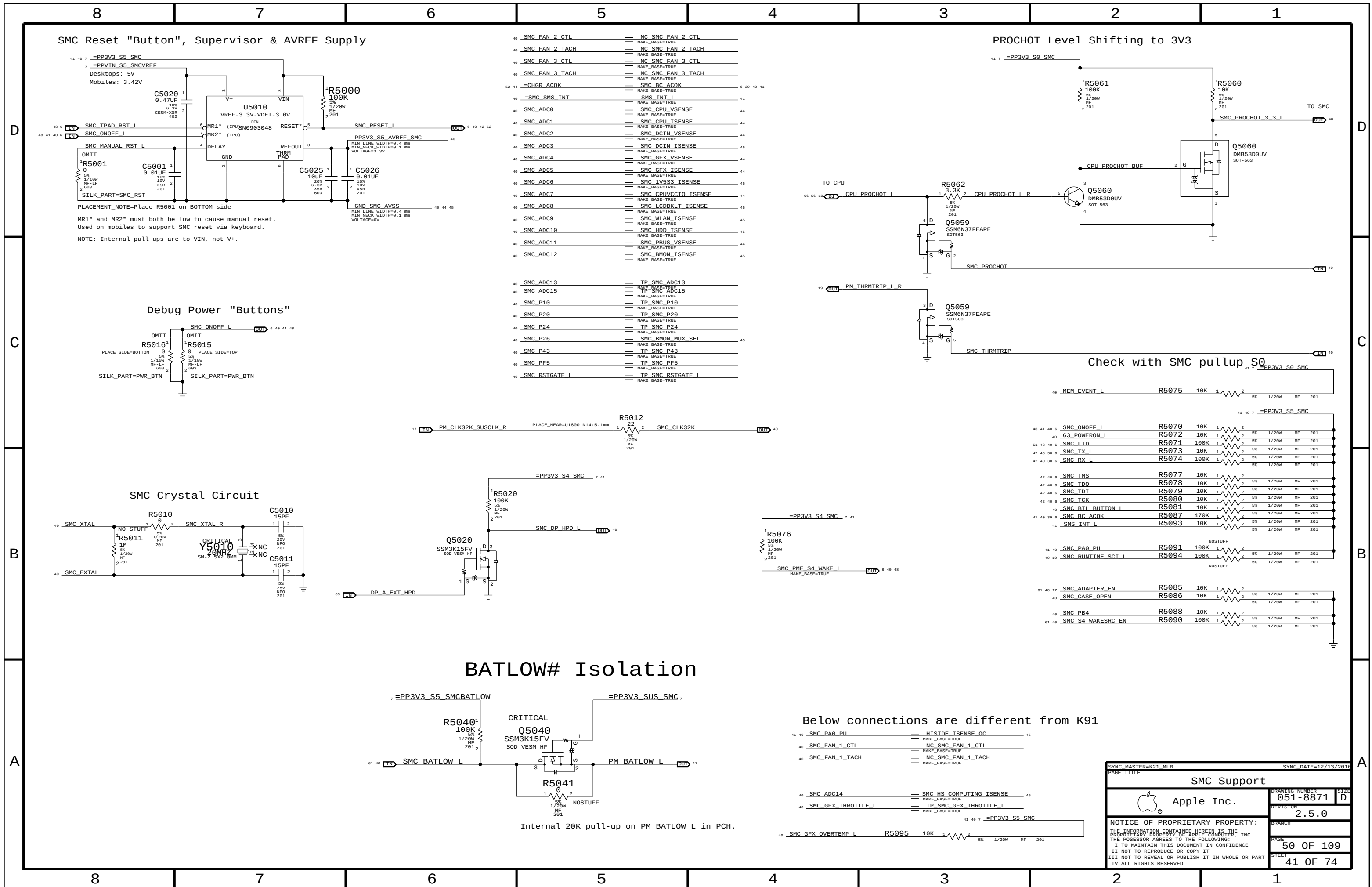


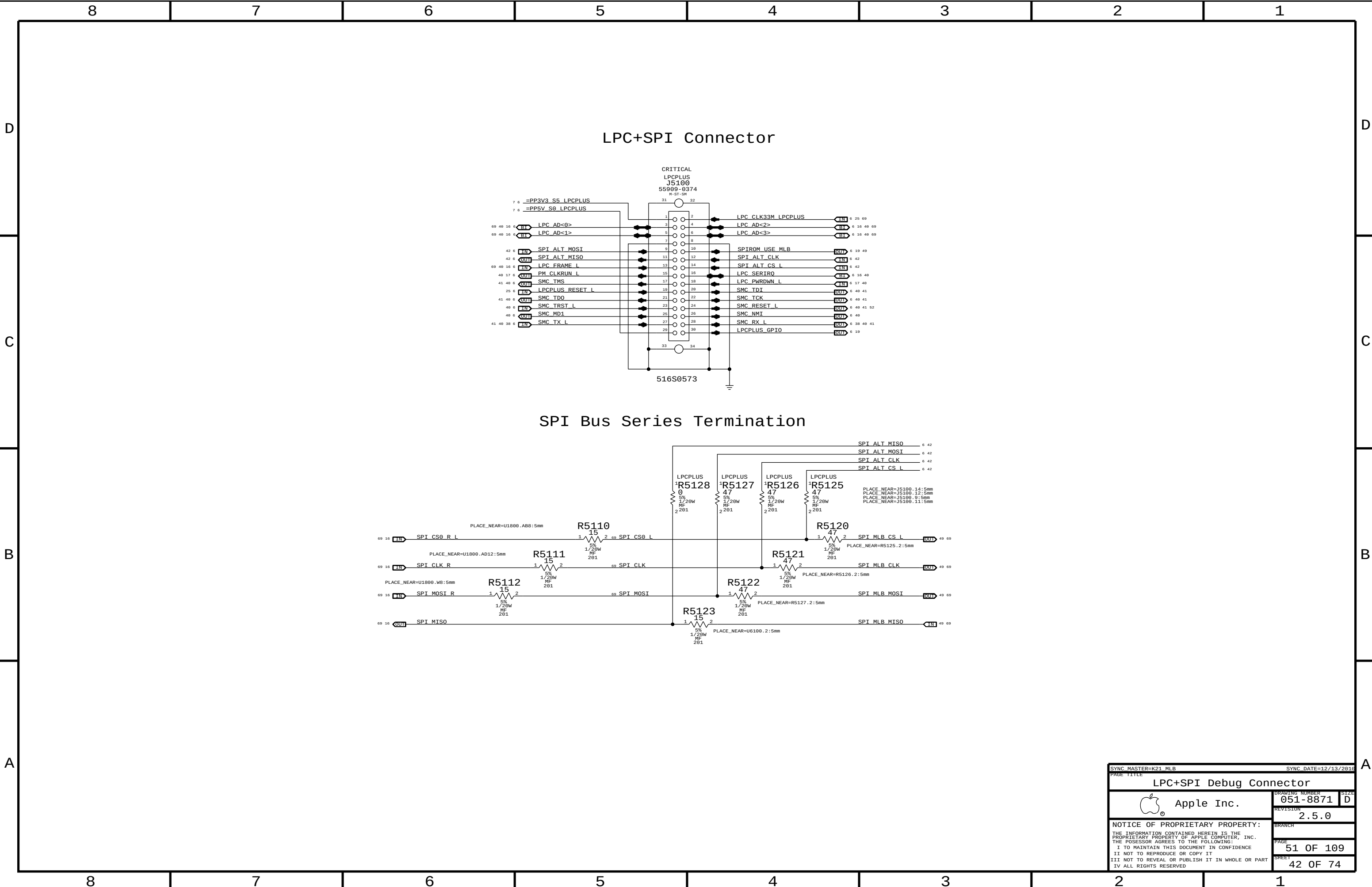


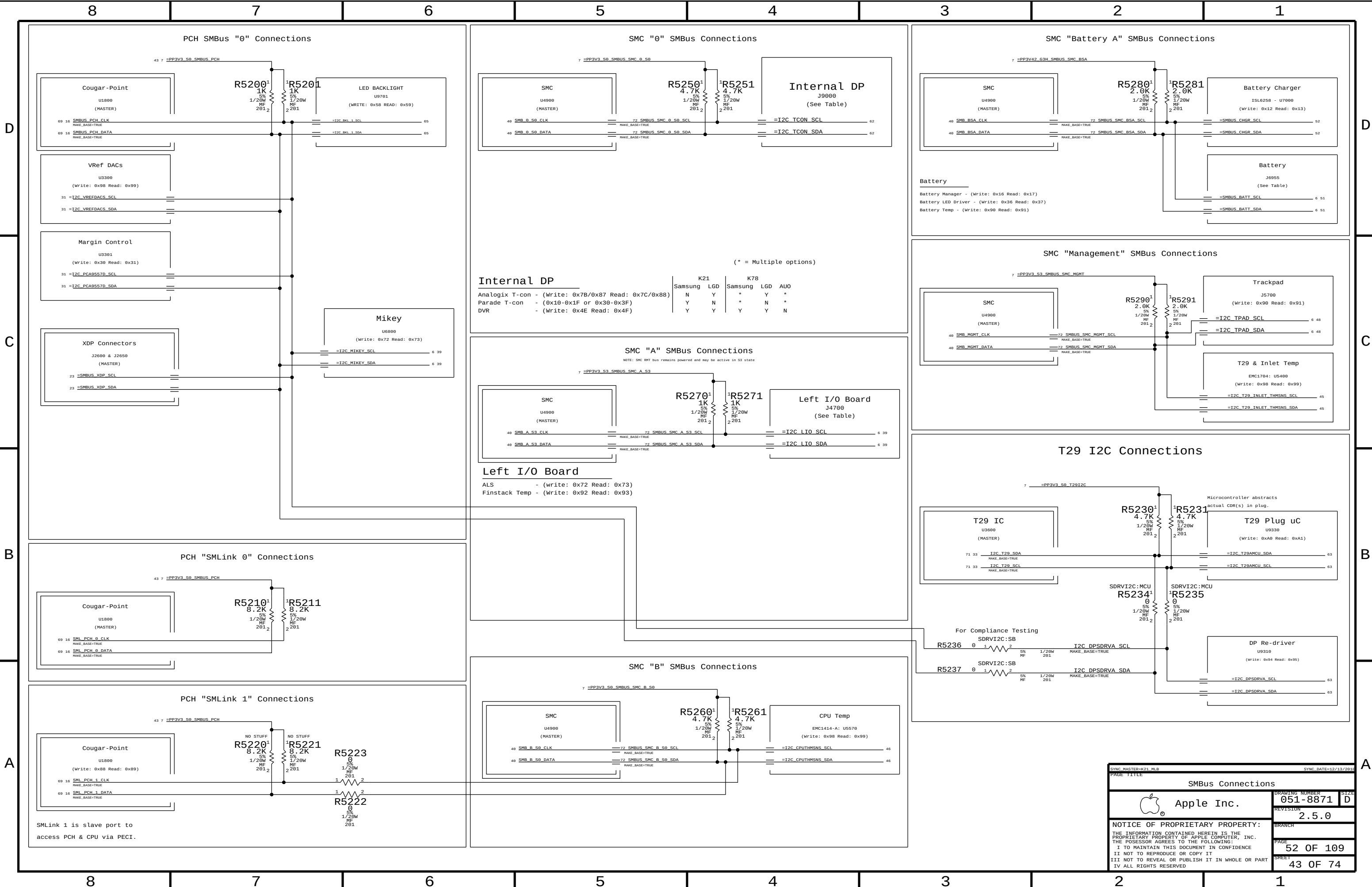
NOTE: Unused pins have "SMC_Pxx" names. Unused pins designed as outputs can be left floating, those designated as inputs require pull-ups.

NOTE: SMS Interrupt can be active high or low, rename net accordingly. If SMS interrupt is not used, pull up to SMC rail.

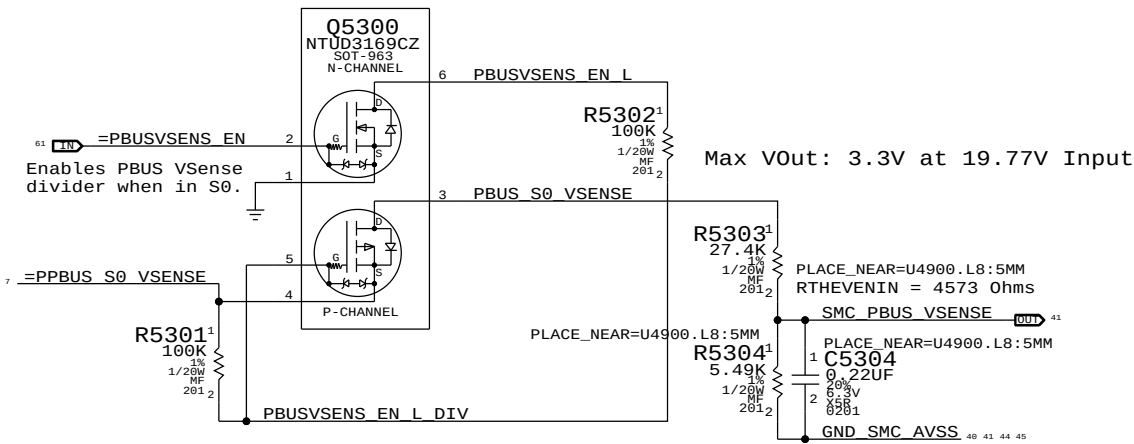
SYNC MASTER=K21 MLB		SYNC DATE=12/13/2010	
PAGE TITLE			
SMC		DRAWING NUMBER	051-8871
Apple Inc.		REVISION	2.5.0
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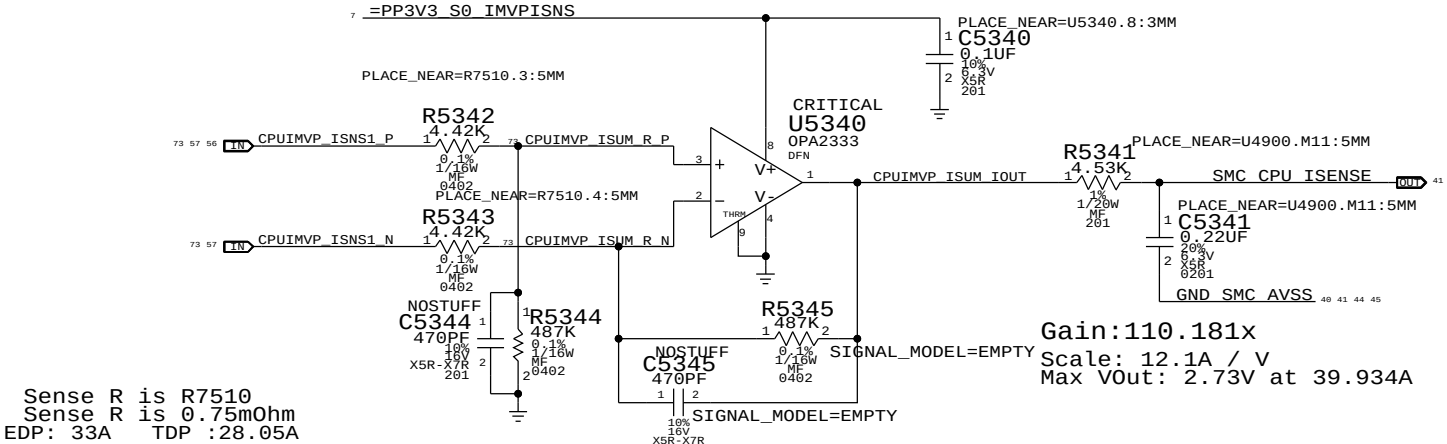




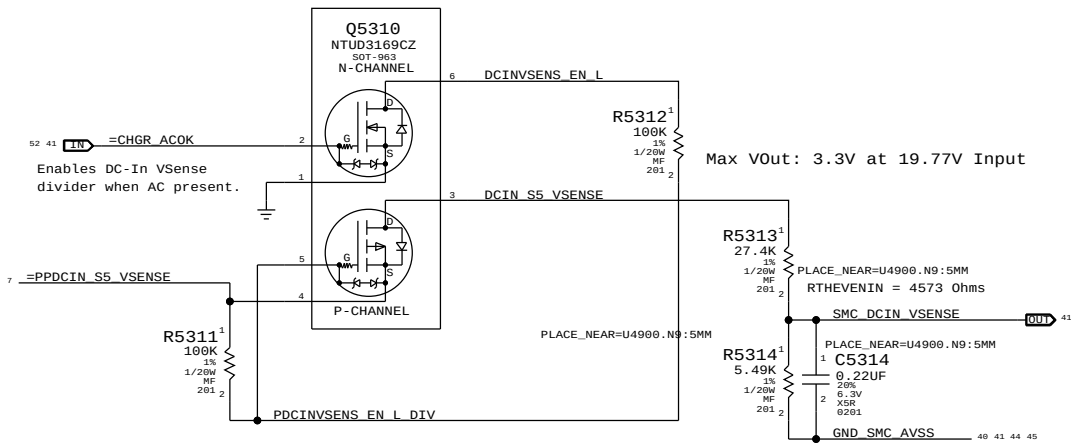
PBUS Voltage Sense Enable & Filter



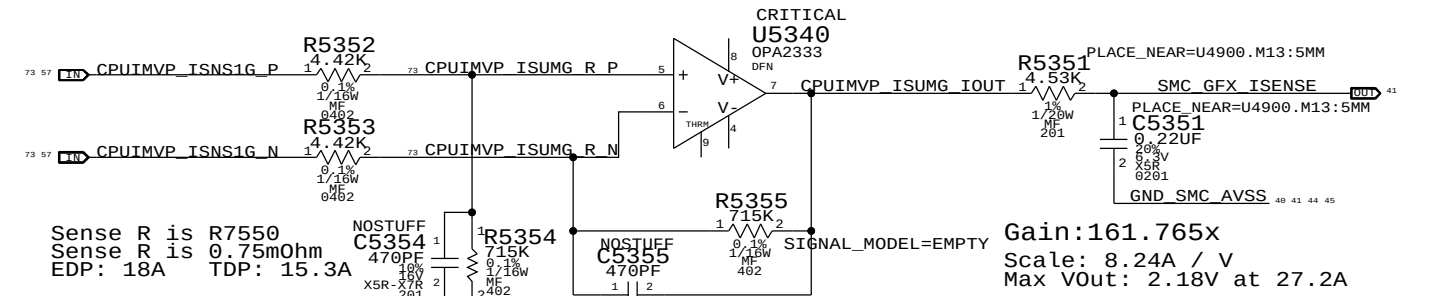
CPU VCore Load Side Current Sense / Filter



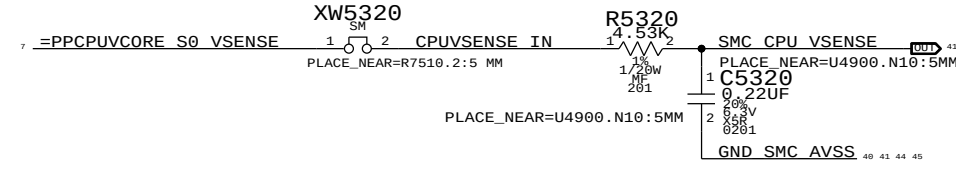
DC-In Voltage Sense Enable & Filter



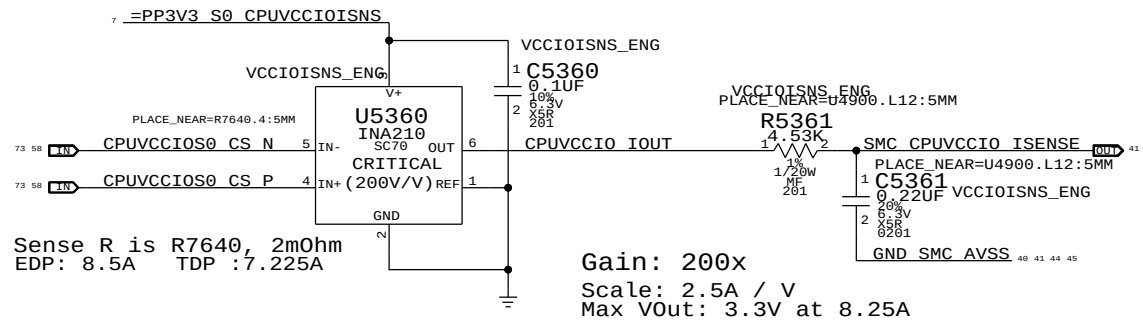
GFX/IG VCore Load Side Current Sense / Filter



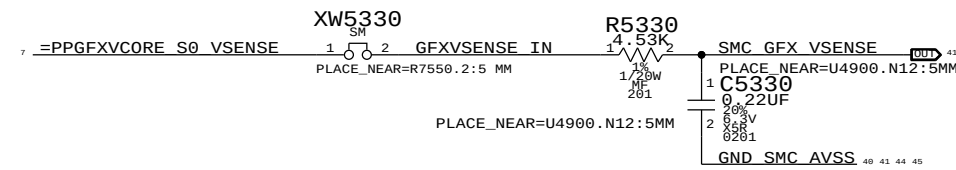
CPU Vcore Voltage Sense / Filter



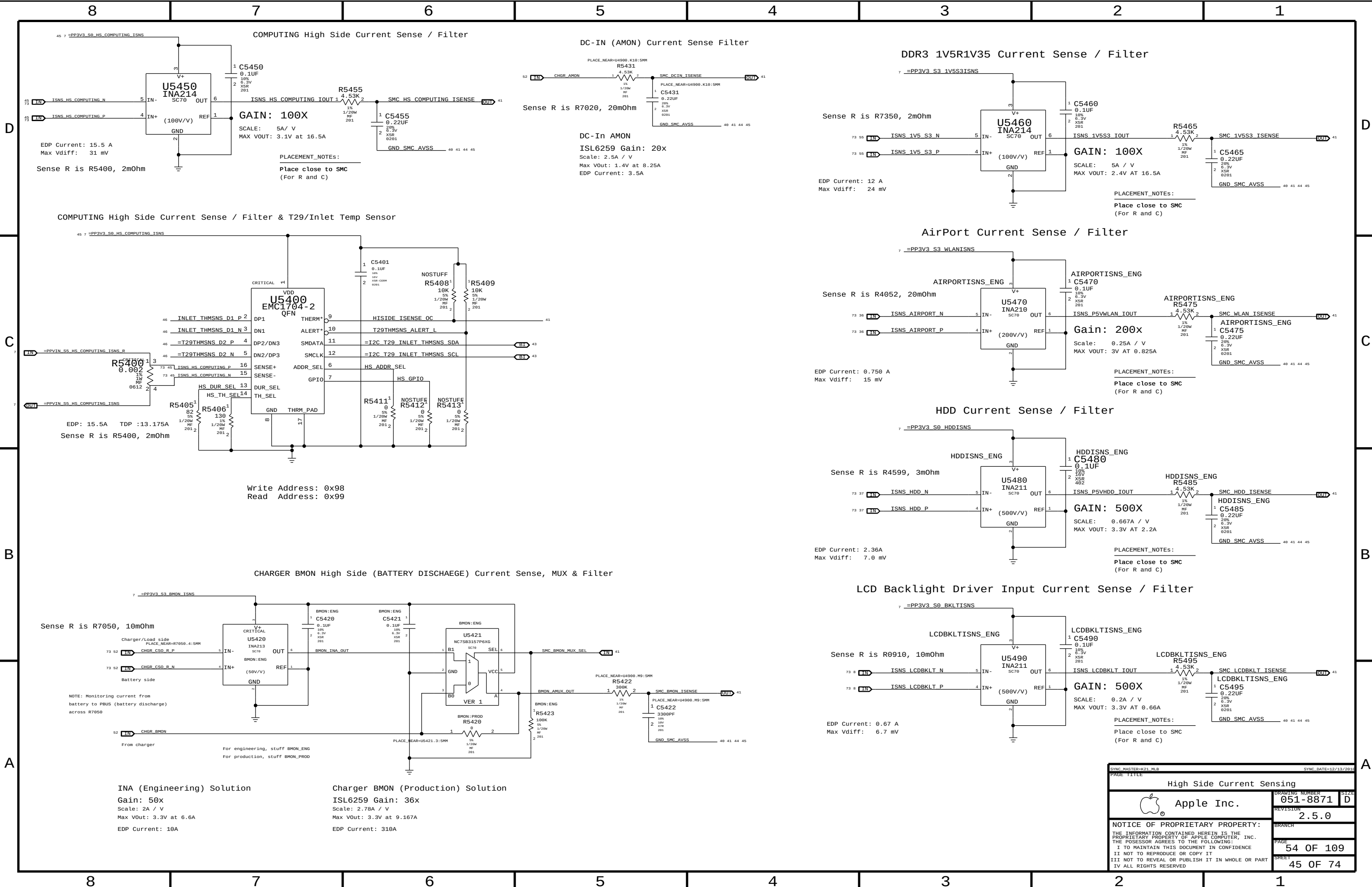
CPU 1.05V VCCIO Current Sense / Filter



GFX/IG Vcore Voltage Sense / Filter



SYNC MASTER=K21 MLB		SYNC DATE=12/13/2016	
PAGE TITLE		Voltage & Load Side Current Sensing	
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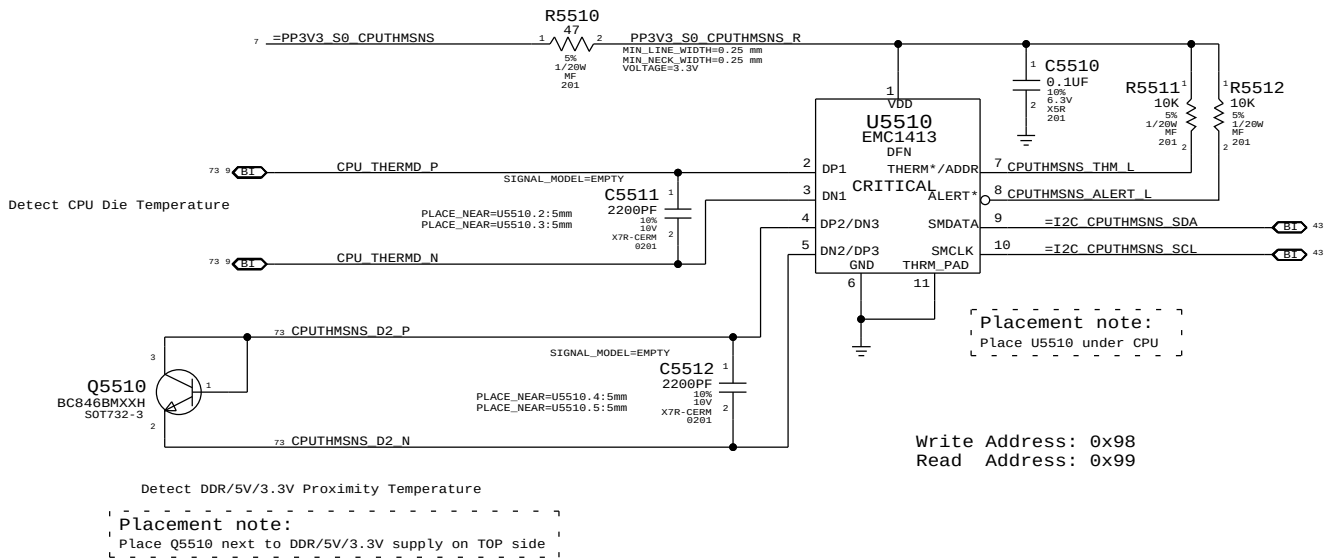
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A

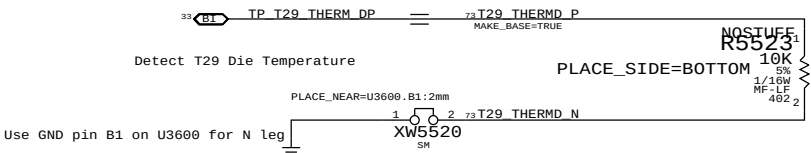
A

SYNC MASTER=K21_MLB		SYNC DATE=12/13/2011	
PAGE TITLE			
High Side Current Sensing			
	DRAWING NUMBER		SIZE
	051-8871		D
	REVISION		2.5.0
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CPU Proximity Sensor



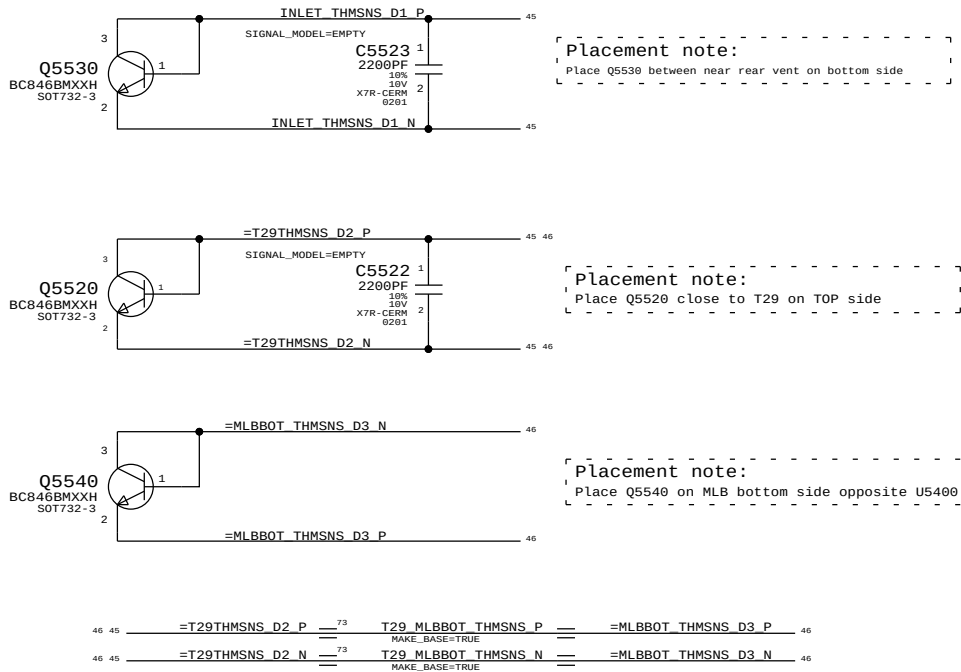
T29 Die



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	1	RES, MF, 1/20W, 100K OHM, 5, 0201, SMD	C5361		VCCIOISNS_PROD
117S0008	1	RES, MF, 1/20W, 100K OHM, 5, 0201, SMD	C5475		AIRPORTISNS_PROD
117S0008	1	RES, MF, 1/20W, 100K OHM, 5, 0201, SMD	C5485		HDDISNS_PROD
117S0008	1	RES, MF, 1/20W, 100K OHM, 5, 0201, SMD	C5495		LCDBKLTISNS_PROD

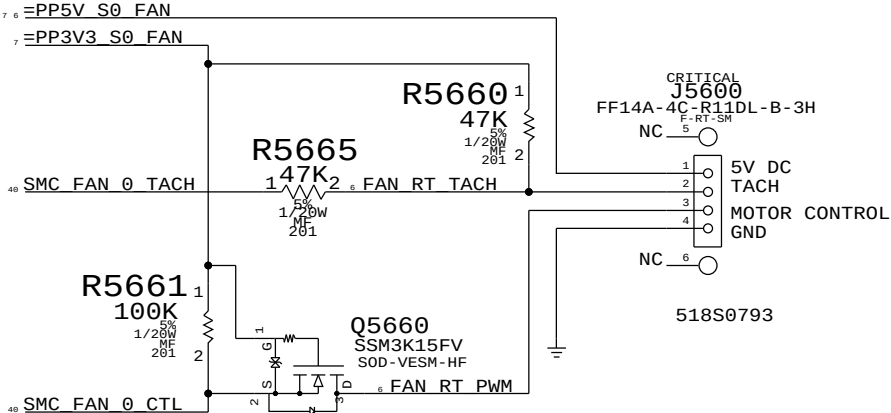
Replacing caps with 100K PD on ISENSE SMC inputs

T29, MLB Bottom & Inlet Proximity Sensors



SYNC MASTER=K21 MLB		SYNC DATE=12/13/2010	
PAGE TITLE		Thermal Sensors	
Apple Inc.		DRAWING NUMBER	051-8871
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FAN CONNECTOR



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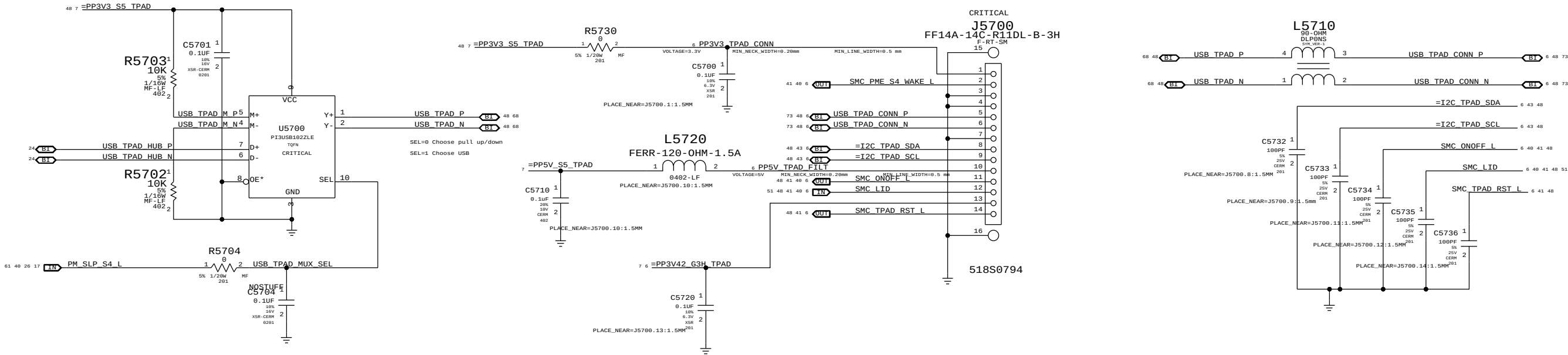
D

C

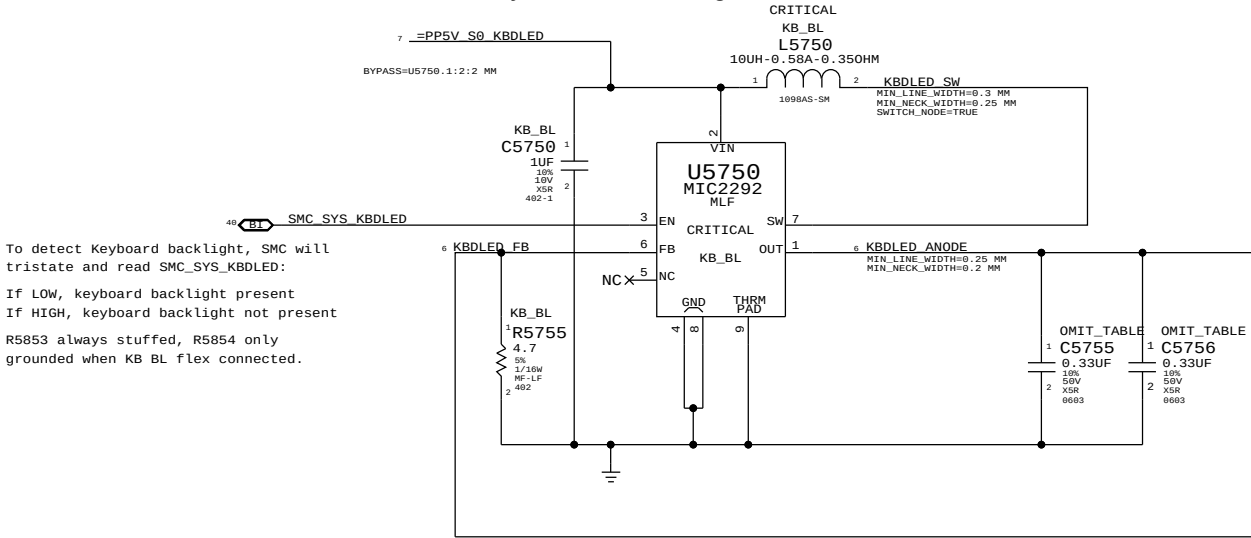
B

A

IPD Flex Connector

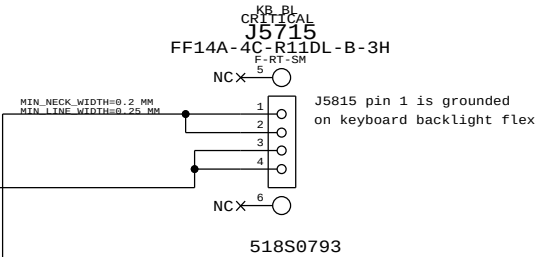


Keyboard Backlight Driver & Detection



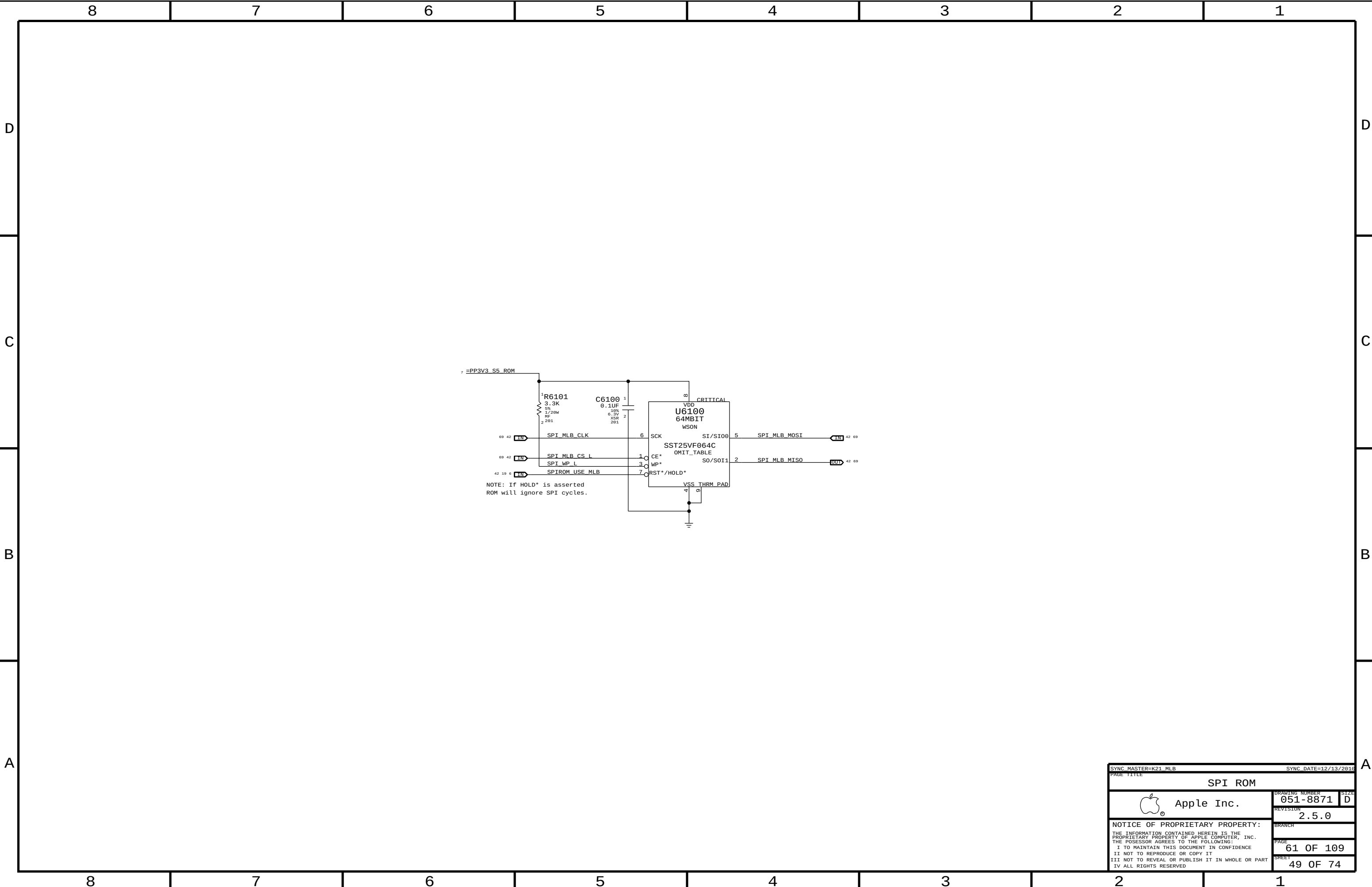
To detect Keyboard backlight, SMC will tristate and read SMC_SYS_KBDLED:
If LOW, keyboard backlight present
If HIGH, keyboard backlight not present
R5853 always stuffed, R5854 only grounded when KB BL flex connected.

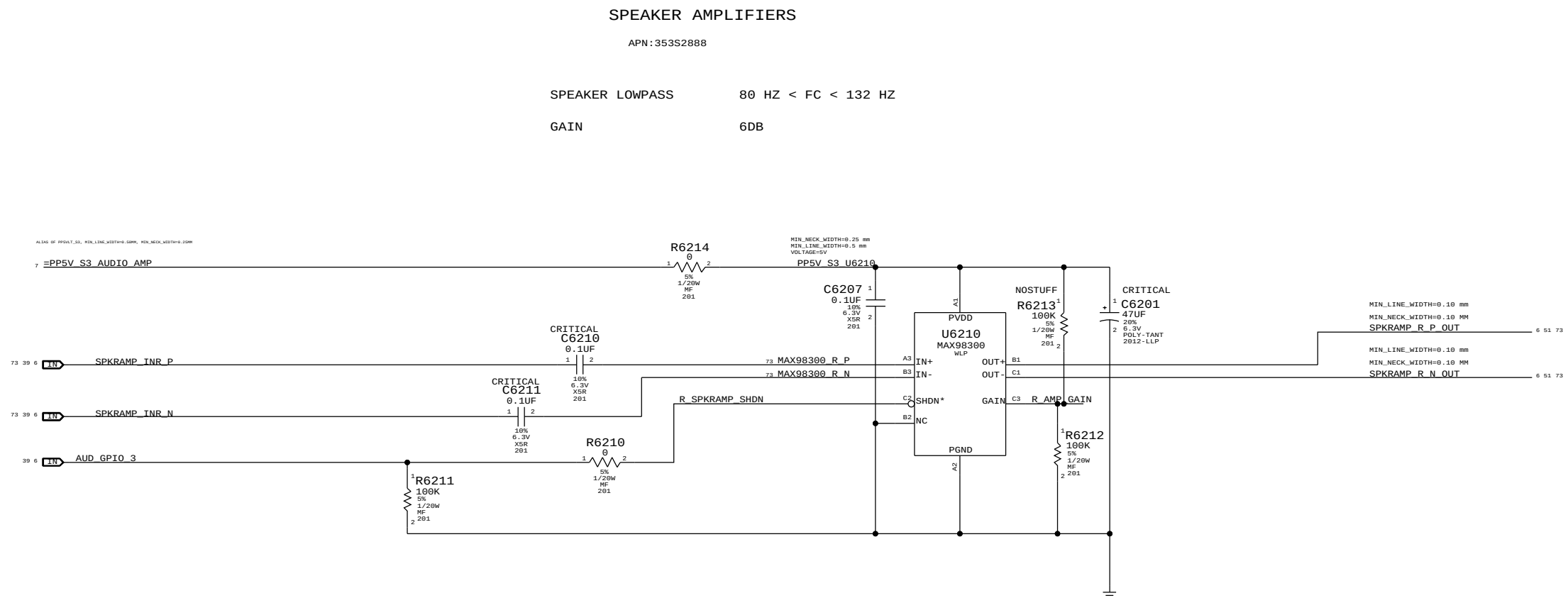
Keyboard Backlight Connector

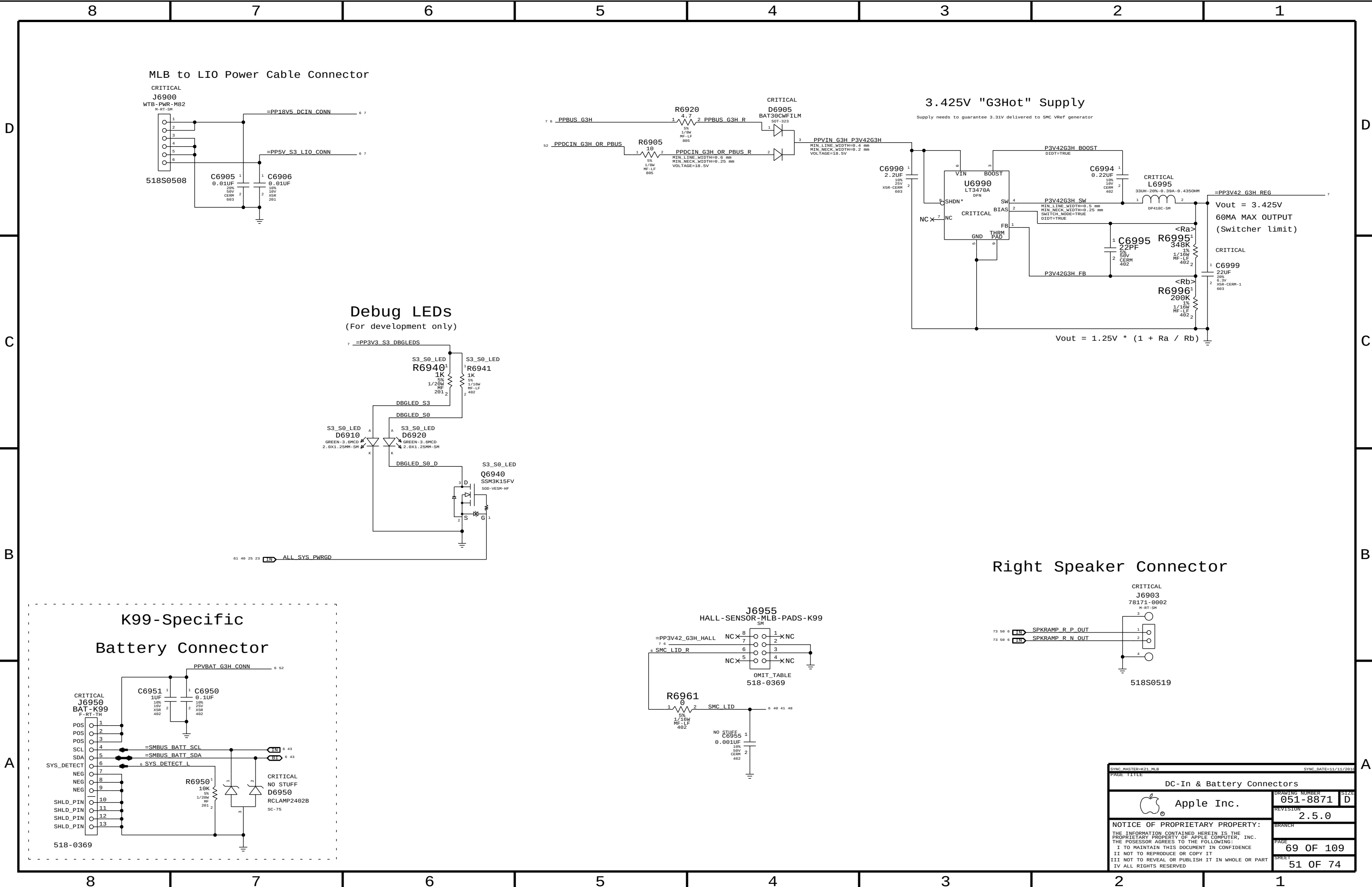


PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0704	2	CAP, CER, 0.22UF, 10%, 50V, XSR, 0603	C5756, C5755		KB_BL

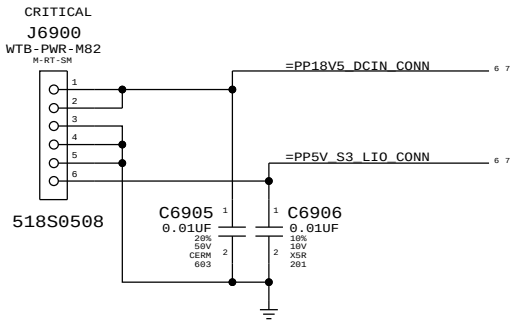
SYNC MASTER=K21 MLB		SYNC DATE=12/13/2010	
PAGE TITLE			
IPD / KBD Backlight			
 Apple Inc.		DRAWING NUMBER	051-8871
		SIZE	D
		REVISION	2.5.0
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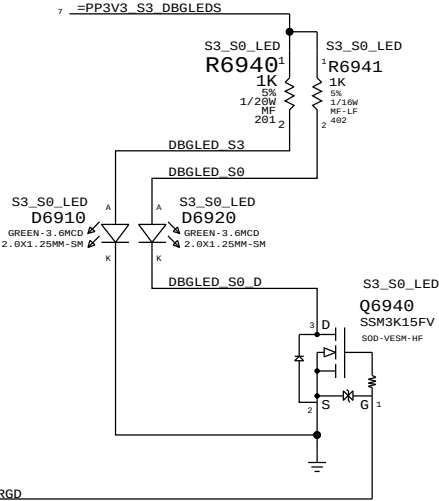




MLB to LIO Power Cable Connector

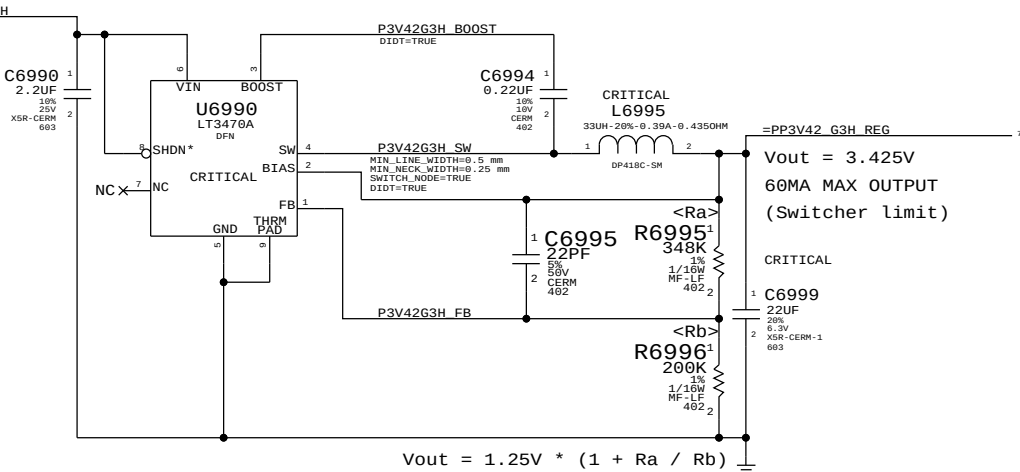


Debug LEDs
(For development only)

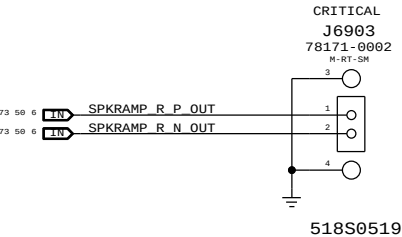


3.425V "G3Hot" Supply

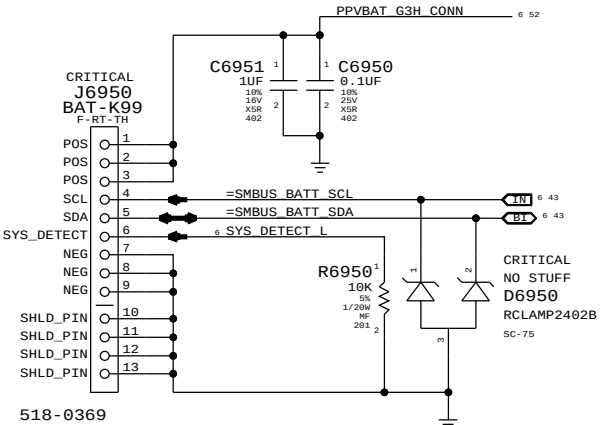
Supply needs to guarantee 3.31V delivered to SMC Vref generator



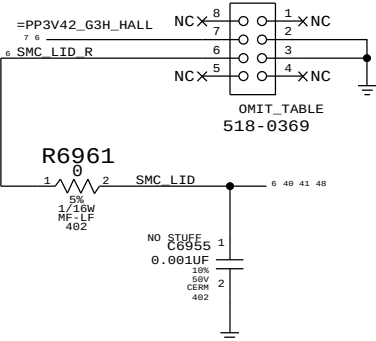
Right Speaker Connector



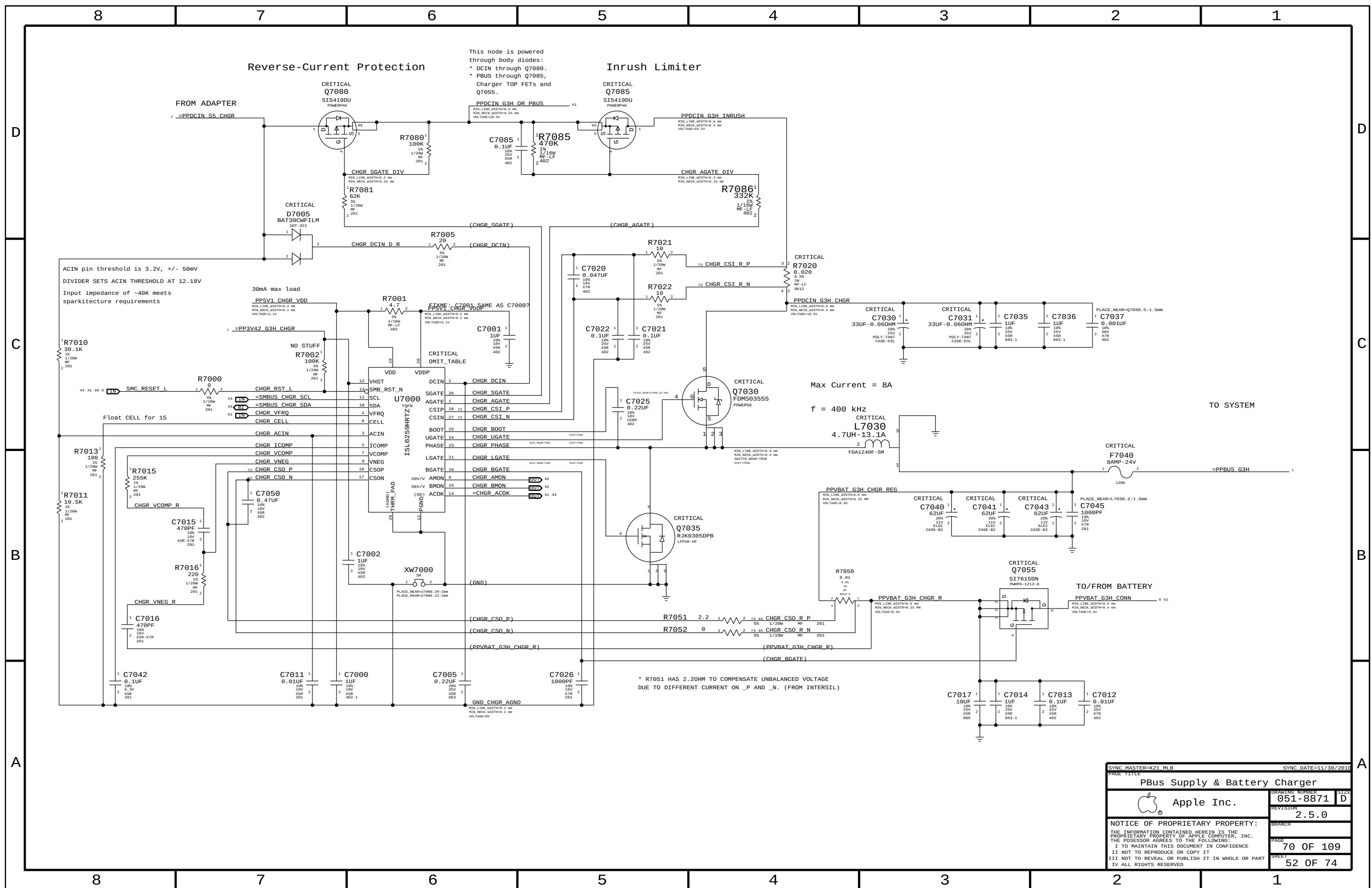
K99-Specific
Battery Connector

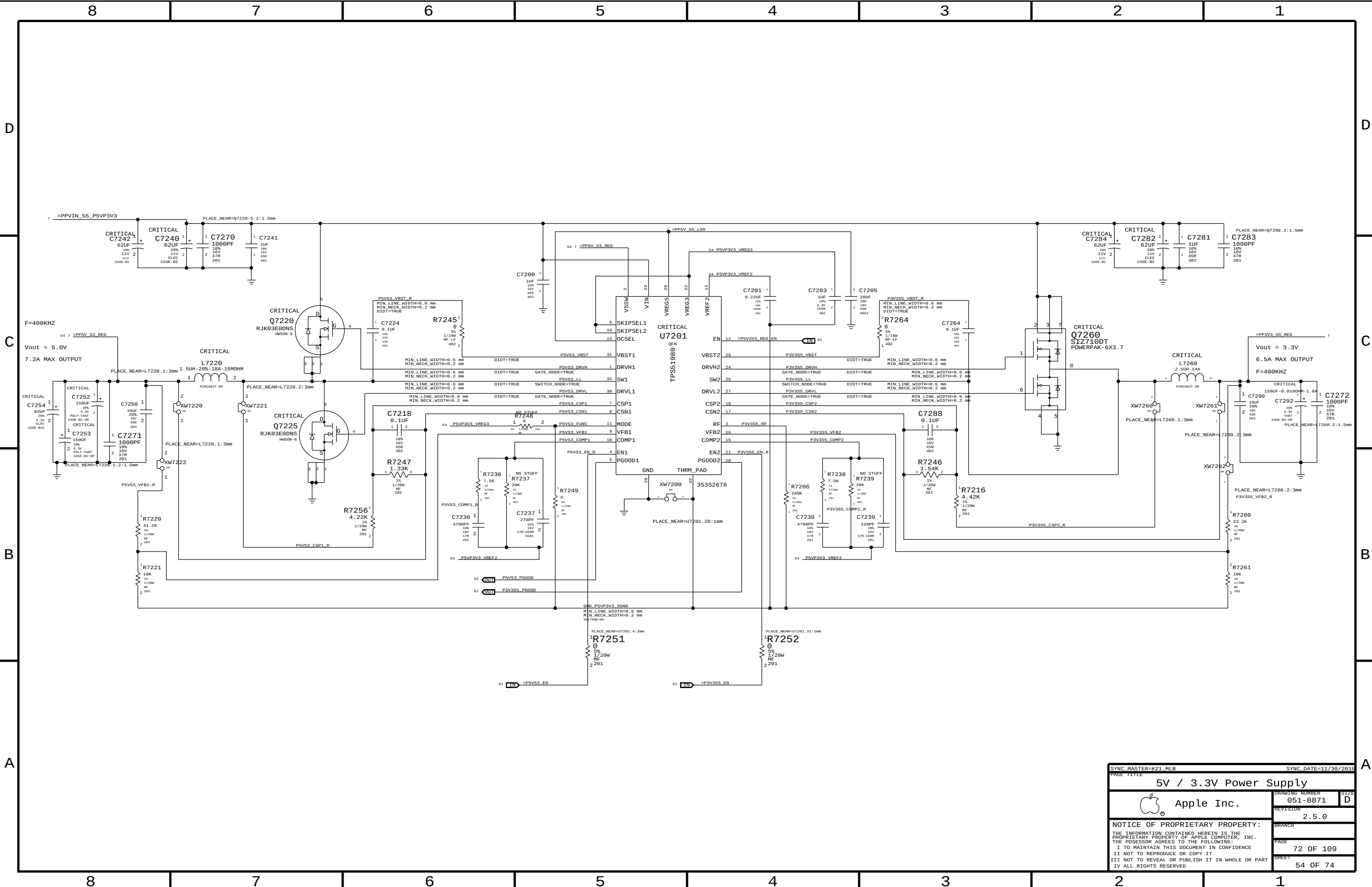


J6955
HALL-SENSOR-MLB-PADS-K99

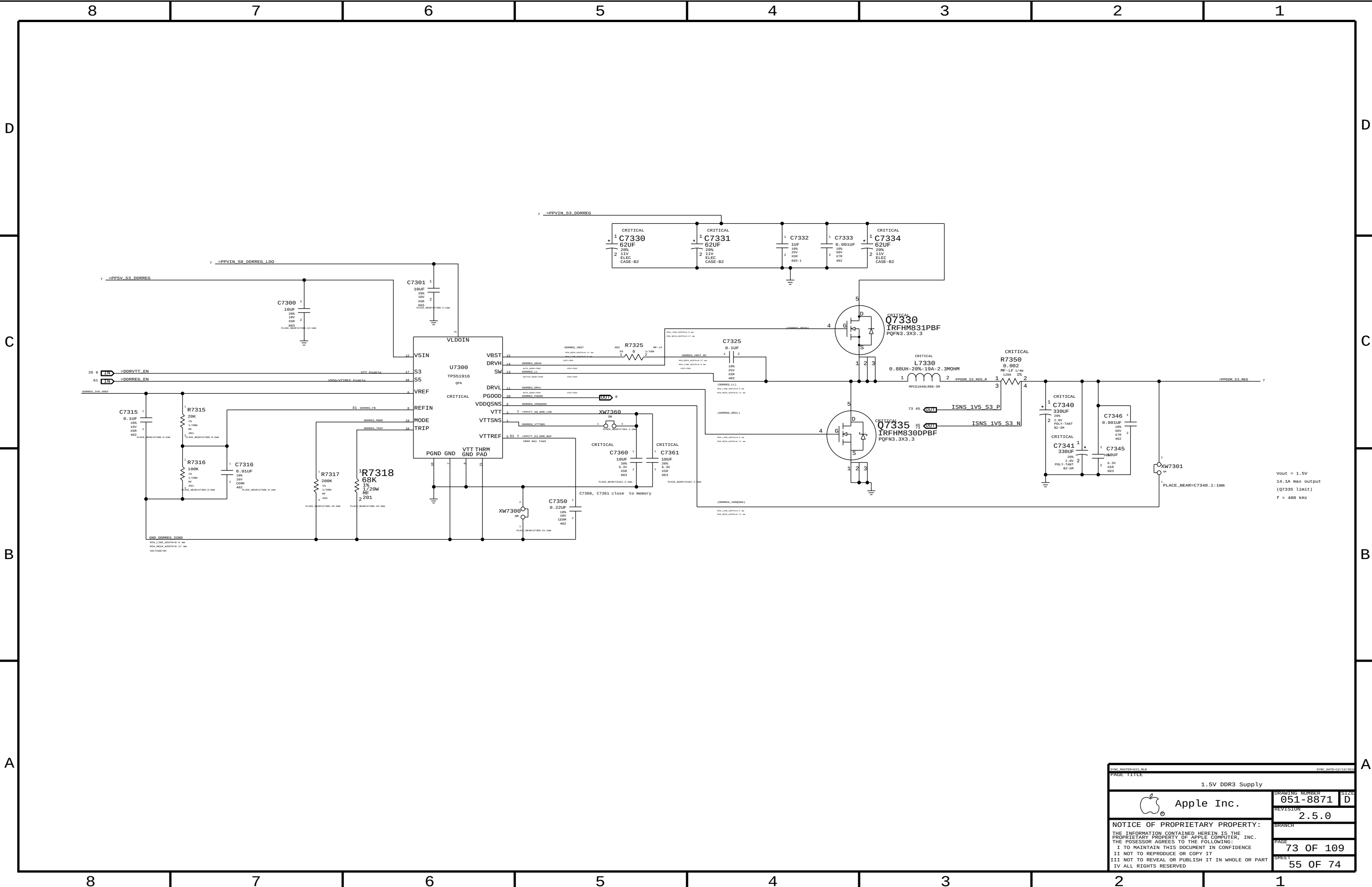


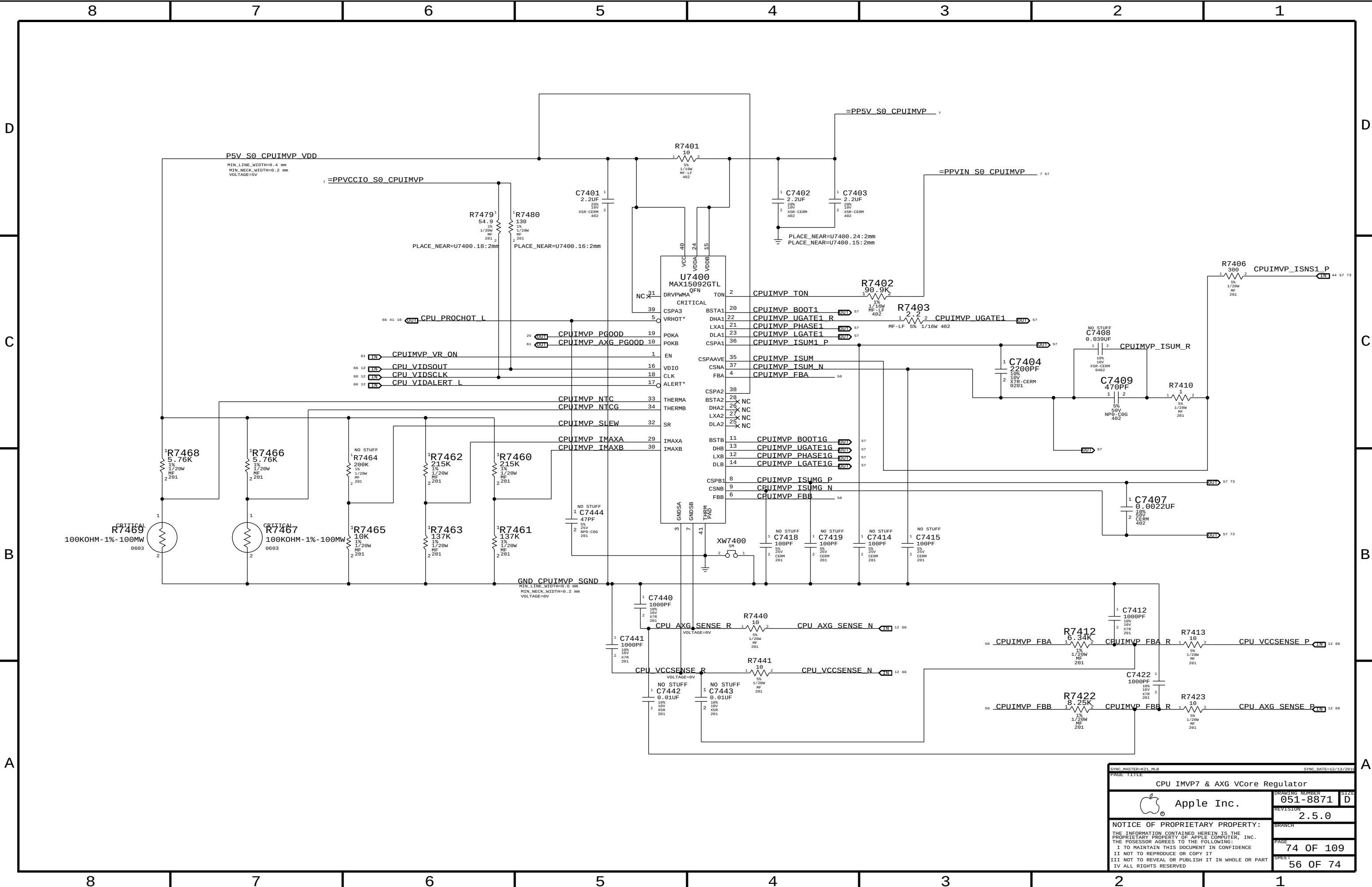
SYNC_MASTER=K21_MLB		SYNC_DATE=11/11/2011	
PAGE TITLE			
DC-In & Battery Connectors			
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SYNC MASTER=K21 MLB		SYNC DATE=11/30/2010	
PAGE TITLE			
5V / 3.3V Power Supply			
 Apple Inc.		DRAWING NUMBER	051-8871
		SIZE	D
		REVISION	2.5.0
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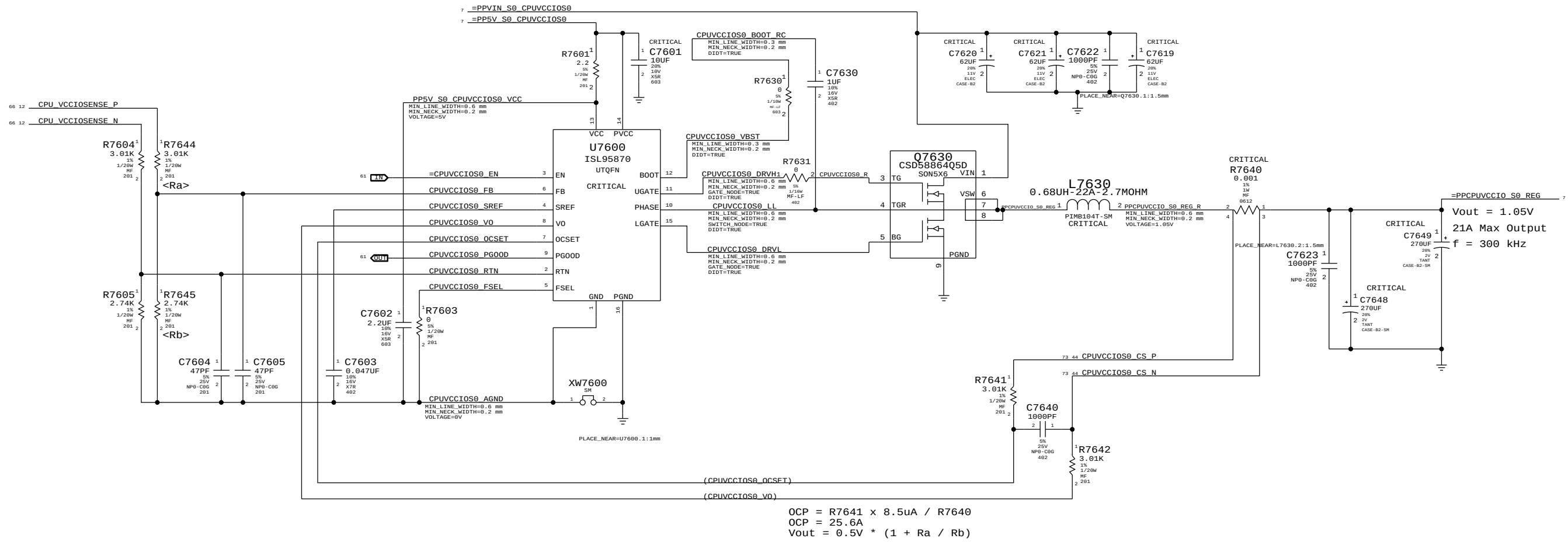




8	7	6	5	4	3	2	1
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DCBA

CPU VCCIO (1.05V S0) Regulator



D

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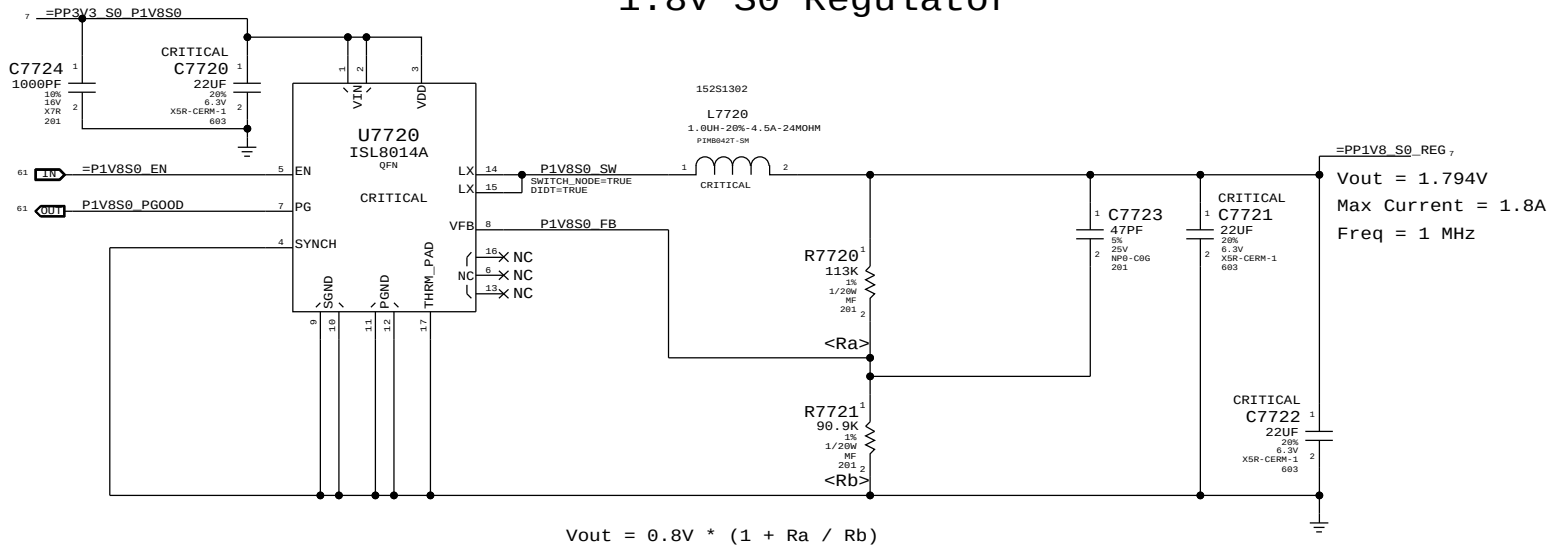
D

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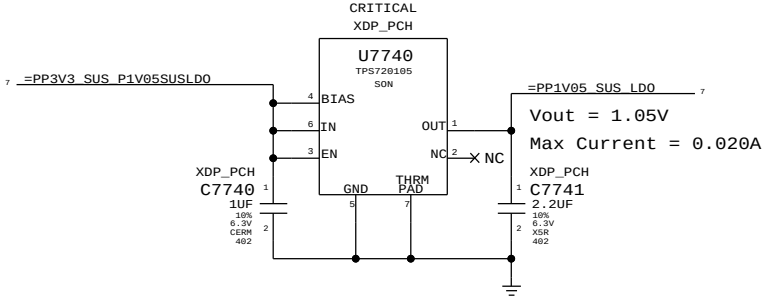
A

1.8V S0 Regulator

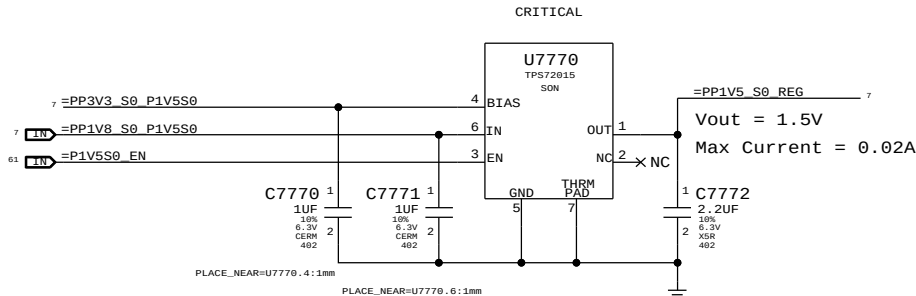


1.05V SUS LDO

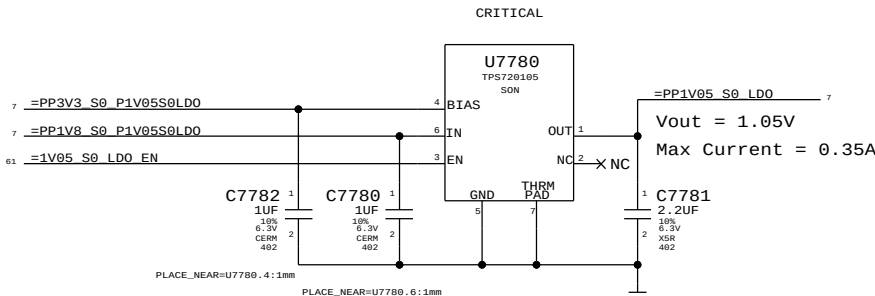
Cougar Point requires JTAG pull-ups to be powered at 1.05V when SUS suspend well is active. Pull-ups (3) must be 51 ohms to support XDP (not required in production). 70mA is required to support pull-ups. Alternative is strong voltage dividers (200/100) to 3.3V S5, which burns 100mW in all S-states.



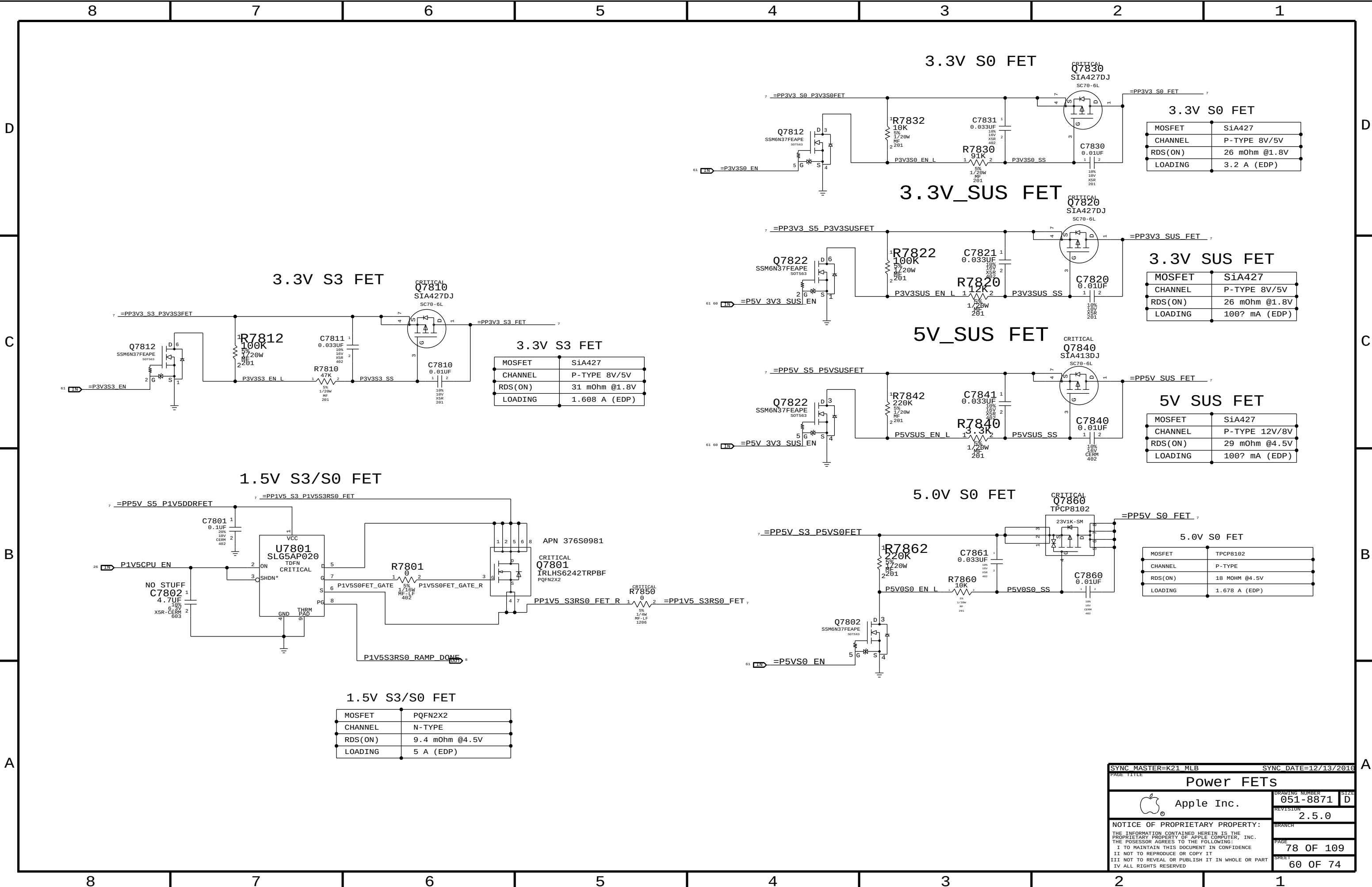
1.5V S0 LDO



1.05V S0 LDO



SYNC MASTER=K21 MLB		SYNC DATE=12/13/2010	
PAGE TITLE		PAGE	
Misc Power Supplies		DRAWING NUMBER	
Apple Inc.		051-8871	
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MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	3.2 A (EDP)

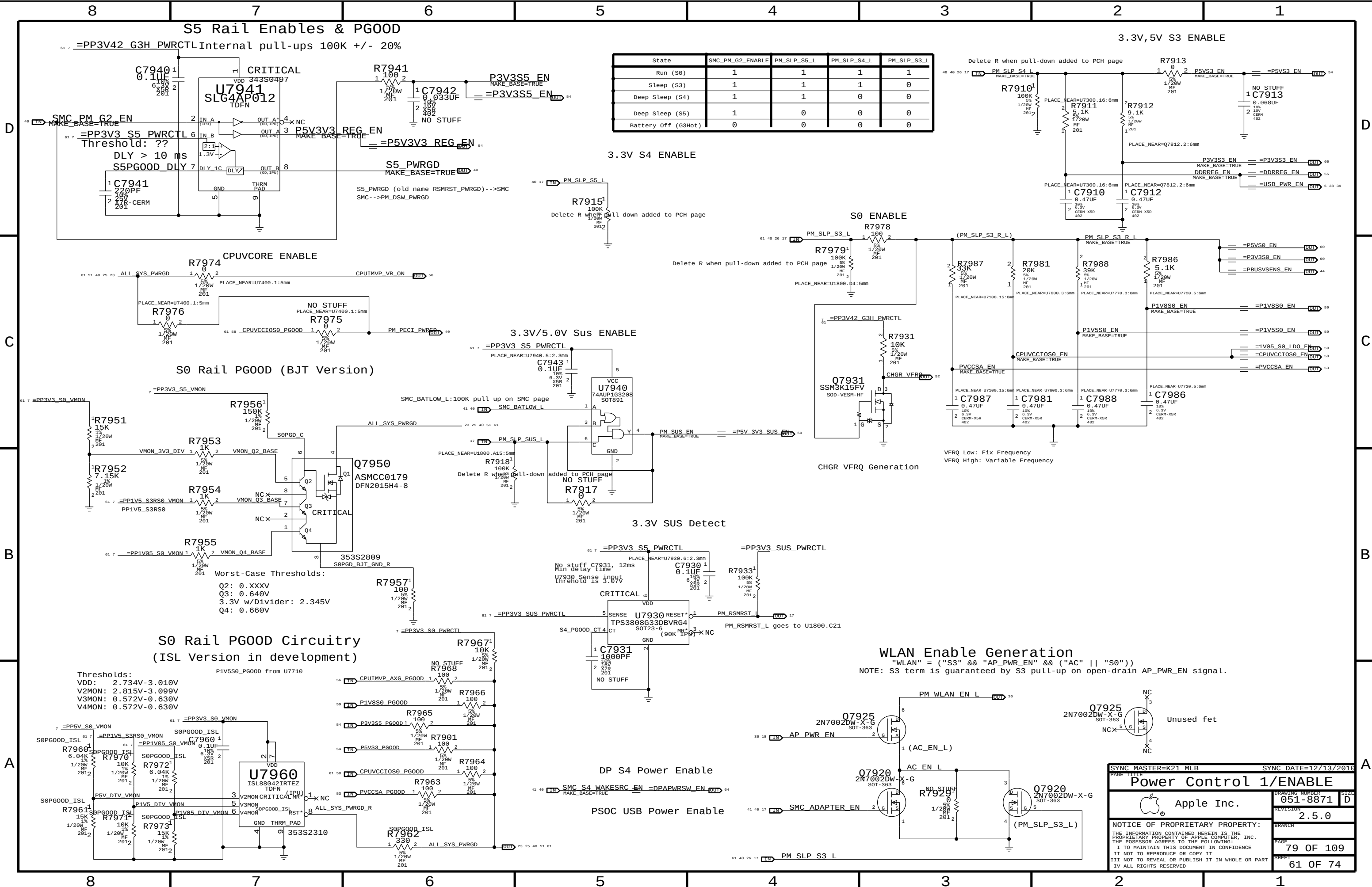
MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	100? mA (EDP)

MOSFET	SiA427
CHANNEL	P-TYPE 12V/8V
RDS(ON)	29 mOhm @4.5V
LOADING	100? mA (EDP)

MOSFET	TPCP8102
CHANNEL	P-TYPE
RDS(ON)	18 MOHM @4.5V
LOADING	1.678 A (EDP)

MOSFET	PQFN2X2
CHANNEL	N-TYPE
RDS(ON)	9.4 mOhm @4.5V
LOADING	5 A (EDP)

SYNC MASTER=K21 MLB		SYNC DATE=12/13/2016	
PAGE TITLE			
Power FETs			
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		PAGE	78 OF 109
		SHEET	60 OF 74



State	SMC_PM_G2_ENABLE	PM_SLP_S5_L	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	1	1	1	1
Sleep (S3)	1	1	1	0
Deep Sleep (S4)	1	1	0	0
Deep Sleep (S5)	1	0	0	0
Battery Off (G3Hot)	0	0	0	0

3.3V S4 ENABLE

3.3V/5.0V Sus ENABLE

3.3V SUS Detect

DP S4 Power Enable

PSOC USB Power Enable

WLAN Enable Generation

"WLAN" = ("S3" && "AP_PWR_EN" && ("AC" || "S0"))
NOTE: S3 term is guaranteed by S3 pull-up on open-drain AP_PWR_EN signal.

SYNC MASTER=K21 MLB

SYNC DATE=12/13/2016

Power Control 1/ENABLE

Apple Inc.

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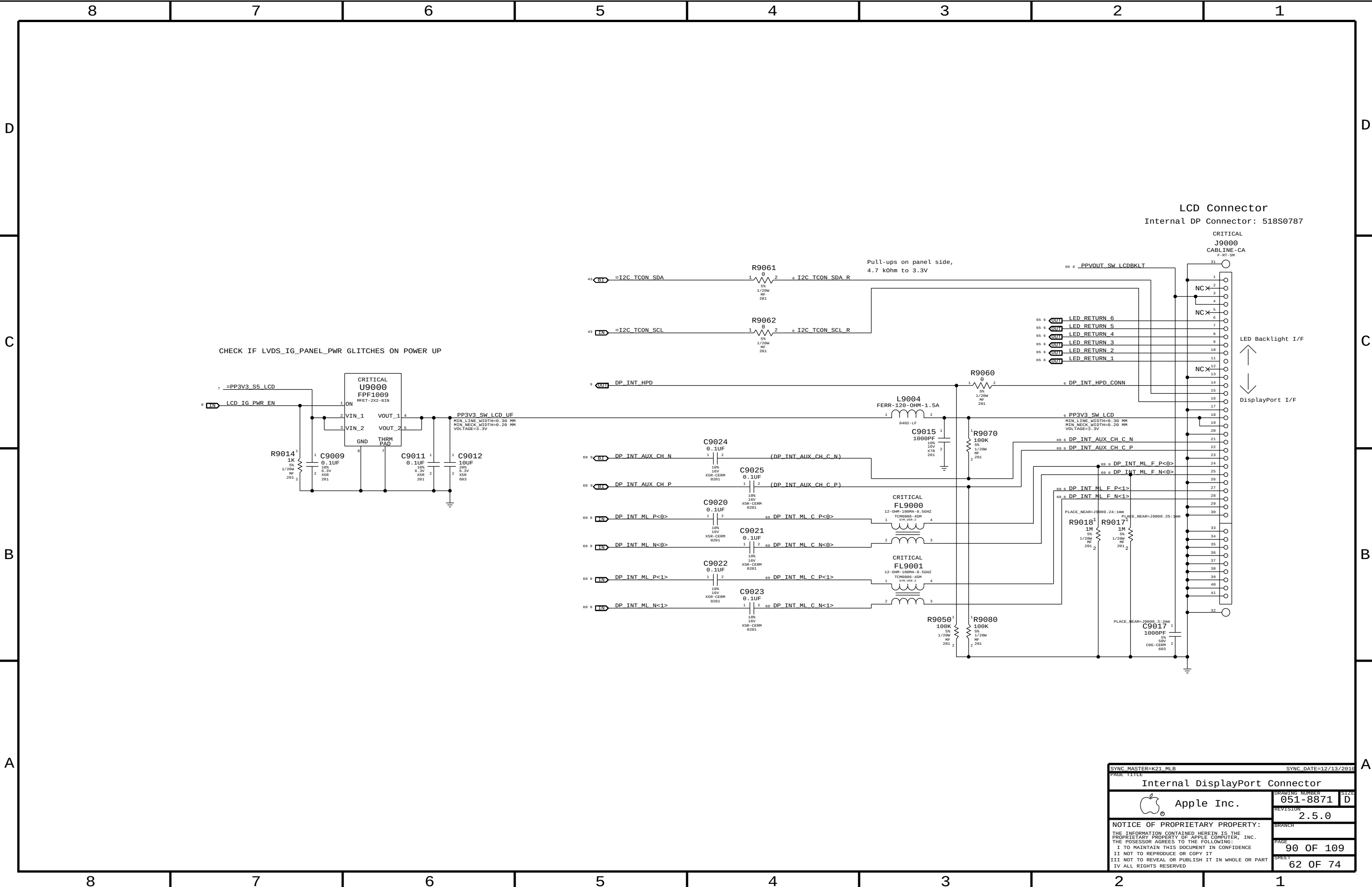
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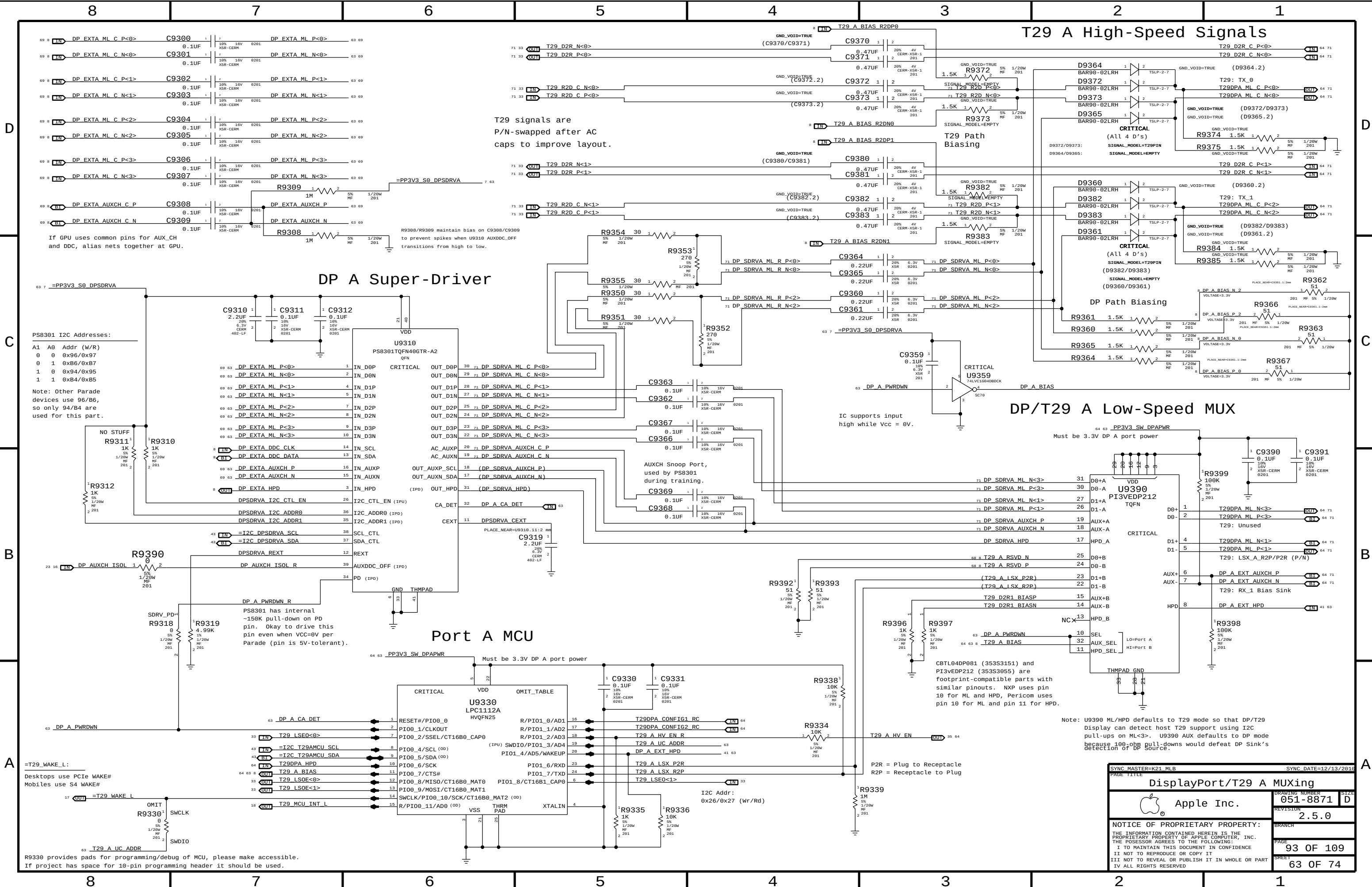


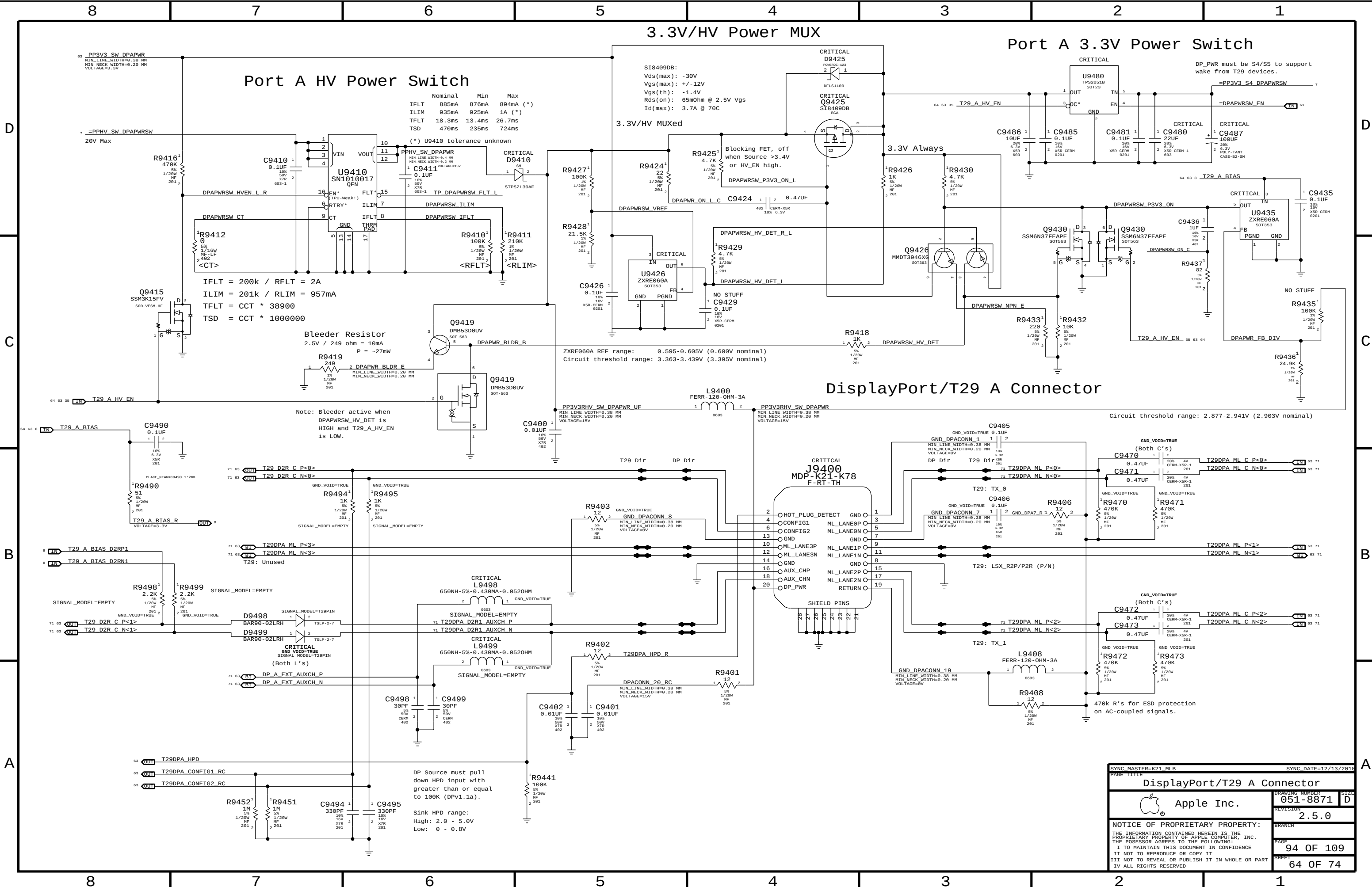
LCD Connector
Internal DP Connector: 518S0787

CRITICAL
J9000
CABLINE-CA
F-RT-SM

LED Backlight I/F
↑
↓
DisplayPort I/F

SYNC MASTER=K21 MLB		SYNC DATE=12/13/2010	
PAGE TITLE			
Internal DisplayPort Connector			
 Apple Inc.		DRAWING NUMBER	051-8871
		SHEET NUMBER	D
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Port A HV Power Switch

Port A 3.3V Power Switch

3.3V/HV Power MUX

DisplayPort/T29 A Connector

	Nominal	Min	Max
IFLT	885mA	876mA	894mA (*)
ILIM	935mA	925mA	1A (*)
TFLT	18.3ms	13.4ms	26.7ms
TSD	470ms	235ms	724ms

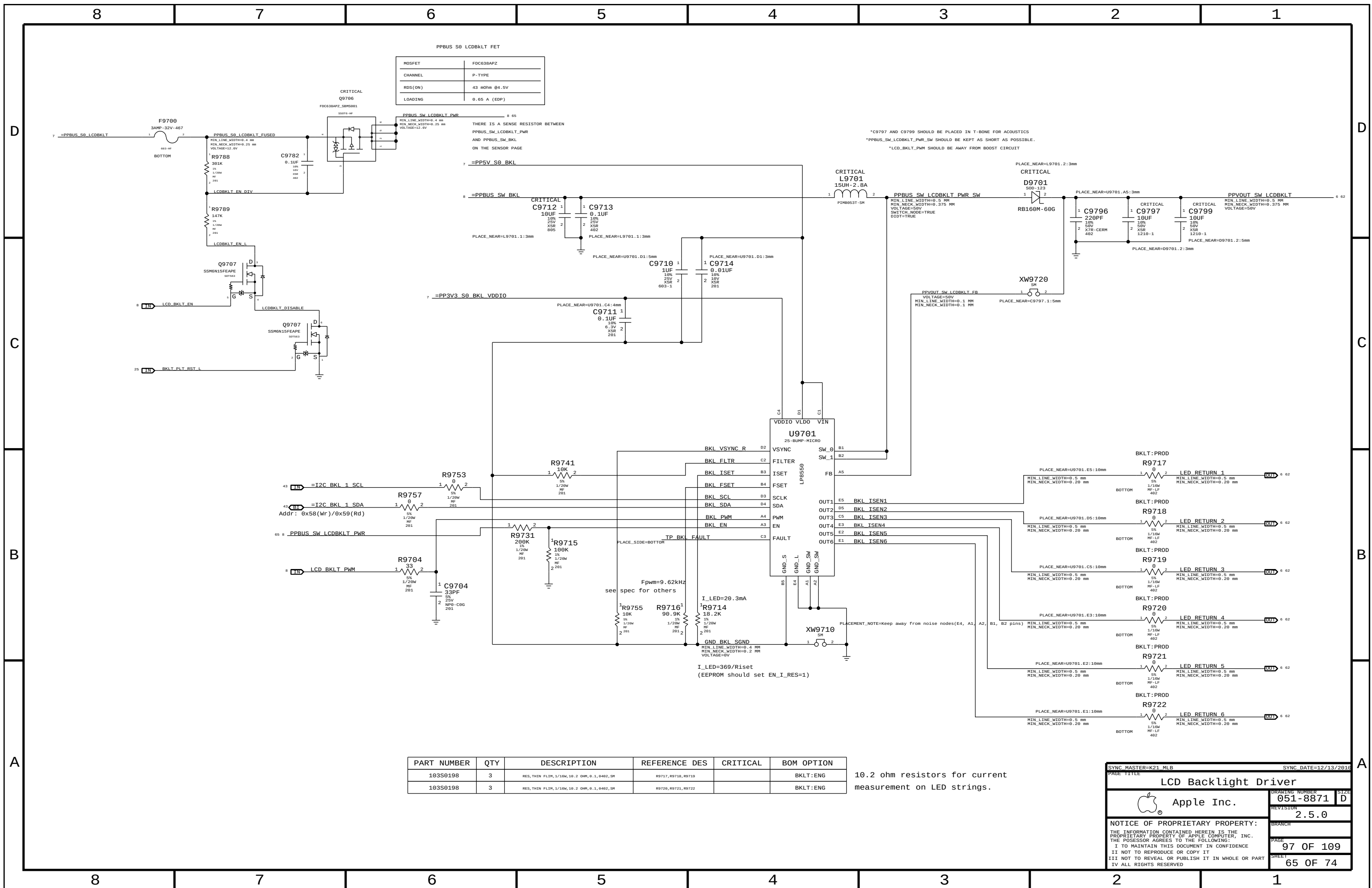
IFLT = 200k / RFLT = 2A
ILIM = 201k / RLIM = 957mA
TFLT = CCT * 38900
TSD = CCT * 100000

Bleeder Resistor
2.5V / 249 ohm = 10mA
P = -27mW

ZXRE060A REF range: 0.595-0.605V (0.600V nominal)
Circuit threshold range: 3.363-3.439V (3.395V nominal)

Circuit threshold range: 2.877-2.941V (2.903V nominal)

SYNC MASTER=K21 MLB		SYNC DATE=12/13/2010	
PAGE TITLE			
DisplayPort/T29 A Connector		DRAWING NUMBER	051-8871
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Memory Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
MEM_A_CLK	MEM_72D	MEM_CLK	MEM A CLK P<5..0>	8 11 27 28 32
MEM_A_CLK	MEM_72D	MEM_CLK	MEM A CLK N<5..0>	8 11 27 28 32
MEM_A_CTRL	MEM_55S	MEM_CTRL	MEM A CKE<3..0>	8 11 27 28 32
MEM_A_CTRL	MEM_55S	MEM_CTRL	MEM A CS L<3..0>	8 11 27 28 32
MEM_A_CTRL	MEM_55S	MEM_CTRL	MEM A ODT<3..0>	8 11 27 28 32
MEM_A_CMD	MEM_55S	MEM_CMD	MEM A A<15..0>	8 11 27 28 32
MEM_A_CMD	MEM_55S	MEM_CMD	MEM A BA<2..0>	11 27 28 32
MEM_A_CMD	MEM_55S	MEM_CMD	MEM A RAS L	11 27 28 32
MEM_A_CMD	MEM_55S	MEM_CMD	MEM A CAS L	11 27 28 32
MEM_A_CMD	MEM_55S	MEM_CMD	MEM A WE L	11 27 28 32
MEM_A_DQ_BYTE0	MEM_50S	MEM_DATA	MEM A DQ<7..0>	11 27
MEM_A_DQ_BYTE1	MEM_50S	MEM_DATA	MEM A DQ<15..8>	11 27
MEM_A_DQ_BYTE2	MEM_50S	MEM_DATA	MEM A DQ<23..16>	11 27
MEM_A_DQ_BYTE3	MEM_50S	MEM_DATA	MEM A DQ<31..24>	11 27
MEM_A_DQ_BYTE4	MEM_50S	MEM_DATA	MEM A DQ<39..32>	11 28
MEM_A_DQ_BYTE5	MEM_50S	MEM_DATA	MEM A DQ<47..40>	11 28
MEM_A_DQ_BYTE6	MEM_50S	MEM_DATA	MEM A DQ<55..48>	11 28
MEM_A_DQ_BYTE7	MEM_50S	MEM_DATA	MEM A DQ<63..56>	11 28
MEM_A_DQS0	MEM_85D	MEM_DQS	MEM A DQS P<0>	11 27
MEM_A_DQS0	MEM_85D	MEM_DQS	MEM A DQS N<0>	11 27
MEM_A_DQS1	MEM_85D	MEM_DQS	MEM A DQS P<1>	11 27
MEM_A_DQS1	MEM_85D	MEM_DQS	MEM A DQS N<1>	11 27
MEM_A_DQS2	MEM_85D	MEM_DQS	MEM A DQS P<2>	11 27
MEM_A_DQS2	MEM_85D	MEM_DQS	MEM A DQS N<2>	11 27
MEM_A_DQS3	MEM_85D	MEM_DQS	MEM A DQS P<3>	11 27
MEM_A_DQS3	MEM_85D	MEM_DQS	MEM A DQS N<3>	11 27
MEM_A_DQS4	MEM_85D	MEM_DQS	MEM A DQS P<4>	11 28
MEM_A_DQS4	MEM_85D	MEM_DQS	MEM A DQS N<4>	11 28
MEM_A_DQS5	MEM_85D	MEM_DQS	MEM A DQS P<5>	11 28
MEM_A_DQS5	MEM_85D	MEM_DQS	MEM A DQS N<5>	11 28
MEM_A_DQS6	MEM_85D	MEM_DQS	MEM A DQS P<6>	11 28
MEM_A_DQS6	MEM_85D	MEM_DQS	MEM A DQS N<6>	11 28
MEM_A_DQS7	MEM_85D	MEM_DQS	MEM A DQS P<7>	11 28
MEM_A_DQS7	MEM_85D	MEM_DQS	MEM A DQS N<7>	11 28
MEM_B_CLK	MEM_72D	MEM_CLK	MEM B CLK P<5..0>	8 11 29 30 32
MEM_B_CLK	MEM_72D	MEM_CLK	MEM B CLK N<5..0>	8 11 29 30 32
MEM_B_CTRL	MEM_55S	MEM_CTRL	MEM B CKE<3..0>	8 11 29 30 32
MEM_B_CTRL	MEM_55S	MEM_CTRL	MEM B CS L<3..0>	8 11 29 30 32
MEM_B_CTRL	MEM_55S	MEM_CTRL	MEM B ODT<3..0>	8 11 29 30 32
MEM_B_CMD	MEM_55S	MEM_CMD	MEM B A<15..0>	8 11 29 30 32
MEM_B_CMD	MEM_55S	MEM_CMD	MEM B BA<2..0>	11 29 30 32
MEM_B_CMD	MEM_55S	MEM_CMD	MEM B RAS L	11 29 30 32
MEM_B_CMD	MEM_55S	MEM_CMD	MEM B CAS L	11 29 30 32
MEM_B_CMD	MEM_55S	MEM_CMD	MEM B WE L	11 29 30 32
MEM_B_DQ_BYTE0	MEM_50S	MEM_DATA	MEM B DQ<7..0>	11 29
MEM_B_DQ_BYTE1	MEM_50S	MEM_DATA	MEM B DQ<15..8>	11 29
MEM_B_DQ_BYTE2	MEM_50S	MEM_DATA	MEM B DQ<23..16>	11 29
MEM_B_DQ_BYTE3	MEM_50S	MEM_DATA	MEM B DQ<31..24>	11 29
MEM_B_DQ_BYTE4	MEM_50S	MEM_DATA	MEM B DQ<39..32>	11 30
MEM_B_DQ_BYTE5	MEM_50S	MEM_DATA	MEM B DQ<47..40>	11 30
MEM_B_DQ_BYTE6	MEM_50S	MEM_DATA	MEM B DQ<55..48>	11 30
MEM_B_DQ_BYTE7	MEM_50S	MEM_DATA	MEM B DQ<63..56>	11 30
MEM_B_DQS0	MEM_85D	MEM_DQS	MEM B DQS P<0>	11 29
MEM_B_DQS0	MEM_85D	MEM_DQS	MEM B DQS N<0>	11 29
MEM_B_DQS1	MEM_85D	MEM_DQS	MEM B DQS P<1>	11 29
MEM_B_DQS1	MEM_85D	MEM_DQS	MEM B DQS N<1>	11 29
MEM_B_DQS2	MEM_85D	MEM_DQS	MEM B DQS P<2>	11 29
MEM_B_DQS2	MEM_85D	MEM_DQS	MEM B DQS N<2>	11 29
MEM_B_DQS3	MEM_85D	MEM_DQS	MEM B DQS P<3>	11 29
MEM_B_DQS3	MEM_85D	MEM_DQS	MEM B DQS N<3>	11 29
MEM_B_DQS4	MEM_85D	MEM_DQS	MEM B DQS P<4>	11 30
MEM_B_DQS4	MEM_85D	MEM_DQS	MEM B DQS N<4>	11 30
MEM_B_DQS5	MEM_85D	MEM_DQS	MEM B DQS P<5>	11 30
MEM_B_DQS5	MEM_85D	MEM_DQS	MEM B DQS N<5>	11 30
MEM_B_DQS6	MEM_85D	MEM_DQS	MEM B DQS P<6>	11 30
MEM_B_DQS6	MEM_85D	MEM_DQS	MEM B DQS N<6>	11 30
MEM_B_DQS7	MEM_85D	MEM_DQS	MEM B DQS P<7>	11 30
MEM_B_DQS7	MEM_85D	MEM_DQS	MEM B DQS N<7>	11 30
		MEM_PWR	PP1V5 S3RS0	6 7
		MEM_PWR	PP1V5 S3	6 7
		MEM_PWR	PP0V75 S3 MEM VREFCA A	27 28 29 30 31
		MEM_PWR	PP0V75 S3 MEM VREFDQ A	9 27 28 29 30 31

Need to support MEM_*-style wildcards!

DDR3: Sandybridge SFF 2C when routed on Type-3 (Through hole) should follow rPGA guidelines per Huron River SFF D6 rev1.0 (#438297).
DQS intra-pair matching should be within 0.127mm, no inter-pair matching requirement.
DQ to DQS matching per byte lane should be within 0.127mm.
DQS to clock matching should be within [CLK-63.5mm] and [CLK+38.1mm].
CLK intra-pair matching should be within 0.127mm, inter-pair matching should be within 0.9508mm.
CONTROL signals should be matched within [CLK-2.54mm] to [CLK+0.6mm] of CLK pairs.
A/BA/CMD signals should be matched within [CLK-12.7mm] to [CLK+12.7mm] of CLK pairs A/BA/CMD signals to each other should match within 5.08mm.
DQ/DQS/A/BA/cmd signal spacing is 4x dielectric, CLK is 5x dielectric.
Maximum length of any signal from die pad to SODIMM pad is 119.83mm, from procesor ball to SODIMM pad is 88.9mm.
SOURCE: Huron River Platform DG, Rev 1.01 (#436735), Section 2.5

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Memory Constraints

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Digital Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_850	*	=85_0HM_DIFF	=85_0HM_DIFF	=85_0HM_DIFF	=85_0HM_DIFF	=85_0HM_DIFF	=85_0HM_DIFF
LVDS_900	*	=90_0HM_DIFF	=90_0HM_DIFF	=90_0HM_DIFF	=90_0HM_DIFF	=90_0HM_DIFF	=90_0HM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	*	=3x_DIELECTRIC	?
LVDS	*	=3x_DIELECTRIC	?

LVDS intra-pair matching should be 5 mils. Pairs should be within 100 mils of clock length.
DisplayPort/TMDS intra-pair matching should be 5 ps. Inter-pair matching should be within 150 ps.
DisplayPort/AUX CH intra-pair matching should be 5 ps. No relationship to other signals.
Max length of LVDS/DisplayPort/TMDS traces: 12 inches.
SOURCE: MCP719 Interface DG (DG-03328-001-V00), Sections 2.5.3 & 2.5.4.

SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_900	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA	*	=4x_DIELECTRIC	?
SATA_1COMP	*	8 MIL	?

SOURCE: MCP79 Interface DG (DG-03328-001_v00), Section 2.7.1.

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_USB_BIAS	*	=STANDARD	8 MIL	8 MIL	=STANDARD	=STANDARD	=STANDARD
USB_85D	*	=BS_OHM_DIFF	=BS_OHM_DIFF	=BS_OHM_DIFF	=BS_OHM_DIFF	=BS_OHM_DIFF	=BS_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=2x_DIELECTRIC	?	USB	TOP, BOTTOM	=4x_DIELECTRIC	?

SOURCE: Calpella Platform Design Guide for Ixex Peak M (DG-398905-398905_v1.5), Section 3.8

PCH Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE	
	PHYSICAL	SPACING	
DP_MI	DP_85D	DTSPAYPORT	DP IG ML P<3..0>
DP_MI	DP_85D	DISPLAYPORT	DP IG ML N<3..0>
DP_EXT_AUXCH	DP_85D	DTSPAYPORT	DP IG AUX CH P
DP_EXT_AUXCH	DP_85D	DISPLAYPORT	DP IG AUX CH N
LVDS_IG_A_CLK	LVDS_90D	LVDS	LVDS IG A CLK P
LVDS_IG_A_CLK	LVDS_90D	LVDS	LVDS IG A CLK N
LVDS_IG_A_DATA	LVDS_90D	LVDS	LVDS IG A DATA P<2..0>
LVDS_IG_A_DATA	LVDS_90D	LVDS	LVDS IG A DATA N<2..0>
	LVDS_90D	LVDS	LVDS IG A DATA P<3>
	LVDS_90D	LVDS	LVDS IG A DATA N<3>
	LVDS_90D	LVDS	LVDS IG B DATA P<3..0>
	LVDS_90D	LVDS	LVDS IG B DATA N<3..0>
	LVDS_90D	LVDS	LVDS IG B CLK P
	LVDS_90D	LVDS	LVDS IG B CLK N
	SATA_90D	SATA	SATA HDD R2D C P
	SATA_90D	SATA	SATA HDD R2D C N
SATA_HDD_R2D	SATA_90D	SATA	SATA HDD R2D P
SATA_HDD_R2D	SATA_90D	SATA	SATA HDD R2D N
SATA_HDD_D2R	SATA_90D	SATA	SATA HDD D2R P
SATA_HDD_D2R	SATA_90D	SATA	SATA HDD D2R N
	SATA_90D	SATA	SATA HDD D2R C P
	SATA_90D	SATA	SATA HDD D2R C N
	SATA_90D	SATA	SATA ODD R2D C P
	SATA_90D	SATA	SATA ODD R2D C N
SATA_ODD_R2D	SATA_90D	SATA	SATA ODD R2D P
SATA_ODD_R2D	SATA_90D	SATA	SATA ODD R2D N
SATA_ODD_D2R	SATA_90D	SATA	SATA ODD D2R P
SATA_ODD_D2R	SATA_90D	SATA	SATA ODD D2R N
	SATA_90D	SATA	SATA HDD R2D RC P
	SATA_90D	SATA	SATA HDD R2D RC N
	SATA_90D	SATA	SATA HDD D2R RC P
	SATA_90D	SATA	SATA HDD D2R RC N
PCH_SATA_ICOMP		SATA_ICOMP	PCH SATAICOMP
USB_HUB1_UP	USB_85D	USB	USB HUB1 UP P
	USB_85D	USB	USB HUB1 UP N
USB_HUB2_UP	USB_85D	USB	USB HUB2 UP P
	USB_85D	USB	USB HUB2 UP N
USB_EXT_A	USB_85D	USB	USB EXT_A P
USB_EXT_A	USB_85D	USB	USB EXT_A N
USB_EXTB	USB_85D	USB	USB EXT_B P
	USB_85D	USB	USB EXT_B N
USB_EXT_C	USB_85D	USB	USB EXT_C P
	USB_85D	USB	USB EXT_C N
USB_EXTD	USB_85D	USB	USB EXT_D P
	USB_85D	USB	USB EXT_D N
USB_EXTD	USB_85D	USB	USB T29A P
	USB_85D	USB	USB T29A N
	USB_85D	USB	T29 A RSVD P
	USB_85D	USB	T29 A RSVD N
USB_CAMERA	USB_85D	USB	USB CAMERA P
	USB_85D	USB	USB CAMERA N
USB_CAMERA	USB_85D	USB	USB CAMERA CONN P
	USB_85D	USB	USB CAMERA CONN N
USB_BT	USB_85D	USB	USB BT P
USB_BT	USB_85D	USB	USB BT N
USB_TPAD	USB_85D	USB	USB TPAD P
	USB_85D	USB	USB TPAD N
USB_IR	USB_85D	USB	USB IR P
	USB_85D	USB	USB IR N
USB_SD CARD	USB_85D	USB	USB SD CARD P
	USB_85D	USB	USB SD CARD N
USB_BRCRYPT	USB_85D	USB	USB BRCRYPT P
	USB_85D	USB	USB BRCRYPT N
PCH_USB_RBTAS	PCH_USB_RBTAS		PCH USB RBIAS
PCH_DIEFCLK_UNUSED	CLK_PCIE_90D	CLK_PCIE	PCIE CLK100M PCH P
PCH_DIEFCLK_UNUSED	CLK_PCIE_90D	CLK_PCIE	PCIE CLK100M PCH N
	CLK_PCIE_90D	CLK_PCIE	FSB CLK133M PCH P
	CLK_PCIE_90D	CLK_PCIE	FSB CLK133M PCH N
PCH_DIEFCLK_UNUSED	CLK_PCIE_90D	CLK_PCIE	PCH CLK96M DOT P
PCH_DIEFCLK_UNUSED	CLK_PCIE_90D	CLK_PCIE	PCH CLK96M DOT N
PCH_DIEFCLK_UNUSED	CLK_PCIE_90D	CLK_PCIE	PCH CLK100M SATA P
PCH_DIEFCLK_UNUSED	CLK_PCIE_90D	CLK_PCIE	PCH CLK100M SATA N
	CPU_56S	CLK_PCIE	PCH CLK14P3M REFCLK
IPC_CLK33M	CPU_56S	CLK_PCIE	PCH CLK33M PCIIIN
GEX_CLK_DPLLSS	CLK_PCIE_90D	CLK_PCIE	GFX CLK120M DPPLLSS P
GEX_CLK_DPLLSS	CLK_PCIE_90D	CLK_PCIE	GFX CLK120M DPPLLSS N

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LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	6 MIL	?
CLK_LPC	*	8 MIL	?

SOURCE: Calpella Platform Design Guide for Ixex Peak M (DG-398905-398905_v1.5), Section 3.15

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SM8_S05	*	=50_OHR_SE	=50_OHR_SE	=50_OHR_SE	=50_OHR_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

SOURCE: Calpella Platform Design Guide for Ibex Peak M (DG-398905-398905_v1.5), Section 3.15

SIO Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_55S	*	=55_OH1_SE	=55_OH1_SE	=55_OH1_SE	=55_OH1_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	8 MIL	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPT_555	*	=55_OHR_SE	=55_OHR_SE	=55_OHR_SE	=55_OHR_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	8 MIL	?

DisplayPort Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_BSD	*	=BS_OHM_DIFF	=BS_OHM_DIFF	=BS_OHM_DIFF	=BS_OHM_DIFF	=BS_OHM_DIFF	=BS_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	*	=3x_DIELECTRIC	?

PCI-Express Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_BSD	*	=BS_OHM_DIFF	=BS_OHM_DIFF	=BS_OHM_DIFF	=BS_OHM_DIFF	=BS_OHM_DIFF	=BS_OHM_DIFF
CLK_PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	*	=3X_DIELECTRIC	?
CLK_PCIE	*	20 MIL	?

System Clock Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_25M_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	=2X_DIELECTRIC	?
CLK_25M	*	=5X_DIELECTRIC	?

NOTE: 25MHZ system clocks very sensitive to noise.

PCH Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
I	LPC_AD	LPC_50S	LPC	LPC_AD<3..0>	6 16 40 42
	LPC_FRAME_L	LPC_50S	LPC	LPC_FRAME_L	6 16 40 42
	LPC_RESET_L	LPC_50S	LPC	LPC_RESET_L	25
I	LPC_CLK33M	CLK_LPC_50S	CLK_LPC	LPC_CLK33M_SMC_R	38 25
	LPC_CLK33M	CLK_LPC_50S	CLK_LPC	LPC_CLK33M_SMC	25 40
	LPC_CLK33M	CLK_LPC_50S	CLK_LPC	LPC_CLK33M_LPCPLUS	6 25 42
I	SMBUS_PCH_CLK	SMB_50S	SMB	SMBUS_PCH_CLK	16 43
	SMBUS_PCH_DATA	SMB_50S	SMB	SMBUS_PCH_DATA	16 43
	SMBUS_PCH_0_CLK	SMB_50S	SMB	SMB_PCH_0_CLK	16 43
	SMBUS_PCH_0_DATA	SMB_50S	SMB	SMB_PCH_0_DATA	16 43
	SMBUS_PCH_1_CLK	SMB_50S	SMB	SMB_PCH_1_CLK	16 43
	SMBUS_PCH_1_DATA	SMB_50S	SMB	SMB_PCH_1_DATA	16 43
I	HDA_BIT_CLK	HDA_50S	HDA	HDA_BIT_CLK	6 16 39
		HDA_50S	HDA	HDA_BIT_CLK_R	16
	HDA_SYNC	HDA_50S	HDA	HDA_SYNC	6 16 39
		HDA_50S	HDA	HDA_SYNC_R	16
	HDA_RST_L	HDA_50S	HDA	HDA_RST_R_L	16
		HDA_50S	HDA	HDA_RST_L	6 16 39
I	HDA_SDIN0	HDA_50S	HDA	HDA_SDIN0	6 16 39
		HDA_50S	HDA	AUD_SDI_R	
I	HDA_SDOUT	HDA_50S	HDA	HDA_SDOUT	6 16 39
		HDA_50S	HDA	HDA_SDOUT_R	16
I	PM_SUS_CLK	CLK_SLOW_55S	CLK_SLOW	PM_CLK32K_SUSCLK	
I	SPT_CLK	SPT_55S	SPT	SPI_CLK_R	16 42
		SPT_55S	SPT	SPI_CLK	42
	SPT_MOSI	SPT_55S	SPT	SPI_MOSI_R	16 42
		SPT_55S	SPT	SPT_MOSI	42
	SPT_MISO	SPT_55S	SPT	SPT_MISO	16 42
	SPT_CS0	SPT_55S	SPT	SPT_CS0_R_L	16 42
		SPT_55S	SPT	SPI_CS0_L	42
		SPT_55S	SPT	SPI_MLB_CLK	42 49
		SPT_55S	SPT	SPI_MLB_MOSI	42 49
		SPT_55S	SPT	SPI_MLB_MISO	42 49
		SPT_55S	SPT	SPI_MLB_CS_L	42 49
I		PCIE_85D	PCIE	PCIE_ENET_R2D_P	
		PCIE_85D	PCIE	PCIE_ENET_R2D_N	
	PCIE_ENET_R2D	PCIE_85D	PCIE	PCIE_ENET_R2D_C_P	
		PCIE_85D	PCIE	PCIE_ENET_R2D_C_N	
	PCIE_ENET_D2R	PCIE_85D	PCIE	PCIE_ENET_D2R_P	
		PCIE_85D	PCIE	PCIE_ENET_D2R_N	
I		PCIE_85D	PCIE	PCIE_ENET_D2R_C_P	
		PCIE_85D	PCIE	PCIE_ENET_D2R_C_N	
		PCIE_85D	PCIE	PCIE_AP_R2D_P	6 36
		PCIE_85D	PCIE	PCIE_AP_R2D_N	6 36
	PCIE_AP_R2D	PCIE_85D	PCIE	PCIE_AP_R2D_C_P	16 36
		PCIE_85D	PCIE	PCIE_AP_R2D_C_N	16 36
I	PCIE_AP_D2R	PCIE_85D	PCIE	PCIE_AP_R2D_C_N	6 16 36
		PCIE_85D	PCIE	PCIE_AP_D2R_N	6 16 36
		PCIE_85D	PCIE	PCIE_FW_R2D_P	
		PCIE_85D	PCIE	PCIE_FW_R2D_N	
	PCIE_FW_R2D	PCIE_85D	PCIE	PCIE_FW_R2D_C_P	
		PCIE_85D	PCIE	PCIE_FW_R2D_C_N	
I		PCIE_85D	PCIE	PCIE_FW_R2D_C_N	
		PCIE_85D	PCIE	PCIE_FW_D2R_P	
	PCIE_FW_D2R	PCIE_85D	PCIE	PCIE_FW_D2R_P	
		PCIE_85D	PCIE	PCIE_FW_D2R_N	
		PCIE_85D	PCIE	PCIE_FW_D2R_C_P	
		PCIE_85D	PCIE	PCIE_FW_D2R_C_N	
I	PCIE_AP_D2R	PCIE_85D	PCIE	CONN_PCIE_AP_D2R_P	
		PCIE_85D	PCIE	CONN_PCIE_AP_D2R_N	
	PCIE_AP_R2D	PCIE_85D	PCIE	CONN_PCIE_AP_R2D_P	
		PCIE_85D	PCIE	CONN_PCIE_AP_R2D_N	
		CLK_PCIE_90D	CLK_PCIE	PEG_CLK100M_P	8 16
		CLK_PCIE_90D	CLK_PCIE	PEG_CLK100M_N	8 16
I	PCIE_CLK100M_ENET	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_ENET_P	
		CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_ENET_N	
	MCP_PF1_REFCLK	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_AP_P	6 16 36
		CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_AP_N	6 16 36
	MCP_PF2_REFCLK	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_FW_P	
		CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_FW_N	
I		CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_EXCARD_P	8 16
		CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_EXCARD_N	8 16
	CPU1_27P4S	CPU1_COMP		PCH_VSS_NCTF<1>	6
	CPU1_27P4S	CPU1_COMP		PCH_VSS_NCTF<2>	6
	CPU1_27P4S	CPU1_COMP		PCH_VSS_NCTF<5>	6
I	CPU1_27P4S	CPU1_COMP		TP_PCH_VSS_NCTF<7>	6
	CPU1_27P4S	CPU1_COMP		PCH_VSS_NCTF<9>	6 69
	CPU1_27P4S	CPU1_COMP		PCH_VSS_NCTF<9>	6 69
	CPU1_27P4S	CPU1_COMP		PCH_VSS_NCTF<11>	6
	CPU1_27P4S	CPU1_COMP		PCH_VSS_NCTF<12>	6
	CPU1_27P4S	CPU1_COMP		PCH_VSS_NCTF<15>	6
	CPU1_27P4S	CPU1_COMP		PCH_VSS_NCTF<17>	6
	CPU1_27P4S	CPU1_COMP		PCH_VSS_NCTF<19>	6
	CPU1_27P4S	CPU1_COMP		PCH_VSS_NCTF<21>	6
	CPU1_27P4S	CPU1_COMP		PCH_VSS_NCTF<22>	6
	CPU1_27P4S	CPU1_COMP		PCH_VSS_NCTF<25>	6
	CPU1_27P4S	CPU1_COMP		PCH_VSS_NCTF<27>	6
CPU1_27P4S	CPU1_COMP		PCH_VSS_NCTF<29>	6	

Chipset Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
DP_EXT_A ML	DP_85D	DISPLAYPORT	DP_EXT_A ML C P<3..0>	8 63
DP_EXT_A ML	DP_85D	DISPLAYPORT	DP_EXT_A ML C N<3..0>	8 63
	DP_85D	DISPLAYPORT	DP_EXT_A ML P<3..0>	63
	DP_85D	DISPLAYPORT	DP_EXT_A ML N<3..0>	63
DP_EXT_A AUXCH	DP_85D	DISPLAYPORT	DP_EXT_A AUXCH C	8 63
DP_EXT_A AUXCH	DP_85D	DISPLAYPORT	DP_EXT_A AUXCH C N	8 63
	DP_85D	DISPLAYPORT	DP_EXT_A AUXCH P	63
	DP_85D	DISPLAYPORT	DP_EXT_A AUXCH N	63
DP_INT_ML	DP_85D	DISPLAYPORT	DP_INT_ML C P<3..0>	62
DP_INT_ML	DP_85D	DISPLAYPORT	DP_INT_ML C N<3..0>	62
	DP_85D	DISPLAYPORT	DP_INT_ML P<3..0>	9 62
	DP_85D	DISPLAYPORT	DP_INT_ML N<3..0>	9 62
	DP_85D	DISPLAYPORT	DP_INT_ML F P<3..0>	6 62
	DP_85D	DISPLAYPORT	DP_INT_ML F N<3..0>	6 62
DP_INT_AUXCH	DP_85D	DISPLAYPORT	DP_INT_AUX_CH C P	6 62
DP_INT_AUXCH	DP_85D	DISPLAYPORT	DP_INT_AUX_CH C N	6 62
	DP_85D	DISPLAYPORT	DP_INT_AUX_CH P	9 62
	DP_85D	DISPLAYPORT	DP_INT_AUX_CH N	9 62
	PCIE_85D	PCIE	PCIE_T29_R2D C P<3..0>	8 33
	PCIE_85D	PCIE	PCIE_T29_R2D C N<3..0>	8 33
PCIE_PEG_D2R_LANE3	PCIE_85D	PCIE	PCIE_T29_D2R P<3>	8 33
PCIE_PEG_D2R_LANE2	PCIE_85D	PCIE	PCIE_T29_D2R P<2>	8 33
PCIE_PEG_D2R_LANE1	PCIE_85D	PCIE	PCIE_T29_D2R P<1>	8 33
PCIE_PEG_D2R_LANE0	PCIE_85D	PCIE	PCIE_T29_D2R P<0>	8 33
PCIE_PEG_D2R_LANE3	PCIE_85D	PCIE	PCIE_T29_D2R N<3>	8 33
PCIE_PEG_D2R_LANE2	PCIE_85D	PCIE	PCIE_T29_D2R N<2>	8 33
PCIE_PEG_D2R_LANE1	PCIE_85D	PCIE	PCIE_T29_D2R N<1>	8 33
PCIE_PEG_D2R_LANE0	PCIE_85D	PCIE	PCIE_T29_D2R N<0>	8 33
PCIE_PEG_R2D_LANE3	PCIE_85D	PCIE	PCIE_T29_R2D P<3>	33
PCIE_PEG_R2D_LANE2	PCIE_85D	PCIE	PCIE_T29_R2D P<2>	33
PCIE_PEG_R2D_LANE1	PCIE_85D	PCIE	PCIE_T29_R2D P<1>	33
PCIE_PEG_R2D_LANE0	PCIE_85D	PCIE	PCIE_T29_R2D P<0>	33
PCIE_PEG_R2D_LANE3	PCIE_85D	PCIE	PCIE_T29_R2D N<3>	33
PCIE_PEG_R2D_LANE2	PCIE_85D	PCIE	PCIE_T29_R2D N<2>	33
PCIE_PEG_R2D_LANE1	PCIE_85D	PCIE	PCIE_T29_R2D N<1>	33
PCIE_PEG_R2D_LANE0	PCIE_85D	PCIE	PCIE_T29_R2D N<0>	33
	PCIE_85D	PCIE	PCIE_T29_D2R C P<3..0>	33
	PCIE_85D	PCIE	PCIE_T29_D2R C N<3..0>	33
PCIE_CLK100M_T29	CLK_PCFE_90D	CLK_PCFE	PCIE_CLK100M_T29 P	16 33
PCIE_CLK100M_T29	CLK_PCFE_90D	CLK_PCFE	PCIE_CLK100M_T29 N	16 33

Clock Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
	SYSCLK_CLK32K_RTC	CLK_SLOW_55S	CLK_SLOW	SYSCLK_CLK32K_RTC	10 25
	SYSCLK_CLK25M_SB	CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_SB	10 25
		CLK_25M_55S	CLK_25M_SB_R	SYSCLK_CLK25M_SB_R	16 25
		CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_ENET	
		CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_ENET_R	
	SYSCLK_CLK25M_T29	CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_T29	25 33
		CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_T29_R	33

CAESAR IV (Ethernet) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_505	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_3X	*	=3:1_SPACING	?

SOURCE: Broadcom 5764-DS04-RDS Page 38

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_CR_DATA	*	SMIL	?

CAESAR IV (Ethernet PHY) Constraints

[illegible]

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_MDI	*	0.6 MM	?

SOURCE: Broadcom 5764-DS04-RDS Page 38

FireWire Interface Constraints

[illegible]

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FW_TP	*	=3:1_SPACING	?

Ethernet Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
		FNET 50S	FNET 3X	BCM5764 CLK25M XTALI
		FNET 50S	FNET 3X	BCM5764 CLK25M XTALO
		FNET 50S	FNET 3X	FNET RESET_L
	FNET_MDI	FNET 100D	FNET_MDI	FNET MDI P<3..0>
		FNET 100D	FNET_MDI	FNET MDI N<3..0>
				
	CR_DATA	FNET 50S	FNET_CR_DATA	FNET_CR_DATA<7..0>
	CR_DATA	FNET 50S	FNET_CR_DATA	FNET_CR_CMD
	CR_CLK	FNET 50S	FNET_CR_DATA	FNET_CR_CLK
	CR_DATA	FNET 50S	FNET_CR_DATA	SDCONN_DATA<7..0>
	CR_DATA	FNET 50S	FNET_CR_DATA	SDCONN_CMD
	CR_CLK	FNET 50S	FNET_CR_DATA	SDCONN_CLK

FireWire Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
FW_P0_TPA		FW_110D	FW_TP	FW_P0_TPA_P
FW_P0_TPA		FW_110D	FW_TP	FW_P0_TPA_N
FW_P0_TPB		FW_110D	FW_TP	FW_P0_TPB_P
FW_P0_TPB		FW_110D	FW_TP	FW_P0_TPB_N
FW_P1_TPA		FW_110D	FW_TP	FW_P1_TPA_P
FW_P1_TPA		FW_110D	FW_TP	FW_P1_TPA_N
FW_P1_TPB		FW_110D	FW_TP	FW_P1_TPB_P
FW_P1_TPB		FW_110D	FW_TP	FW_P1_TPB_N
Port 2 Not Used				

DisplayPort Signal Constraints

NOTE: DisplayPort Physical/Spacing Constraints provided by Chipset or GPU page.

T29 I2C Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
T29_T2C_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29_I2C	*	=2x_DIELECTRIC	?

T29 SPI Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
T29_SPT_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29_SPI	*	=2x_DIELECTRIC	?

DP/T29 Connector Signal Constraints

[illegible]

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T290P	*	=5x_DIELECTRIC	?	T290P	TOP, BOTTOM	=7x_DIELECTRIC	?

SOURCE: Bill Cornelius's T29 Routing Notes

T29 Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
DP		DP_85D	DISPLAYPORT	DP T29SNK0 ML C P<3..0>
DP		DP_85D	DISPLAYPORT	DP T29SNK0 ML C N<3..0>
DP	DP_T29SNK0_MI	DP_85D	DISPLAYPORT	DP T29SNK0 ML P<3..0>
DP	DP_T29SNK0_MI	DP_85D	DISPLAYPORT	DP T29SNK0 ML N<3..0>
DP		DP_85D	DISPLAYPORT	DP T29SNK0 AUXCH C P
DP		DP_85D	DISPLAYPORT	DP T29SNK0 AUXCH C N
DP	DP_T29SNK0_AUXCH	DP_85D	DISPLAYPORT	DP T29SNK0 AUXCH P
DP	DP_T29SNK0_AUXCH	DP_85D	DISPLAYPORT	DP T29SNK0 AUXCH N
DP		DP_85D	DISPLAYPORT	DP T29SNK1 ML C P<3..0>
DP		DP_85D	DISPLAYPORT	DP T29SNK1 ML C N<3..0>
DP	DP_T29SNK1_MI	DP_85D	DISPLAYPORT	DP T29SNK1 ML P<3..0>
DP	DP_T29SNK1_MI	DP_85D	DISPLAYPORT	DP T29SNK1 ML N<3..0>
DP		DP_85D	DISPLAYPORT	DP T29SNK1 AUXCH C P
DP		DP_85D	DISPLAYPORT	DP T29SNK1 AUXCH C N
DP	DP_T29SNK1_AUXCH	DP_85D	DISPLAYPORT	DP T29SNK1 AUXCH P
DP	DP_T29SNK1_AUXCH	DP_85D	DISPLAYPORT	DP T29SNK1 AUXCH N
DP				
DP		T29_T2C_55S	T29_T2C	T2C T29_SCL
DP		T29_T2C_55S	T29_T2C	T2C T29_SDA
DP	T29_SPT_CLK	T29_SPT_55S	T29_SPT	T29_SPT_CLK
DP	T29_SPT_M0ST	T29_SPT_55S	T29_SPT	T29_SPT_M0ST
DP	T29_SPT_MISO	T29_SPT_55S	T29_SPT	T29_SPT_MISO
DP	T29_SPT_CS_1	T29_SPT_55S	T29_SPT	T29_SPT_CS_1
DP				
DP		T29DP_80D	T29DP	T29_R2D C P<3..0>
DP		T29DP_80D	T29DP	T29_R2D C N<3..0>
DP		T29DP_80D	T29DP	T29_D2R P<3..0>
DP		T29DP_80D	T29DP	T29_D2R N<3..0>
DP				
DP	T29_R2D0	T29DP_80D	T29DP	T29_R2D P<0>
DP	T29_R2D0	T29DP_80D	T29DP	T29_R2D N<0>
DP	T29_R2D1	T29DP_80D	T29DP	T29_R2D P<1>
DP	T29_R2D1	T29DP_80D	T29DP	T29_R2D N<1>
DP		T29DP_80D	T29DP	T29_R2D C F P<1..0>
DP		T29DP_80D	T29DP	T29_R2D C F N<1..0>
DP	T29_D2R0	T29DP_80D	T29DP	T29_D2R C P<0>
DP	T29_D2R0	T29DP_80D	T29DP	T29_D2R C N<0>
DP	T29_D2R1	T29DP_80D	T29DP	T29_D2R C P<1>
DP	T29_D2R1	T29DP_80D	T29DP	T29_D2R C N<1>
DP		T29DP_80D	T29DP	T29DPA D2R1 AUXCH P
DP		T29DP_80D	T29DP	T29DPA D2R1 AUXCH N
DP		T29DP_80D	T29DP	DP SDRVA ML C P<3..0>
DP		T29DP_80D	T29DP	DP SDRVA ML C N<3..0>
DP		T29DP_80D	T29DP	DP SDRVA ML R P<3..0>
DP		T29DP_80D	T29DP	DP SDRVA ML R N<3..0>
DP	DP_SDRVA_MI_EVEN	T29DP_80D	T29DP	DP SDRVA ML P<0>
DP	DP_SDRVA_MI_EVEN	T29DP_80D	T29DP	DP SDRVA ML N<0>
DP	DP_SDRVA_MI_ODD	T29DP_80D	T29DP	DP SDRVA ML P<1>
DP	DP_SDRVA_MI_ODD	T29DP_80D	T29DP	DP SDRVA ML N<1>
DP	DP_SDRVA_MI_EVEN	T29DP_80D	T29DP	DP SDRVA ML P<2>
DP	DP_SDRVA_MI_EVEN	T29DP_80D	T29DP	DP SDRVA ML N<2>
DP	DP_SDRVA_MI_ODD	T29DP_80D	T29DP	DP SDRVA ML P<3>
DP	DP_SDRVA_MI_ODD	T29DP_80D	T29DP	DP SDRVA ML N<3>
DP	DP_SDRVA_AUXCH	T29DP_80D	T29DP	DP SDRVA AUXCH P
DP	DP_SDRVA_AUXCH	T29DP_80D	T29DP	DP SDRVA AUXCH N
DP		T29DP_80D	T29DP	DP SDRVA AUXCH C P
DP		T29DP_80D	T29DP	DP SDRVA AUXCH C N
DP				
DP	T29DPA_MI_ODD			T29DPA ML P<1>
DP	T29DPA_MI_ODD			T29DPA ML N<1>
DP	T29DPA_MI_ODD			T29DPA ML P<3>
DP	T29DPA_MI_ODD			T29DPA ML N<3>
DP		T29DP_80D	T29DP	T29DPA ML P<3..0>
DP		T29DP_80D	T29DP	T29DPA ML N<3..0>
DP		T29DP_80D	T29DP	T29DPA ML C P<3..0>
DP		T29DP_80D	T29DP	T29DPA ML C N<3..0>
DP	DP_A_EXT_AUXCH	T29DP_80D	T29DP	DP A EXT AUXCH P
DP	DP_A_EXT_AUXCH	T29DP_80D	T29DP	DP A EXT AUXCH N

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SENSE_I701_555	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
THERM_I701_555	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
DIFFPAIR	*	=1:1_DIFFPAIR			=1:1_DIFFPAIR	=1:1_DIFFPAIR	=1:1_DIFFPAIR

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE	*	=2:1_SPACING	7
THERM	*	=2:1_SPACING	7
AUDIO	*	=2:1_SPACING	7

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_COMP	GND	*	GND_P2MM
CPU_VCCSENSE	GND	*	GND_P2MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENETCONN	*	25 MILS	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_MDI	GND	*	GND_P2MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	GND	*	GND_P2MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.20 MM	1000
PWR_P2MM	*	0.20 MM	1000

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	GND	*	GND_P2Mm
PCIE	GND	*	GND_P2Mm
SATA	GND	*	GND_P2Mm
USB	GND	*	GND_P2Mm
CLK_PCIE	SB_POWER	*	PWR_P2Mm
SATA	SB_POWER	*	PWR_P2Mm
USB	SB_POWER	*	PWR_P2Mm

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	GND	*	GND_P2MH
MEM_CMD	GND	*	GND_P2MH
MEM_CTRL	GND	*	GND_P2MH
MEM_DATA	GND	*	GND_P2MH
MEM_DQS	GND	*	GND_P2MH

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
LVDS	GND	*	GND_P2MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_405 OVERRIDE	+	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	400 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_720 OVERRIDE	+	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	400 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_375 OVERRIDE	+	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	400 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_850 OVERRIDE	+	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	400 MIL OVERRIDE	OVERRIDE	OVERRIDE
PCIE_950 OVERRIDE	+	OVERRIDE	OVERRIDE	0.075 MM OVERRIDE	10 MM OVERRIDE	OVERRIDE	OVERRIDE
USB_850 OVERRIDE	TOP	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
CPU_27P4S OVERRIDE	TOP	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	400 MIL OVERRIDE	OVERRIDE	OVERRIDE
CLK_PCIE_900 OVERRIDE	TOP	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	400 MIL OVERRIDE	OVERRIDE	OVERRIDE

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
	PHYSICAL		SYMBOLIC	
	FNET 180D	FNETCONN	ENETCONN P<3..0>	
	FNET 180D	FNETCONN	ENETCONN N<3..0>	
	SATA 90D	SATA	SATA ODD D2R UF P	
	SATA 90D	SATA	SATA ODD D2R UF N	
	SATA 90D	SATA	SATA HDD D2R RDRVR OUT P	
	SATA 90D	SATA	SATA HDD D2R RDRVR OUT N	
	SATA 90D	SATA	SATA HDD R2D RDRVR IN P	
	SATA 90D	SATA	SATA HDD R2D RDRVR IN N	
R225	SATA 90D	SATA	SATA HDD D2R RDRVR IN P	
R226	SATA 90D	SATA	SATA HDD D2R RDRVR IN N	
R227	SATA 90D	SATA	SATA HDD R2D RDRVR OUT P	
R228	SATA 90D	SATA	SATA HDD R2D RDRVR OUT N	
	SENSE_DIFFPAIR	THERM 1T01 55S THERM	CPUTHMSNS D2 P	
		THERM 1T01 55S THERM	CPUTHMSNS D2 N	
	CPU_THERMD	THERM 1T01 55S THERM	CPU THERMD P	
		THERM 1T01 55S THERM	CPU THERMD N	
	SENSE_DIFFPAIR	THERM 1T01 55S THERM	T29 THERMD P	
		THERM 1T01 55S THERM	T29 THERMD N	
	SENSE_DIFFPAIR	THERM 1T01 55S THERM	T29 MLBB0T THMSNS P	
		THERM 1T01 55S THERM	T29 MLBB0T THMSNS N	
	SENSE_DIFFPAIR	SENSE 1T01 55S SENSE	ISNS HS COMPUTING N	
		SENSE 1T01 55S SENSE	ISNS HS COMPUTING P	
	SENSE_DIFFPAIR	SENSE 1T01 55S SENSE	ISNS HS OTHER N	
		SENSE 1T01 55S SENSE	ISNS HS OTHER P	
	SENSE_DIFFPAIR	SENSE 1T01 55S SENSE	CPUVCCIOS0 CS N	
		SENSE 1T01 55S SENSE	CPUVCCIOS0 CS P	
	SENSE_DIFFPAIR	SENSE 1T01 55S SENSE	CPUIMVP ISNS1 P	
		SENSE 1T01 55S SENSE	CPUIMVP ISNS1 N	
	SENSE_DIFFPAIR	SENSE 1T01 55S SENSE	CPUIMVP ISNS2 P	
		SENSE 1T01 55S SENSE	CPUIMVP ISNS2 N	
	SENSE_DIFFPAIR	SENSE 1T01 55S SENSE	CPUIMVP ISNS1G P	
		SENSE 1T01 55S SENSE	CPUIMVP ISNS1G N	
	SENSE_DIFFPAIR	SENSE 1T01 55S SENSE	CPUIMVP ISUM R P	
		SENSE 1T01 55S SENSE	CPUIMVP ISUM R N	
	SENSE_DIFFPAIR	SENSE 1T01 55S SENSE	CPUIMVP ISUMG R P	
		SENSE 1T01 55S SENSE	CPUIMVP ISUMG R N	
	SENSE_DIFFPAIR	SENSE 1T01 55S SENSE	CPUIMVP ISNS P	
		SENSE 1T01 55S SENSE	CPUIMVP ISNS N	
	SENSE_DIFFPAIR	SENSE 1T01 55S SENSE	VCCSA50 CS P	
		SENSE 1T01 55S SENSE	VCCSA50 CS N	
	SENSE_DIFFPAIR	SENSE 1T01 55S SENSE	CPUIMVP ISUMG P	
		SENSE 1T01 55S SENSE	CPUIMVP ISUMG N	
	SENSE_DIFFPAIR	SENSE 1T01 55S SENSE	ISNS CPU N	
		SENSE 1T01 55S SENSE	ISNS CPU P	
	SENSE_DIFFPAIR	SENSE 1T01 55S SENSE	ISNS HDD N	
		SENSE 1T01 55S SENSE	ISNS HDD P	
	SENSE_DIFFPAIR	SENSE 1T01 55S SENSE	ISNS HDD R N	
		SENSE 1T01 55S SENSE	ISNS HDD R P	
	SENSE_DIFFPAIR	SENSE 1T01 55S SENSE	ISNS LCDBLKT N	
		SENSE 1T01 55S SENSE	ISNS LCDBLKT P	
	SENSE_DIFFPAIR	SENSE 1T01 55S SENSE	ISNS ODD N	
		SENSE 1T01 55S SENSE	ISNS ODD P	
	SENSE_DIFFPAIR	SENSE 1T01 55S SENSE	ISNS ODD R N	
		SENSE 1T01 55S SENSE	ISNS ODD R P	
	SENSE_DIFFPAIR	SENSE 1T01 55S SENSE	ISNS 1V5 S3 N	
		SENSE 1T01 55S SENSE	ISNS 1V5 S3 P	
	SENSE_DIFFPAIR	SENSE 1T01 55S SENSE	ISNS P1V8GPU R N	
		SENSE 1T01 55S SENSE	ISNS P1V8GPU R P	
R835	SENSE_DIFFPAIR	SENSE 1T01 55S SENSE	ISNS AIRPORT N	
R836		SENSE 1T01 55S SENSE	ISNS AIRPORT P	
R229	LVDS_90D	LVDS	LVDS_CONN A CLK F N	
R230	LVDS_90D	LVDS	LVDS_CONN A CLK F P	

Audio Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
	SPKRAMP_INR	DIFFEPAIR	AUDIO	SPKRAMP_INR_P	6 39 50 73
		DIFFEPAIR	AUDIO	SPKRAMP_INR_N	6 39 50 73
	MAX98300_R	DIFFEPAIR	AUDIO	MAX98300_R_P	50
		DIFFEPAIR	AUDIO	MAX98300_R_N	50

ELECTRICAL_CONSTRAINT_SET		NET TYPE	
	PINVALUE		SPINCODE
PCIE CLK100M AP	CLK PCIE 98D	CLK PCIE	PCIE CLK100M AP CONN P
	CLK PCIE 98D	CLK PCIE	PCIE CLK100M AP CONN N
	1T01_DIFFPAIR		CHGR CSI R P
	1T01_DIFFPAIR		CHGR CSI R N
	1T01_DIFFPAIR		CHGR CS0 R P
	1T01_DIFFPAIR		CHGR CS0 R N
(USB_EXTN)	USB_R5D	USB	USB2 EXTA MUXED P
(USB_EXTN)	USB_R5D	USB	USB2 EXTA MUXED N
(USB_EXTN)	USB_R5D	USB	USB2 LT1 P
(USB_EXTN)	USB_R5D	USB	USB2 LT1 N
	USB_R5D	USB	CONN USB2 BT P
	USB_R5D	USB	CONN USB2 BT N
	USB_R5D	USB	USB LT2 P
	USB_R5D	USB	USB LT2 N
	DP_R5D	DISPLAYPORT	DP IG AUX CH C P
	DP_R5D	DISPLAYPORT	DP IG AUX CH C N
SPK_OUT	DIFFPAIR	AUDIO	SPKRAMP L P OUT
SPK_OUT	DIFFPAIR	AUDIO	SPKRAMP L N OUT
SPK_OUT	DIFFPAIR	AUDIO	SPKRAMP SUB P OUT
SPK_OUT	DIFFPAIR	AUDIO	SPKRAMP SUB N OUT
SPK_OUT	DIFFPAIR	AUDIO	SPKRAMP R P OUT
SPK_OUT	DIFFPAIR	AUDIO	SPKRAMP R N OUT
AUD_DIFF	1T01_DIFFPAIR	AUDIO	SSM2315 SUB N
AUD_DIFF	1T01_DIFFPAIR	AUDIO	SSM2315 SUB P
AUD_DIFF	1T01_DIFFPAIR	AUDIO	SSM2315 L N
AUD_DIFF	1T01_DIFFPAIR	AUDIO	SSM2315 L P
AUD_DIFF	1T01_DIFFPAIR	AUDIO	SSM2315 R N
AUD_DIFF	1T01_DIFFPAIR	AUDIO	SSM2315 R P
AUD_DIFF	1T01_DIFFPAIR	AUDIO	AUD L02 N R
AUD_DIFF	1T01_DIFFPAIR	AUDIO	AUD L02 P R
AUD_DIFF	1T01_DIFFPAIR	AUDIO	AUD L01 N R
AUD_DIFF	1T01_DIFFPAIR	AUDIO	AUD L01 P R
AUD_DIFF	1T01_DIFFPAIR	AUDIO	AUD L02 N L
AUD_DIFF	1T01_DIFFPAIR	AUDIO	AUD L02 P L
AUD_DIFF	1T01_DIFFPAIR	AUDIO	SPKRAMP INR P
AUD_DIFF	1T01_DIFFPAIR	AUDIO	SPKRAMP INL N
SPKRAMP_INR	DIFFPAIR	AUDIO	SPKRAMP INR P
AUD_DIFF	DIFFPAIR	AUDIO	SPKRAMP INR N
AUD_DIFF	1T01_DIFFPAIR	AUDIO	SPKRAMP INSUB P
AUD_DIFF	1T01_DIFFPAIR	AUDIO	SPKRAMP INSUB N
	USB_R5D	USB	USB TPAD R P
	USB_R5D	USB	USB TPAD R N
		SB_POWER	PP3V3 S5
		SB_POWER	PP3V3 S0
		GND	GND

Misc Net Properties

		NET_TYPE		
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING		
USEW (USB_EXTN)	USB_85D	USB	USB_EXTN_MUXED_P	
USEB (USB_EXTN)	USB_85D	USB	USB_EXTN_MUXED_N	
USEB (USB_EXTN)	USB_85D	USB	USB LT1_P	
USEB (USB_EXTN)	USB_85D	USB	USB LT1_N	
USEB (USB_TPAD)	USB_85D	USB	USB_TPAD_CONN_P	6 48
USEB (USB_TPAD)	USB_85D	USB	USB_TPAD_CONN_N	6 48
USEB SMBUS_SMC_MGMT_SDA	SMB_55S	SMB	I2C SMC SMS_SDA_R	
USEB SMBUS_SMC_MGMT_SCL	SMB_55S	SMB	I2C SMC SMS_SCL_R	
USEB	SMB_55S	SMB	I2C TCON_SCL	
USEB	SMB_55S	SMB	I2C TCON_SDA	
USEB	SMB_55S	SMB	I2C TCON_SCL_CONN	
USEB	SMB_55S	SMB	I2C TCON_SDA_CONN	

Memory Constraint Relaxations

Allow 0.127 mm necks for >0.127 mm lines for ARD fanout.

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_72D	BOTTOM			0.127 MM	6.35 MM		
MEM_85D	TOP			0.1 MM	6.35 MM		

SYNC MASTER-CONSTRAINTS		SYNC DATE=04/08/2013	
PAGE TITLE			
Project Specific Constraints			
	DRAWING NUMBER		SIZE
	051-8871		
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		2.5.0	
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8	7	6	5	4	3	2	1
K90i Board-Specific Spacing & Physical Constraints							
BOARD LAYERS				BOARD AREAS		BOARD UNITS (MIL OR MM)	ALLEGRO VERSION
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, ISL8, ISL9, ISL10, ISL11, BOTTOM				NO_TYPE, BGA		MM	15.5.1
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=50_OHM_SE	=50_OHM_SE	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP, BOTTOM	Y	0.110 MM	0.090 MM			
50_OHM_SE	*	Y	0.090 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHM_SE	TOP, BOTTOM	Y	0.170 MM	0.170 MM			
40_OHM_SE	ISL3, ISL4, ISL9, ISL10	Y	0.140 MM	0.140 MM	=STANDARD	=STANDARD	=STANDARD
40_OHM_SE	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
37_OHM_SE	TOP, BOTTOM	Y	0.195 MM	0.1 MM			
37_OHM_SE	ISL3, ISL4, ISL9, ISL10	Y	0.160 MM	0.1 MM	=STANDARD	=STANDARD	=STANDARD
37_OHM_SE	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27P4_OHM_SE	TOP, BOTTOM	Y	0.310 MM	0.2 MM			
27P4_OHM_SE	*	Y	0.250 MM	0.2 MM	=STANDARD	=STANDARD	=STANDARD
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	TOP, BOTTOM	Y	0.090 MM	0.090 MM			
55_OHM_SE	*	Y	0.076 MM	0.076 MM	=STANDARD	=STANDARD	=STANDARD
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
72_OHM_DIFF	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
72_OHM_DIFF	ISL3, ISL10	Y	0.135 MM	0.135 MM		0.130 MM	0.130 MM
72_OHM_DIFF	ISL4, ISL9	Y	0.155MM	0.155 MM		0.130 MM	0.130 MM
72_OHM_DIFF	TOP, BOTTOM	Y	0.165 MM	0.165 MM		0.130 MM	0.130 MM
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
85_OHM_DIFF	ISL3, ISL10	Y	0.095 MM	0.1 MM		0.170 MM	0.170 MM
85_OHM_DIFF	ISL4, ISL9	Y	0.115 MM	0.115 MM		0.170 MM	0.170 MM
85_OHM_DIFF	TOP, BOTTOM	Y	0.130 MM	0.130 MM		0.195 MM	0.195 MM
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	ISL3, ISL10	Y	0.089 MM	0.089 MM		0.210 MM	0.210 MM
90_OHM_DIFF	ISL4, ISL9	Y	0.105 MM	0.105 MM		0.210 MM	0.210 MM
90_OHM_DIFF	TOP, BOTTOM	Y	0.115 MM	0.115 MM		0.210 MM	0.210 MM
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
100_OHM_DIFF	ISL3, ISL10	Y	0.074 MM	0.074 MM		0.250 MM	0.250 MM
100_OHM_DIFF	ISL4, ISL9	Y	0.085 MM	0.085 MM		0.250 MM	0.250 MM
100_OHM_DIFF	TOP, BOTTOM	Y	0.091 MM	0.091 MM		0.200 MM	0.200 MM
NOTE: 110_DIFF is 110-ohms differential impedance on outer layers and 105-ohms on inner layers.							
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
110_OHM_DIFF	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
110_OHM_DIFF	ISL3, ISL10	N	0.070 MM	0.070 MM		0.330 MM	0.330 MM
110_OHM_DIFF	ISL4, ISL9	Y	0.071 MM	0.071 MM		0.300 MM	0.300 MM
110_OHM_DIFF	TOP, BOTTOM	Y	0.077 MM	0.077 MM		0.280 MM	0.280 MM
NOTE: These are Intel recommended impedances for PEG, unused on K90i.							
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
48_OHM_SE	TOP, BOTTOM	Y	0.120 MM	0.165 MM			
48_OHM_SE	*	Y	0.097 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
80_OHM_DIFF	ISL3, ISL10	Y	0.110 MM	0.110 MM		0.170 MM	0.170 MM
80_OHM_DIFF	ISL4, ISL9	Y	0.129 MM	0.129 MM		0.170 MM	0.170 MM
80_OHM_DIFF	TOP, BOTTOM	Y	0.145 MM	0.145 MM		0.180 MM	0.180 MM
SPACING_RULE_SET LAYER LINE-TO-LINE SPACING WEIGHT							
DEFAULT	*	0.1 MM	?				
STANDARD	*	=DEFAULT	?				
BGA_P1MM	*	=DEFAULT	?				
BGA_P2MM	*	=DEFAULT	?				
NET_SPACING_TYPE1 NET_SPACING_TYPE2 AREA_TYPE SPACING_RULE_SET							
*	*	BGA	BGA_P1MM				
MEM_CLK	*	BGA	BGA_P2MM				
CLK_PCIE	*	BGA	BGA_P2MM				
CLK_SLOW	*	BGA	BGA_P2MM				
SPACING_RULE_SET LAYER LINE-TO-LINE SPACING WEIGHT							
1.5:1_SPACING	*	0.15 MM	?				
2:1_SPACING	*	0.2 MM	?				
2.5:1_SPACING	*	0.25 MM	?				
3:1_SPACING	*	0.3 MM	?				
4:1_SPACING	*	0.4 MM	?				
SPACING_RULE_SET LAYER LINE-TO-LINE SPACING WEIGHT							
2X_DIELECTRIC	*	0.140 MM	?				
3X_DIELECTRIC	*	0.210 MM	?				
4X_DIELECTRIC	*	0.280 MM	?				
5X_DIELECTRIC	*	0.350 MM	?				
7X_DIELECTRIC	*	0.490 MM	?				
PHYSICAL_RULE_SET LAYER ALLOW_ROUTE ON LAYER? MINIMUM LINE WIDTH MINIMUM NECK WIDTH MAXIMUM NECK LENGTH DIFFPAIR PRIMARY GAP DIFFPAIR NECK GAP							
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM
PHYSICAL_RULE_SET LAYER ALLOW_ROUTE ON LAYER? MINIMUM LINE WIDTH MINIMUM NECK WIDTH MAXIMUM NECK LENGTH DIFFPAIR PRIMARY GAP DIFFPAIR NECK GAP							
85_DIFF_BGA	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
85_DIFF_BGA	ISL3, ISL4	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM
85_DIFF_BGA	ISL9, ISL10	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM
NOTE: 85_DIFF_BGA is 85-ohms differential impedance on outer layers and 80-ohms on inner layers.							
PHYSICAL_RULE_SET LAYER ALLOW_ROUTE ON LAYER? MINIMUM LINE WIDTH MINIMUM NECK WIDTH MAXIMUM NECK LENGTH DIFFPAIR PRIMARY GAP DIFFPAIR NECK GAP							
90_DIFF_BGA	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
90_DIFF_BGA	ISL3, ISL4	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM
90_DIFF_BGA	ISL9, ISL10	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM
NOTE: 90_DIFF_BGA is 90-ohms differential impedance on outer layers and 85-ohms on inner layers.							
PHYSICAL_RULE_SET LAYER ALLOW_ROUTE ON LAYER? MINIMUM LINE WIDTH MINIMUM NECK WIDTH MAXIMUM NECK LENGTH DIFFPAIR PRIMARY GAP DIFFPAIR NECK GAP							
100_DIFF_BGA	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
100_DIFF_BGA	ISL3, ISL4	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM
100_DIFF_BGA	ISL9, ISL10	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM
NOTE: 100_DIFF_BGA is 100-ohms differential impedance on outer layers and 95-ohms on inner layers.							
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